

# Mathematical inference

science and logic

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Master Thesis Artificial Intelligence

2026



Open Universiteit



*“I got bored one day and I put everything on a bagel...  
**everything:** All my hopes and dreams, my old report  
cards, every breed of dog, every last personal ad on  
Craigslist, sesame, poppy seed, salt... and it collapsed  
in on itself. ’Cause you see, when you really put  
everything on a bagel, it becomes this: The truth.”*

*Jobu Tupaki in Everything Everywhere All At Once*



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## CHAPTER 1

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### **Prelude: Andante sostenuto**

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*“Science, including philosophy, is intrinsically polemical, and the polemical cannot be separated from the agonistic. In epochs in which great new things arise, the agonistic factor usually comes to the fore”*

Johan Huizinga in *Homo Ludens*

### 1.1 Inspiration

When I was an undergraduate studying physics, probability theory often felt oddly under-defined. In statistical physics, and even more sharply in quantum physics, probability seemed to appear sometimes as ignorance, sometimes as frequency, sometimes as a physical tendency, and sometimes merely as a useful rule for calculation. The mathematical formalism worked well enough; that was not the issue. The problem was that there didn't seem to be a coherent interpretation, so the underlying logical story remained inconsistent.

After my studies I discovered E. T. Jaynes' *Probability Theory: The Logic of Science* [Jay03]. That book was, for me, the book I would have liked to have read much earlier. Jaynes, also trained as a physicist, refused to treat probability as a bag of unrelated conventions. He presented it, instead, as a disciplined extension of logic: a way of saying what one should believe, given what one already knows. The attraction of this view was not only technical: it also restored a kind of intellectual honesty to the subject, since hidden assumptions and convenient interpretations could no longer be smuggled in without being noticed.

Yet, Jaynes' book stops before the story is finished. Part of this is literal: the book was posthumously released on the basis of an unfinished manuscript. Part of it is conceptual: Jaynes gave a powerful solution for probability, following and extending Cox's programme [Cox46; Cox61], but several of his broader complaints about mathematics, subjectivity, causality and scientific practice remain only partially resolved. This thesis is therefore both a tribute and an attempt at continuation. It asks what Jaynes' programme might look like if it were done in an epistemological, constructive, type-theoretic and causal setting.

### 1.2 Introduction

The central concern of this work is the process by which new knowledge is added to existing information. I will call this process "mathematical inference". It is not a marginal technicality; it is, or at the very least should be, the core of scientific reasoning. Jaynes was therefore not simply concerned with the correctness of a few specific statistical formulae: he thought that the core of science itself was at stake, because large parts of mathematical and statistical practice were built on foundations that were too permissive, too tolerant. His polemical tone makes sense in this context: if the problem is pervasive and foundational, then a mere technical correction would miss part of the point.

The urgency that is felt in Jaynes' text also perfectly fits the opening quotation from Huizinga on the polemical and agonistic nature of defending new scientific paradigms [Hui49]. This urgency becomes even more justified if one looks at two interrelated phenomena: independent analysts reaching divergent conclusions from the same data [Bot+20; Bre+22; Sch+21; Sil+18], and large replication projects that reveal failures to replicate important and accepted scientific findings [Cam+16; Kle+14; Ope15]. Based on these phenomena, it becomes abundantly clear that a well-defined theory and formalism for mathematical inference is not a mere theoretical exercise, but an urgent matter.

Three concerns run through this thesis:

1. Mathematics should have foundations that make assumptions explicit and prevent uncontrolled use of ideal objects like infinity;
2. Mathematical inference should be treated as a principled extension of logic, while also being honest about the limits of that derivation;
3. Scientific practice should be organised in such a way that good inference is encouraged rather than punished.

Jaynes addresses all three, but solves only part of the second. The present work takes these concerns together, since separating them would miss part of the problem.

The formal answer developed here starts from Martin-Löf-style type theory [DP24; Mar84; NPS90]. This is not chosen for foundational purity: it is chosen because constructive type theory forces one to be explicit about what is being constructed, in which context, and by which rules. It also connects mathematics to computation, so that the reconstruction is not just a philosophical exercise, but computationally checkable. For this reason, it is also a natural environment for rebuilding finite probability theory, especially in light of finitary refinements of Cox-style arguments [CW17].

Moreover, if probability is called subjective, then subjectivity should not merely mean that one may freely choose a prior – as is the case in Jaynes’ story. A subjective theory should also be able to represent which questions are in scope, which facts are remembered, which possibilities are being imagined, and how relations between concepts change. Also for this reason, Martin-Löf-style type theory proves to be an excellent basis, as its intuitionistic and epistemological foundations naturally support these operations on contexts [DP24; SN25]. However, to explicate that specific operations on the context change someone’s knowledge, someone’s *epistemic state*, modes and modalities are introduced. This introduction is made possible by modal dependent type theory, which extends Martin-Löf-style type theory so that these modes can be described natively [Gra+21; Gra23; GSB19].

The probabilistic part is not the whole story: Jaynes presents probability theory as a universal logic of science, but the mature formal theory of causal models was developed largely outside his work. To extend Jaynes’ probabilistic framework beyond prediction, Pearl’s structural causal models and do-calculus therefore have to be incorporated into the framework [Pea09; PM18].

Finally, inference is not performed in a vacuum. McElreath’s presentation of Bayesian modelling is valuable not only because it teaches calculations, but also because it keeps statistical inference close to modelling choices, scientific questions and workflow [McE20]. The reproducibility crisis and the diverging conclusions from competent researchers mentioned earlier illustrate that even technically competent inference can become unreliable inside a bad scientific pipeline [GL13; Ioa05]. Good science needs good inference, but also a good environment in which inference takes place.

## Scope and assumptions

This is a thesis in artificial intelligence, where artificial intelligence is understood broadly as the formalisation and implementation of inference. In this sense probability theory, logic, causality and proof checking all fall naturally within its

## 1. Prelude: Andante sostenuto

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scope. The aim is not to build a large applied AI system, but to investigate what kind of mathematical inference such systems should ideally perform when they reason under uncertainty, act under causal assumptions, or revise an epistemic state.

The thesis looks at inference on both the mathematical and sociological level. Mathematically, it asks which formal rules should ideally govern uncertain, subjective and causal inference. Sociologically, it asks in what scientific environment mathematical and statistical inferences are generally made. This distinction matters, because the proposed formal machinery may not fit comfortably inside every existing scientific habit; and if that turns out to be the case, the problem may lie – at least partly – with those habits.

The mathematical scope is deliberately restricted: finite probability spaces; finite causal models; explicitly representable contexts and epistemic states; and central objects that are constructible and implementable. Thus the technical development is discrete throughout, but more narrowly so: the results below are finite results, not a general theory of arbitrary discrete spaces. This restriction is not merely a convenience: it follows Jaynes’ suspicion of uncontrolled infinities; it follows the constructive spirit of Martin-Löf-style type theory; and it fits the computational aspect of artificial intelligence.

Selected formal parts are implemented in Lean 4 [MU21] where this is useful and feasible. Lean is used here as a proof checker for representative definitions, proofs and examples, not as a promise that every philosophical claim in the thesis will be mechanised: the point is more modest and more important. The central mathematical constructions should be expressible in a form precise enough for independent mechanical checking. Recent discussions of digitised proofs and AI-assisted mathematics make clear that this is becoming an increasingly relevant part of mathematical practice [Cas26; Kak26; Orn26; Slo26].

Several limitations follow directly from this scope: this work does not reconstruct all of classical probability theory; it does not provide a general theory of continuous probability spaces; it does not attempt to solve measure theory; it does not claim that Martin-Löf-style type theory is the only acceptable foundation for mathematics; it does not claim that Pearl’s framework captures everything that can meaningfully be called causality; and it does not pretend to give a complete empirical sociology of science. The aim is narrower: to construct a coherent finite framework in which probability, subjectivity and causality can be represented with as little hand-waving as possible, and to fit this framework within its practical context.

### Research questions

The thesis is guided by the following research questions:

1. Can Jaynes’ probabilistic framework be cast in the language of Martin-Löf-style type theory?
2. Can Jaynes’ probabilistic framework be extended so as to natively include subjectivity?

3. Can Jaynes' probabilistic framework be extended so as to include Pearl-style causal reasoning?
4. Can Jaynes' commentary on bad scientific practice be placed in a wider institutional context?

### Expected contributions

My first hope is to clarify Jaynes' programme. Jaynes' probability theory will be presented as an attempt to treat probability as extended logic, while the thesis carefully separates what he actually derives from what he claims. The distinction matters, because some of Jaynes' strongest claims – around universality, subjectivity and completeness – are exactly where later objections become sharpest [Col04; CW17; Hal99a; Hal99b; Sha04; Van03].

Second, the thesis offers a reconstruction of the relevant slice of Jaynes' programme inside a Martin-Löf-style type-theoretic setting. In this setting, probability assignments should be representable as explicit objects, and the familiar rules of probability should either be derived or obeyed by construction. Probability types are the natural vehicle for this, and they connect naturally to finitary versions of Cox-style reasoning [CW17].

A constructive account of causality follows. Pearl-style structural causal models are not replaced; they are embedded in a broader type-theoretic and epistemic setting. Interventions, a representative model-level counterfactual calculation and changes of causal structure are then treated as changes between explicitly represented epistemic states.

Alongside causality comes a more explicit account of subjectivity. Subjectivity is not reduced to the choice of a prior – instead, epistemic position is encoded through modes and transitions between them. This makes it possible to speak formally about learning, forgetting, interest and hypothesising, which matters especially for artificial intelligence, where agents may have bounded resources, limited memory and different contexts [HP07; NR19; Sim55].

A limited Lean 4 formalisation accompanies the formal chapters. This is not solely decorative; it is a guard against hidden assumptions. When a definition or proof cannot be made precise, even in the finite setting, this indicates not an implementation nuisance, but mathematical information.

The thesis closes with a diagnosis of scientific practice that connects back to the formal work. Hidden assumptions in mathematics and hidden incentives in science are different phenomena, but they are intertwined: both make it possible for conclusions to look cleaner than the process that produced them. The thesis therefore ends with practical suggestions concerning policies, incentives and research culture.

Compactly put: the formal part succeeds if finite Jaynes-style probabilistic updating and finite Pearl-style causal reasoning can be represented in the chosen Martin-Löf-style framework, with representative constructions checked in Lean 4. The wider thesis succeeds if this formal work can be convincingly connected to the claim that good science needs both good inference and a good environment in which this inference takes place.

### Relevance

The relevance for artificial intelligence is direct. If artificial intelligence is, at least in part, the formalisation and implementation of inference, then its reliability depends on the quality of the reasoning methods on which it is built. A finite, constructive and computational account of probability and causality would provide a clearer route from abstract reasoning to executable systems.

The causal extension is especially important. Modern intelligent systems are increasingly expected not only to predict, but to answer questions about intervention, responsibility and alternative outcomes. Prediction alone is too weak for that task. Pearl’s causal framework shows why prediction has to be supplemented by intervention and counterfactual reasoning [Pea09; PM18]. This thesis asks how such reasoning can be embedded in a more explicitly constructive and subjective framework.

The broader scientific-practice component is also relevant to artificial intelligence. AI systems trained on, used within, or used to accelerate science inherit the weaknesses of the scientific pipeline. If the surrounding environment rewards fragile results, opaque reasoning or publication gamesmanship, then faster inference may simply produce bad science more efficiently. The formal and institutional questions therefore have to be considered together.

### 1.3 Outline

The thesis is structured as follows:

The first movement expounds Jaynes’ central claims: it discusses his demand for solid foundations in mathematics, solid foundations in statistics, and a good and honest attitude for doing these sciences. It also introduces the mathematical manifesto that will guide the rest of the work.

The second movement discusses Jaynes’ solution for statistics and the problems surrounding it: it reconstructs the Jaynes–Cox view, separates derivation from interpretation, and discusses objections concerning consistency, reasonableness, universality, subjectivity and causality.

The third movement proposes a more suitable mathematical setting: it discusses alternative logics and type theory, then presents the Martin-Löf-style type theory used in the constructive parts of the thesis.

The fourth movement uses this mathematical setting to address the problems raised earlier: it develops context manipulation, modalities, probability types and structural causal models inside the chosen type-theoretic framework. It also collects the formal results.

The fifth movement returns to scientific practice: it expands Jaynes’ complaints about bad attitudes into a wider analysis of perverse incentives in modern science. It also proposes practical changes.

The appendices contain definitions, proofs, arguments and code-oriented material that would interrupt the main line of the text. They are not less important, but they are placed where they can be checked without interrupting the narrative.

### 1.4 Idiosyncrasies

Before continuing, a few habits of this text deserve some attention:

The structure is intentionally unconventional. The thesis is arranged as a symphony, with movements rather than ordinary parts. This is inspired by Huizinga’s insistence on the playful character of culture and science, and by the suspicion that some of this playfulness has been replaced by a more anxious and institutionalised competition. The musical structure is therefore not only decorative: it is a reminder that scientific thought is not just a sequence of deliverables, but also a playful activity. The tempo markings are mood cues rather than a strict musical programme: the Prelude is deliberate, the first movement polemical, the second reflective, the third slow and foundational, the fourth broad and constructive, and the fifth quick and heated.

Acronyms will be avoided where they do not genuinely help. Science already contains enough abbreviations that sound useful mainly because they sound fundable. Likewise, ordinary terms will not be visually inflated with unnecessary capital letters. This means there will be no grand acronym for “Mathematical Inference as a Logical Extension” (MILE); there will simply be “mathematical inference”.

The thesis is written with active AI collaboration: ChatGPT is being used for source discovery and general framing; Lumo is being used for stylistic assistance; Claude and ChatGPT are used for coding and proof-checking work; Claude is also co-responsible for the figures and diagrams; ChatGPT made the front-cover. Having agents assisting mathematical projects belongs to the wider movement towards AI-assisted formal mathematics [Cas26; Kak26; Orn26; Sku26; Slo26]. The whole storyline, critical analysis of the literature and verification and improvement of prose and proofs, is entirely done by me. This thesis in AI is therefore, rather appropriately, 100% AI-made, 100% human-reviewed. Every paragraph was (re-)written, weighed and revised by me; the intellectual responsibility, including responsibility for errors, remains completely mine.

The intended audience is broad, and this is risky. The thesis lies at the crossing of logic, mathematics, artificial intelligence, philosophy of probability, causality and the sociology of science. Readers from any one of these domains may sometimes want more detail exactly where another reader needs less. The compromise adopted here is narrative-first: the main text preserves the flow of ideas, while technical definitions, proofs and code are moved to the appendices whenever they would otherwise break the story.

These idiosyncrasies are not meant as a decorative ornament: they are part of the method. A thesis about explicit reasoning, constructive mathematics and scientific honesty should itself try to be explicit about the way it reasons and the choices it makes. The reader is invited to follow this movement.



## CHAPTER 2

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### First movement: Allegro con brio

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*“Indeed, an insistence that all of mathematics be stated in that language all of the time can place unnecessary burdens on a theory, particularly one intended for application in the real world. It can also degenerate into an affectation, used only linguistically rather than functionally. To give every old, familiar notion a new, impressive name and symbol unknown to Gauss and Cauchy has nothing to do with rigor. It is, more often than not, a form of gamesmanship whose real purpose is to conceal the Mickey Mouse triviality of what is being done. One would blush to state it in plain English”*

E. T. Jaynes in *Probability Theory: The Logic of Science*

### 2.1 Solid foundations for mathematics

Jaynes' first complaint is directed at the foundations of mathematics. His target is not mathematics as such, and certainly not calculation, experiment, modelling or honest abstraction. His target is a particular kind of mathematical culture: a culture in which formal derivation is too easily confused with scientific usefulness, and in which the ability to write down a consistent-looking construction is mistaken for the ability to explain anything worth explaining, see the opening quote from Jaynes.

The most visible symptom is what is called “infinite paradoxing” by Jaynes. Modern mathematics is riddled with results that actively discover the limits of infinity (pun intended): Banach–Tarski decompositions, non-measurable Vitali sets, the Borel–Kolmogorov paradox, supertasks such as Thomson’s lamp, and Ross–Littlewood-style urn puzzles [AK91; AK95; BT24; GHR17; Tho54; Vit05]. These examples do not all have the same status: some are genuine theorems; some are philosophical stress tests; some are warnings about limiting procedures.

The deeper objection is therefore methodological. Axiomatic set theory, especially in the Zermelo–Fraenkel with choice tradition, easily allows for deeply abstract constructions. Its ease of use is the main reason for its success. By the same token, it is also deceptively easy to create nonsensical or non-applicable objects. The trained mathematician may admire the sheer possibility; the physicist may ask how it can be measured. Jaynes was a classically-trained physicist: start with what can be stated, counted, constructed or approximated, and move to limits only with a clear account of what the limiting process achieves. If the point of mathematics in science is mathematical inference, then a method that converts finite observations into uncontrolled ideal objects should inherently be treated with suspicion. It may still be useful, but usefulness is not conferred by grandeur.

It is worth noting that, although Jaynes actively fulminates against using Zermelo–Fraenkel with choice theory as a basis for mathematics, he uses it himself, albeit in a rather conservative way. This is therefore an objection he extensively discusses, but is unable to ‘solve’. He proposes a more conservative approach to using the theory as it stands, but that approach asks users deliberately not to explore the full capability of the theory – an argument that is rather unconvincing to the mathematical community.

In this thesis I side with the physicists: for the formal reconstruction, the default mathematical environment will be constructive and finitary. This is not primarily motivated by some form of mathematical purity, but much more by efficiency. A framework that forces objects to be constructed, contexts to be explicitly declared and dependencies to be visible is better suited to the kind of scientific reasoning Jaynes wanted. Rather than running the risk of spending valuable resources and research facilities on obtaining results that are questionable to applied research, it can be more efficient to construct everything in a more limiting environment, where results are more likely to be directly applicable and computationally checkable.

This is why the third movement turns to alternative logics and Martin–Löf-style type theory. The point is not to escape into yet another formalism, but to find a setting in which constructions, contexts and dependencies have to be stated openly. The fourth movement then uses this setting for the constructive

reconstruction itself: probability types, epistemic modalities and structural causal models are all built there within this setting.

## 2.2 Solid foundations for statistics

Jaynes’ second complaint is more direct and more devastating. In his view, twentieth-century statistics had become a patchwork of procedures, tests, conventions and reputations. Some procedures worked well in practice; some were defended by tradition; some were inherited because nobody wanted to reopen old discussions. What was clearly missing was a single account of why statistical inference should have the form it had.

The force of Jaynes’ criticism becomes clearer when placed next to its possible consequences. Independent teams can receive the same data and reach sharply different conclusions, as shown in Figure 2.1 [Bot+20; Bre+22; Sch+21; Sil+18]. This poses a serious problem, because it threatens the foundations of statistical claims in economics, psychology, sociology and other sciences, as large replication studies show; see Figure 2.2 [Cam+16; Kle+14; Ope15]. This objection therefore bears even more serious consequences than the previous objection: not only are time and energy potentially wasted, but there is also an acute risk of consciously propagating incorrect conclusions, which is directly contrary to doing science. This clearly illustrates that inference is fragile when assumptions are hidden, workflows are underdescribed and technical choices are treated as harmless details [GL13; Ioa05].

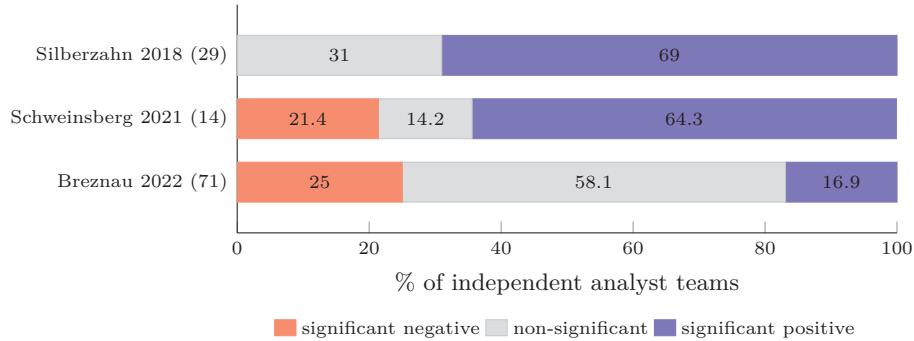


Figure 2.1: Three many-analyst projects in which independent teams analysed one dataset, classified by the *direction* of the reported result. Silberzahn et al. (29 teams): no significant-negative result, 31% non-significant, 69% significant-positive [Sil+18]; Schweinsberg et al. (Hypothesis 1, 14 analysts with known direction): 21.4% / 14.2% / 64.3% [Sch+21]; Breznau et al. (71 teams, estimate-level): about a quarter (25%) significant-negative, a little over half (58.1%) non-significant, 16.9% significant-positive [Bre+22]. Botvinik-Nezer et al.’s neuroimaging study is discussed in the text but omitted here, since its 70 teams answered nine yes/no hypotheses with no directional split [Bot+20].

Cox’s programme offered Jaynes exactly the kind of answer he wanted: if plausibility is represented numerically, if logically equivalent states of information are assigned equivalent plausibilities, and if reasoning is required

## 2. First movement: Allegro con brio

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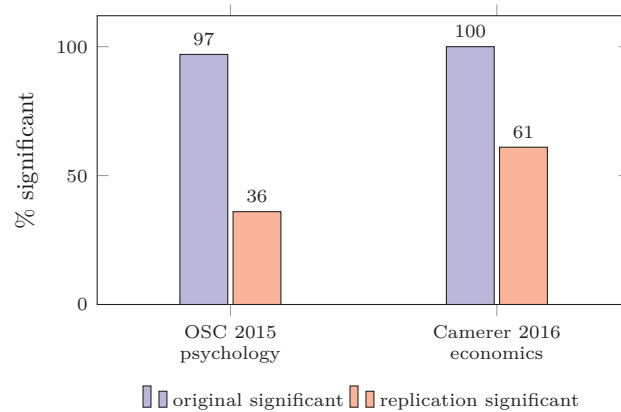


Figure 2.2: Two large replication projects. Of 100 psychology studies, 97% of originals reported a statistically significant result versus 36% of replications [Ope15]; of 18 economics experiments, all reported a positive original finding and 61% (11 of 18) replicated as a significant effect in the same direction [Cam+16].

to be consistent across different routes, then the familiar rules of probability can be recovered after a suitable rescaling [Cox46; Cox61; Jay03]. Probability is then not a bag of recipes, it is extended logic: the calculus of consistent reasoning under uncertainty.

This is the great attraction of the Jaynes–Cox view: it offers a way to say that statistical inference is neither a social convention, nor a computational convenience. It is what one must do if one wants uncertainty to obey the same coherence that ordinary deductive logic obeys in the limiting case of absolute certainty. Keynes, Ramsey and Jeffreys had already made probability philosophically serious as a calculus of partial belief; Cox and Jaynes gave the programme its sharpest logical form [Jef61; Key21; Ram31].

Of the three complaints filed in this movement, this one can be considered the most ‘solved’ and the most complete by Jaynes. I therefore see Jaynes’ solution as both hugely important, but still partly incomplete: important because it gives a principled answer to a real problem (probability as a form of coherent inference); and incomplete because his derivations depend themselves on assumptions that are not completely innocent, and because his derivations lack causality and a native representation of subjectivity.

The second movement will therefore critically assess the Jaynesian programme: what it derives, what it assumes and where its claims become too large. The fourth movement then rederives the finite Bayesian core and expands it such that subjectivity and causality are made explicit within the formal language.

### 2.3 A good and honest attitude for doing these sciences

The third complaint is less formal, but no less central. Jaynes does not only object to bad foundations: he objects to bad habits. His prose is rather polemical

and, at times, even develops into full-blown personal attacks<sup>1</sup>. Since, for Jaynes, the very heart of scientific praxis is at stake – with an impractical basis for mathematics and a non-rigorous basis for statistics – this objection is absolutely the most serious. It directly takes aim at people who, in his view, can be considered personally responsible for the bad state of science as it stands.

The habits Jaynes attacks can be arranged under five headings:

1. **Gamesmanship**: the desire to display technical virtuosity, rather than to clarify a problem;
2. **Cutting corners**: the focus on polished results, without explaining the painstaking work that led to these outcomes;
3. **Hidden assumptions**: the silent transfer of what should have been stated into what is merely presumed;
4. **Hyperfocus**: the habit to only investigate an incredibly specialised subfield, whilst forgetting the wider context in which this research sits;
5. **Theoretical navel-gazing**: the production of mathematics that neither explains, predicts, nor helps anyone build or test anything practical.

For this complaint, too, Jaynes addresses the problem abundantly, but is unable to ‘solve’ it. He treats this complaint rather personally, while I believe it deserves a more institutional view. The vices mentioned by Jaynes are not limited to ‘pure’ mathematics and statistics, and they are, unfortunately, not limited to Jaynes’ opponents<sup>2</sup>. They are temptations built into scholarly life.

The fifth movement returns to these vices in a more institutional setting. I will argue there that they appear whenever professional reward becomes too detached from intellectual responsibility. The formal reconstruction of the intervening movements is therefore not separate from the sociological diagnosis: both are attempts to make hidden assumptions harder to hide.

However, that is not all. In software development, there has been some attempt to establish workflow standards as part of an effort to optimise operations, regardless of the particulars of each separate project. In mathematics, and science in general, there seems to be a general lack of such efforts. In the following, I will put forward a standard that I will adhere to, that is aimed at preventing most of these critiques. This thesis is therefore to be read as an attempt to obey that standard: a mathematical manifesto.

## 2.4 Mathematical manifesto

From these complaints a small mathematical manifesto follows. It is not a new foundation, nor a grand acronym, nor a shiny institutional programme. It borrows its form from the Agile Manifesto and from later workflow frameworks such as Scrum: a short set of comparative preferences, meant to guide practice [Bec+01; SS20]. For the rest of the thesis, the discipline is this manifesto:

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<sup>1</sup>Especially Ronald Fisher, but also William Feller, Nicolas Bourbaki/the Bourbaki group and others take personal blows.

<sup>2</sup>Cutting corners also appears inside Jaynes’ own work: some of his strongest claims about universality, subjectivity and completeness move faster than the derivations can justify. This is not a reason to discard Jaynes: it is a reason to take him seriously enough to apply his own standard to him.

## 2. First movement: Allegro con brio

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I am uncovering better ways of developing mathematical theories by doing it and helping others do it. Through this work I have come to value:

Algorithms over abstractions

Narrative over formulae

Explicit over implicit

Simple over complex

That is, while there is value in the items on the right, I value the items on the left more.

Algorithms over abstractions does not mean that abstraction is bad, but it means that abstraction should remain connected to construction, computation or procedure. If a concept cannot in principle be implemented, checked, traced or approximated, then its scientific role should be handled with care.

Narrative over formulae does not mean that formulae are ornamental, but it means that formulae should serve the movement of thought. A derivation is not a heap of symbols; it is a story about why one statement follows from another. When the story is hidden, the formulae may still be true, but they become harder to inspect.

Explicit over implicit is the most Jaynesian rule: assumptions should be visible, contexts should be named and dependencies should be tracked. If a proof requires a specific domain, a decidable event, a specific logic or a specific causal graph, that requirement should not be secretly smuggled in. The constructive framework used later is chosen in this spirit: not because it is the easiest language at first glance, but because it makes hidden commitments harder to hide.

Simple over complex is not a cult of simplicity: it is a preference for the smallest mechanism that does the job. Complexity is sometimes necessary, but it should be paid for consciously. If a simpler derivation, formula, or even plain English sentence does the job equally well, then the more complex solution requires a justification.

## CHAPTER 3

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### Second movement: Andante con moto

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*“A story is needed before one can set sail. Look at us. For the sake of the story, we are sitting here aboard a water taxi on our way to the Holy Grail, the unknown, and adventure. People think life becomes more enjoyable through stories. That is true enough, but it is not the point. Without stories, it becomes meaningless. Sense consists of sentences. The words that tell the story lay a reassuring relation of cause and effect over random facts and events. People crave plot because a plot reduces the unbearable and unmanageable chaos of the sublunary world to a human scale, and to a chain of initiatives and consequences that a person is able to comprehend. A plot gives a sense of control, provenance and destination, origin and direction”*

Ilja Leonard Pfeijffer in *Grand Hotel Europa*

## 3.1 The Jaynes–Cox programme

### Desiderata

The Jaynes–Cox programme begins with a modest question: what rules should govern reasonable degrees of belief [Cox46; Cox61; Jay03]? The exposition starts with three desiderata that, according to Jaynes, are the minimal, reasonable requirements that a calculus of plausible reasoning ought to satisfy. In Jaynes' own words, in the compact formulation [Jay03]:

- (I) Representation of degrees of plausibility by real numbers;
- (II) Qualitative correspondence with common sense;
- (III) Consistency.

In the ensuing discussion, Jaynes unpacks consistency into three demands: a conclusion should not depend on the route by which it is reached; relevant evidence should not be ignored; and equivalent states of knowledge should receive equivalent plausibility assignments. The first desideratum translates to divisibility and comparability: plausibility should be represented on a single ordered numerical scale, so that one state of information can be compared with another. The second is qualitative agreement with common sense: ordinary Aristotelian logic should appear as the limiting case in which propositions are known with certainty, and small changes in the premises should not produce arbitrary jumps in the conclusions. The third is consistency: if a plausibility can be obtained by two legitimate routes, the two answers should agree. These desiderata are then via functional equations translated in the familiar axioms of probability theory.

### Derived calculus

Cox's formal move is to apply these desiderata to the logical operations of negation and conjunction. If  $A$  and  $B$  are propositions and  $I$  is the background information, then the plausibility of  $\neg A$  should be determined by the plausibility of  $A$ , and the plausibility of  $A \wedge B$  should be determined by the plausibility of  $A$  and the plausibility of  $B$  once  $A$  is assumed. Under the usual regularity assumptions, one can rescale the plausibility function to a probability function  $P$  satisfying

$$P(A \wedge B | I) = P(A | I)P(B | A, I)$$

and

$$P(A | I) + P(\neg A | I) = 1.$$

The first equation is the product rule; the second is the sum rule for complements.

Bayes' theorem then follows by applying the product rule in both orders:

$$P(A | B, I) = \frac{P(B | A, I)P(A | I)}{P(B | I)}$$

whenever  $P(B | I)$  is non-zero. Here  $P(A | I)$  is the prior probability of  $A$  relative to background information  $I$ ,  $P(B | A, I)$  is the likelihood of observing  $B$  under  $A$ ,  $P(B | I)$  is the evidence or normalising term, and  $P(A | B, I)$  is the posterior probability after  $B$  has been taken into account. In general, many probabilities, such as the likelihood, are often clear from the context of the

problem and therefore treated as relatively objective. The prior distribution is usually harder to fix, and is therefore often treated as more subjective. This degree of freedom makes this theory notoriously controversial, but allows the theory to introduce ‘subjectivity’ as the ability to freely choose one’s prior. The finite constructive version of conditional probability and Bayes’ theorem used later is given in Definition A.3.7 and Theorem B.4.10.

With these ingredients, one recovers the ordinary finite probability calculus: non-negativity, normalisation and finite additivity. In the classical measure-theoretic presentation, this calculus is extended to Kolmogorov’s axioms. In this classical reading, a probability function  $P$  satisfies Kolmogorov’s axioms if:

1. *Non-negativity*:  $P(A) \geq 0$  for every  $A \in \mathcal{F}$ ;
2. *Normalisation*:  $P(\Omega) = 1$ ;
3. *Countable additivity*: for every countable pairwise disjoint family  $(A_i)_{i=1}^\infty$  in  $\mathcal{F}$ ,

$$P\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty P(A_i).$$

Here  $\mathcal{F}$  is a  $\sigma$ -algebra of events. The finite analogue used later is defined in Definition A.3.5 and Theorem B.4.6.

This derivation does not mean that Jaynes invented Bayesian reasoning, as Keynes, Jeffreys, Ramsey, Cox and others are key ancestors of the project, and Jaynes explicitly inherits the Coxian line [Cox46; Cox61; Jef61; Key21; Ram31]. Jaynes’ distinct contribution is the clarity of the synthesis: probability as extended logic, statistical inference as disciplined common sense, and scientific disagreement as often traceable to hidden assumptions or silently disregarded desiderata. The philosophical gain is large: what appears to be a specialised derivation sets the norm for coherent reasoning under uncertainty. Probability is then more than a matter of repeated experiments: it is about what follows from information.

Jaynes really maintained that he uncovered a universal and logical theory of doing science and that everything deviating from either Bayes’ rule or his logical desiderata should be considered fundamentally illogical and – even more strongly so – fundamentally unscientific. The attraction of this view is obvious. If correct, the programme gives mathematical inference a foundation. The remaining sections critically assess the programme and ask whether all its assumptions and conclusions can, in fact, be justified.

## 3.2 Objections

The objections to Jaynes do not all have the same focus: some are technical; some are philosophical; some concern scope. I will treat them in an order that moves from the derivation itself to the larger claims built on top of it: consistency, reasonableness, universality, subjectivity and causality.

The guiding distinction is the one already announced in the Prelude: what is derived, and what is claimed. Jaynes derives a powerful representation result under assumptions. He sometimes writes as if he has also derived a universal account of rationality, subjectivity and scientific method. The route through these objections is summarised in Table 3.1.

### 3. Second movement: Andante con moto

| Objection      | Target                            | Problem for Jaynes                                                                                    | Thesis response                                                                                |
|----------------|-----------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Consistency    | Cox-style derivation              | Naive formulations of Cox's assumptions can fail on finite or insufficiently rich domains.            | Treat the counterexamples as design constraints and rebuild the finite assumptions explicitly. |
| Reasonableness | Jaynes' desiderata                | A consistent assumption can still be too technical, too strong or too poorly motivated.               | Use the finite urn principles of Clayton and Waddington as the constructive route.             |
| Universality   | Probability as universal logic    | A theorem inside one background logic is not automatically a logic of all uncertainty or all science. | Put probability inside a constructive and epistemological setting with explicit scope.         |
| Subjectivity   | Priors and background information | The subject is represented mainly by $I$ and the prior, while the agent's position remains implicit.  | Represent epistemic position through contexts, modes and modalities.                           |
| Causality      | Prediction by probability         | Statistical dependence does not by itself distinguish seeing, doing and imagining.                    | Add finite structural causal models and transport the relevant causal results.                 |

Table 3.1: Route through the objections to the Jaynes–Cox programme.

#### Consistency

The first worry is whether Cox-style assumptions are even consistent in the form needed for Jaynes' conclusion. Halpern showed that naive formulations of Cox's theorem can fail – finite domains and insufficiently rich plausibility ranges can produce counterexamples to the intended representation [Hal99a; Hal99b].

Later work repairs the situation. Arnborg and Sjödin treat finite models and argue for additional principles under which Bayesian rules can be recovered or justified [AS00; AS01; AS99]. Van Horn reconstructs the assumptions needed for a continuous Cox-style theorem more carefully, and Snow defends finite-domain reasonableness from a related angle [Sno98; Van03].

These results illustrate that Cox-style derivations require additional assumptions: domain richness, regularity assumptions and a precise account of what counts as a permissible plausibility assignment. For this thesis, these critiques are taken as instructive sources of information: if probability is to be rebuilt constructively, the needed assumptions must be made precise and checkable.

This is the role of the finite reconstruction in Section 5.3: the assumptions are not treated as background atmosphere, but are gathered into explicit finite rules whose proof route is summarised in Table 5.1. Representative finite checks are later collected in Table 5.3.

## Reasonableness

A subtler worry is whether the assumptions are reasonable to begin with. An assumption can be consistent and still be philosophically unattractive. Shafer objects, among other things, to the idea that reasonable expectation must always be represented by a single real number [Sha04], and Van Horn’s reconstruction – although technically precise – does not by itself make every assumption compelling<sup>1</sup> [Van03].

Clayton and Waddington are especially important here because they aim to design explicit logical and intuitive rules that are more in line with Jaynes’ intentions. Instead of simply postulating technical conditions such as universality and strict monotonicity on large domains, they derive the needed structure from urn-based rules [CW17]. The full details are in their paper; the assumptions needed here are:

- *finite urn existence*: symmetric urns of every positive finite size are available;
- *logical equivalence*: equivalent decidable events receive equal plausibility;
- *qualitative coherence*: equality and strict comparison of plausibility behave like an ordered equivalence;
- *relabelling invariance*: permuting finite urn labels does not change plausibility;
- *strict monotonicity*: in a symmetric urn, more favourable cells means greater plausibility;
- *refinement invariance*: splitting every cell uniformly preserves plausibility;
- *product urn compatibility*: rectangular events in product urns behave like products.

Some additional rules and derivations are needed in the original paper, as it operates in the classical axiomatic set-theoretic domain. The thesis does not reproduce Clayton and Waddington’s full route. Instead, the finite assumptions needed later are listed in Definition A.3.6.

As Clayton and Waddington’s account of probability is perfectly in harmony with the mathematical manifesto stated earlier, this thesis bases its derivation on their work. The reconstruction in this thesis only keeps the finite urn principles and changes the order of their derivations somewhat, but the idea remains the same. It is important not to take Jaynes’ desiderata as sacred, immutable tablets, but as a solid basis on which one can sensibly vary. Clayton and Waddington’s account might make the resulting probability theory less grand, but more explicit and, therefore, directly checkable and less evasive.

The constructive proof in Section 5.3 follows this lesson; its route is summarised in Table 5.1 and carried out in Section B.4. It does not try to preserve every grand Coxian claim; it preserves the finite ingredients that can be made explicit, checked and used.

## Universality

Jaynes often writes that his derived probability theory is the universal logic of science. Shafer and Colyvan give useful warnings against this overreach: Cox-style arguments presuppose a background logic, and the choice of that logic matters [Col04; Sha04]. A theorem about plausible inference inside classical

<sup>1</sup>An objection explicitly mentioned by Van Horn himself.

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propositional logic is not automatically a theorem about all forms of uncertainty, all domains of discourse or all scientific reasoning.

The point becomes clearer once one distinguishes ontological and epistemological questions. An ontological logic asks what is the case; an epistemological logic asks what is known, knowable or justified from a given position [Hof23; SN25]. The question “Does God exist?” illustrates the contrast nicely: an ontological view forces one to give an immediate answer; an epistemological stance may answer that the question is simply not answerable.

Jaynes’ framework is epistemological in temperament – as he considers inference as the process of adding knowledge to a given, personal state of information – but formally ontological. He wants probabilities to describe states of knowledge, yet his derivation usually assumes a classical propositional background in which every proposition is immediate (true or false). However, vagueness, fictional discourse, undecided mathematics, limited interest and forgotten information are not well represented by simply assigning a number to such a classical proposition, as also maintained by Van Horn [Van03].

The response is not to abandon probability, but to abandon the claim that probability, derived once in one background logic, is the final logic of everything. The present thesis uses a constructive and epistemological setting, but remains modest about its applications to other domains. Probability itself remains central in this work, but it is placed inside a richer account of contexts, modes and epistemic states. This richer account is, then, better suited for natively representing someone’s knowledge, rather than being an afterthought. The thesis is not about making probability a ‘universal logic’, but about making its domain more explicit and its applications more checkable.

### Subjectivity

In the terminology of Bayes’ theorem above, Jaynes’ subjectivity enters mainly through the background information  $I$  and the prior  $P(A \mid I)$ : different agents may start from different information and therefore assign different probabilities. But the subject is still strangely absent from the formalism. The prior may vary from one agent to another, while the structure of the agent’s position is not itself represented.

For a truly subjective representation, it is essential to know what information, per agent, is available. This means knowing what problems are in scope; what information has been forgotten; what information is of no interest; what connections are believed to hold true; and what can be imagined, based on the available information. These are not merely psychological decorations, but they affect what can be asked, what can be inferred and what counts as an identical problem. These questions should be part of the mathematical formalism itself, rather than being added as an afterthought.

Jaynes rightly objects to treating probability as a property of objects in isolation, a stance often defended by frequentists. In this view, repeating a measurement until one ‘uncovers the True Probability’ simply means, as Jaynes correctly points out, an ill-defined procedure, as one can make a repeated coin flip have any experimental outcome one likes<sup>2</sup>. A coin does not carry

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<sup>2</sup>Jaynes often refers to this phenomenon as the ‘projection fallacy’, where personal missing information is projected on a situation as a whole – like with the coin flip – or even is seen as being *fundamentally* missing – as with quantum mechanics.

a metaphysical label “one half”. A physical procedure, a personal state of information and a model of the experiment are often covertly doing the heavy work. However, stating this in prose does not yet build subjectivity into the mathematics.

Interest is a clear example, somewhat alluded to by Colyvan [Col04]. Classical logic lets every proposition sit in the same universe of discourse. Real agents do not work like that. To a real agent, a proposition may be true, false, unknown, irrelevant, forgotten or outside the present domain of concern. One may not care whether Achilles was considered the fastest among mortals by the ancients, nor what ending Gogol’s *Dead Souls* would have had if he had lived a bit longer, while caring deeply about one’s own medical condition. A logic of subjectivity should be able to represent this asymmetry without pretending that every statement is waiting for some kind of ‘objective’ probability to happen.

The constructive proposal in later sections is to make this epistemic position explicit. Subjectivity is then no longer a label placed on Bayesian calculus after the fact: it becomes part of the machinery that is developed in Sections 5.1 and 5.2.

## Causality

Jaynes’ probability theory can express statistical dependence, but dependence is not causation, as the famous adage goes. That is, a barometer and a storm can be tightly associated without the barometer causing the storm to happen. I give an argument in Section B.1 for why a working theory of scientific reasoning – under which mathematical inference falls – should include causality. The main components of this argument are already present in the opening quote by Pfeijffer, albeit there much more eloquently than I am capable of. This is a philosophical argument from desirability, but it does not yield usable clues on how causality should be mathematically integrated.

Pearl’s framework supplies the missing formal technology. This framework distinguishes three levels of causal inquiry. At the bottom sits association: questions such as  $P(Y | X)$ , where one asks how observations influence each other. Above that comes intervention, with questions such as  $P(Y | \text{do}(X = x))$ , asking what would happen if variables are forced to a specific value. At the top lives counterfactual reasoning — questions such as  $Y_x(u)$ , where one asks what the outcome for a particular unit hypothetically would have been under a different intervention, see Figure 3.1 [Pea09; PM18]. These three levels are not stylistic variations. They require different assumptions, and they answer different questions.

One needs more mathematical machinery for these three levels. At the first level, I can still make do with the Bayesian calculus that Jaynes derived: separate variables that are functionally related. At the second level, I would need connections between variables, as I need to explicitly know what variable is influencing which variable if I were to intervene in a situation. This is neatly represented by causal Bayesian networks and causal graphs, which are directed acyclic graphs representing causal structure. These graphs obey the causal Markov condition: each variable is independent of its non-descendants once its parents are known. Pearl introduces the do-operator to mark the difference between observing a variable and setting one. The formal definitions are collected in Definitions A.3.14, A.3.19, A.3.20, A.3.27, A.3.28 and A.3.33, where Pearl’s

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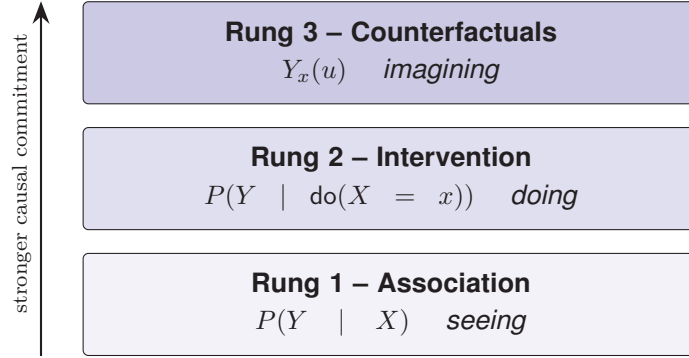


Figure 3.1: Pearl’s ladder of causation: each rung answers a strictly stronger class of question [PM18].

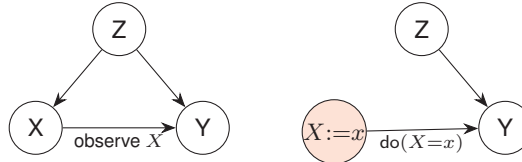


Figure 3.2: Observing  $X$  leaves the confounding path through  $Z$  intact; intervening on  $X$  severs its incoming edges (Definition A.3.27).

three do-calculus rules are the transformation rules that connect observations and interventions under specific conditions [Pea09; Pea12]. Observing leaves the ordinary causal mechanism in place; intervening replaces the mechanism for a variable by an imposed value, see Figure 3.2. This is exactly the distinction Jaynes’ probabilistic calculus cannot express by itself.

At the counterfactual level, a Bayesian network is no longer sufficient. Here, a structural causal model is necessary: a finite collection of endogenous variables, exogenous background variables, structural equations on the endogenous variables and a probability distribution over the exogenous variables. Given evidence, the first step is to update the distribution over the exogenous variables; for a hypothetical, imagined action, a twin network is connected to these exogenous variables and the intervened, endogenous variables are set to a constant. Within this twin network, it is then possible to evaluate a hypothetical outcome in the intervened model, see Figure 3.3. The relevant finitary definitions and remarks are collected in Definitions A.3.24 to A.3.26 and A.3.35 and Remark B.5.12, where the described procedure follows Pearl’s abduction–action–prediction procedure.

For the ordinary interventional fragment, Pearl’s framework is not merely convenient notation: do-calculus, together with ordinary probability manipulations, is complete for identifying the relevant class of causal effects in the standard settings [HV06; Pea12; SP06; TS09].

In my research I consider Pearl’s account of causality a paramount extension of Jaynes’ work. Not only do the structural causal models allow for hypothesising *per se*, but they also provide an ideal representation of a personal state of

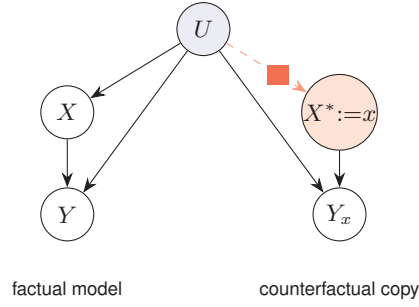


Figure 3.3: A counterfactual copy of a structural causal model. The factual and counterfactual sectors share the same exogenous background  $U$ ; the counterfactual intervention replaces the mechanism for  $X^*$  by the imposed value  $x$ , and  $Y_x$  is then evaluated in the copied model.

information. These models naturally capture what propositions are in scope, and what relations exist between them. Furthermore, by dynamically transitioning between different structural causal models, it is easy to capture epistemological events like forgetting, learning, connecting, hypothesising, and others. Pearl-style causal models will therefore be embedded in the constructive, epistemic setting developed later, which buys one extra freedom: causal structures need not be treated as eternally fixed.

This thesis does not reprove these completeness results. It deduces the relevant finite claims by transport: once the restricted finite structural-causal-model fragment is shown to carry the same variables, graph operations, interventions and probability expressions, the applicable finite results can be moved across the correspondence. This transfer is developed in Section 5.4 and summarised in Table 5.2. It does not imply completeness for every operation made possible by the epistemic construction; nor does it imply full completeness for counterfactual queries. Those stronger claims would require additional work.



## CHAPTER 4

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### Third movement: Adagio

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*“A common mistake that people make when trying to design something completely foolproof is to underestimate the ingenuity of complete fools”*

Douglas Adams in *Mostly Harmless*

## 4. Third movement: Adagio

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This movement prepares the formal language used in the fourth movement: first by sorting the relevant logics, then by moving through type theory, and finally by fixing the Martin-Löf-style fragment on which contexts, modalities, probability types and causal models will rest.

### 4.1 Different types of logic

In light of the overly broad universality claims posited by Jaynes, the next step is to enquire which logic is appropriate for which task. Classical logic is still magnificent when its assumptions fit: propositions are true or false; excluded middle is accepted (as propositions are always either true or false); contradiction is fatal; and the inferential world is crisp as all propositions are always available to everyone. However, reasoning is not always that crisp, and artificial intelligence does not have the luxury of pretending so.

**Fuzzy and many-valued logics** deal with predicates that do not behave as if they have sharp, binary borders. “Young”, “tall”, “similar” and “healthy” are not always best captured as if they only had two options, but rather a whole range of values or a degree of ‘truthness’ [CFN22]. Probability is sometimes misused here: a degree of truth is not the same thing as a degree of belief. Van Horn makes this distinction sharper, emphasising that plausible or probabilistic inference concerns uncertainty about propositions, but does not concern assigning gradients to propositions [Van03].

**Modal logic** adds structure by allowing truth to vary across worlds, states or situations. In Kripke-style semantics, one does not merely ask whether a proposition is true; one asks whether it is true at a specific world, and how different worlds are related [Gar24]. This is already close to the language needed later. An epistemic state can be treated like a world; moving between states can represent learning, forgetting or intervention; modal structure can record which movements are legitimate.

The contrast between **ontological** and **epistemological** logic is more important for this thesis. The above logics are often framed in an ontological setting – although this is not strictly necessary, as will be seen later with modal logic. Ontological logic speaks as if it were describing what *is* or what *exists in reality*; Epistemological logic speaks about *knowledge*: what is known and what can be induced from that [Hof23; SN25]. Jaynes’ language is deeply epistemological, as it frequently refers to the personal knowledge of an agent, but his formal derivation remains close to a classical background.

Within epistemology there are a few candidates to consider. **Temporal** and **dynamic** logics are useful for reasoning about change and actions [BR16; GR25]. Both of these contain valuable components, and the later implementation can be considered a special flavour of a dynamic logic, where modalities govern transitions between epistemic states. **Intuitionism** is an epistemological logic in the constructive tradition: everything needs to be explicitly derived. It therefore does not accept a proposition because its negation cannot be proven, but it needs a positive proof of its truth. It immediately follows that the law of excluded middle is not generally accepted in this logic<sup>1</sup>. Furthermore, real

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<sup>1</sup>Which is based on the premise that all propositions are either true or false. In the classical case, it then follows that if a proposition is explicitly not one, it must be automatically the other, e.g. if something is explicitly *not* false, then it *must* be true. This does not automatically

| Logic             | Type theory                       |
|-------------------|-----------------------------------|
| proposition       | type                              |
| proof             | program / term                    |
| $A \wedge B$      | $A \times B$ (pair)               |
| $A \vee B$        | $A + B$ (sum)                     |
| $A \Rightarrow B$ | $A \rightarrow B$ (function)      |
| $\top, \perp$     | <b>1</b> , <b>0</b> (unit, empty) |
| $\forall x. P(x)$ | $\Pi$ -type                       |
| $\exists x. P(x)$ | $\Sigma$ -type                    |

Figure 4.1: The Curry–Howard correspondence: proofs are programs, propositions are types.

numbers, in the classical sense, are also not accepted, as these are, in axiomatic set theory, completed infinite objects, and not explicitly constructed; only numbers that are completely constructible are accepted [Iem24]. Especially this last point is attractive, as it aligns perfectly with Jaynes’ philosophy of explicit limits and constructions.

The philosophical stance I take here is agnostic about ontology and constructive about epistemology, therefore largely aligning with intuitionism. The explicit argument for this is laid down in Section B.2. In the following, I do not decide what reality is made of; I ask how the knowledge of finite agents is influenced by finite measurements and finite updates.

However, to be completely aligned with the mathematical manifesto, more is needed: an explicit and computational part. This is provided by **type theory**, which deserves its own section, following next.

## 4.2 Type theory

Type theory is the natural technical home for this stance. It connects mathematics to computation through the Curry–Howard correspondence: propositions can be read as types, and proofs as terms inhabiting those types, known as propositions-as-types, see Figure 4.1. To prove a proposition is, in this sense, to construct a proof, or, even more explicitly: compute a proof [Coq25].

A **type** may be understood, broadly, as a hypothesis or, specifically, as a domain of construction. A term  $a : A$  says that  $a$  is an **inhabitant** of the type  $A$ , which means that  $a$  is a “witness” within the domain  $A$ . Within type theory, a **context**,  $\Gamma$ , records the assumptions and variables currently available. Importantly, these contexts are ordered: later entries in the context may depend on previous entries, but not vice versa<sup>2</sup>, see Figure 4.2. What can be derived, given a specific context, is called a **judgement**. For example, a judgement of the form

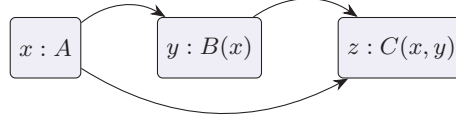
$$\Gamma \vdash a : A$$

follow in the intuitionistic case, where either value needs an explicit confirmation.

<sup>2</sup>This ordering makes these contexts natural vehicles for encoding structural causal models, since these are directed acyclic structures that are only ‘forward-looking’ in terms of dependencies, as will be seen later.

#### 4. Third movement: Adagio

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order matters: later entries may depend on earlier ones

Figure 4.2: A context  $\Gamma = (x : A, y : B(x), z : C(x, y))$  as an ordered telescope of dependencies — the ordering that later mirrors a causal directed acyclic graph.

says that, in context  $\Gamma$ , it can be derived that the term  $a$  has type  $A$ . Furthermore, types themselves can be considered a term of even more general types,  $A : B : C \dots$ , creating a hierarchy, or **universe**, of types. The more formal vocabulary for types, contexts, judgements and universes is given in Definitions A.1.1, A.1.2, A.1.8 and A.1.17. The above already brings subjectivity nearer: what can be derived depends heavily on the specific context considered.

**Substitution** explains how a term depending on a variable changes when that variable is replaced by a particular term. **Function types** transform inhabitants of one type into inhabitants of another. **Lambda abstractions** construct such functions, and **applications** use them. **Pair types** and **sum types** function like containers for two types at once<sup>3</sup>. In the ensuing substitution and application steps, **equality** is not a purely metaphysical relation; it has computational content, for instance through **beta-reduction**. The more formal definitions and machinery are laid down in Definitions A.1.4, A.1.5, A.1.9 to A.1.11 and A.1.20.

A small example can be illuminating, which follows below:

**Example 4.2.1** (A tenure-track offer as a type). Suppose a hiring committee is considering one candidate, Mira, for a tenure-track position. Classically, the proposition

$$\text{Offer}(\text{Mira})$$

is immediately subject to excluded middle:

$$\text{Offer}(\text{Mira}) \vee \neg \text{Offer}(\text{Mira}).$$

Moreover, classical reasoning validates double-negation elimination:

$$\neg \neg \text{Offer}(\text{Mira}) \rightarrow \text{Offer}(\text{Mira}).$$

In the propositions-as-types reading, by contrast,  $\text{Offer}(\text{Mira})$  is a type. A proof that Mira receives the offer is a term

$$\ell : \text{Offer}(\text{Mira}),$$

which may be imagined as the constructed offer letter or the verified decision record. In this constructive reading, negation is interpreted as:

$$\neg A := A \rightarrow \perp$$

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<sup>3</sup>Function types roughly correspond to implication,  $\implies$ , pair types roughly correspond to conjunction,  $\wedge$ , and sum types roughly to disjunction,  $\vee$ .

where  $\perp$  is the empty type; see Definitions A.1.15 and A.1.19 for a formal definition. Thus a proof of  $\neg A$  transforms a witness of  $A$  into contradiction. From a positive witness one can construct double negation:

$$\lambda \ell. \lambda n. n(\ell) : \text{Offer}(\text{Mira}) \rightarrow \neg\neg\text{Offer}(\text{Mira}).$$

The converse direction, double-negation elimination, is valid classically but is not derivable in the constructive base theory used here. Thus, from double negation it is not generally possible to prove that Mira received an offer, unless this is explicitly added to the theory.

Having the mathematical machinery of type theory available is powerful, but does not automatically yield an epistemological theory *per se*. However, an epistemological type theory is available and known as **Martin-Löf-style type theory** [DP24]. This type theory is a combination of intuitionism – explained in the previous section – within the framework of type theory. The following section explains more details about this specific type theory.

### 4.3 Martin-Löf-style type theory

Martin-Löf-style type theory, or intuitionistic type theory, develops type theory as a constructive foundation for mathematics [DP24; Mar84; NPS90]. As with regular type theory, one begins with judgements about contexts, types, terms and equality.

**Dependent types** are a powerful extension: within this theory, a type may depend on a term. If  $A$  is a type and  $B(x)$  is a type depending on  $x : A$ , then one can form dependent function and pair types. Dependent function types,  $\Pi$ , and dependent pair types,  $\Sigma$ , generalise the definitions of their ‘simpler’ versions above<sup>4</sup>. Furthermore, these functions and pairs can then themselves also be considered terms of yet more general types or functions and pairs. It is worth being explicit that this extension makes these functions and pairs extremely flexible<sup>5</sup>. The more formal definitions are presented in Definitions A.1.9 and A.1.10.

The law of excluded middle is not derivable, which is desirable within the intuitionistic discipline. If excluded middle is needed for a particular argument, it must be introduced as an assumption and paid for openly. A small demonstration of non-derivability is made explicit in Proposition B.3.1.

The above are generally accepted for Martin-Löf-style type theory, but there are some specific choices that can be made in the application of this theory, especially with respect to **identity types** and **inductive types**. Identity types operationalise equality, as they allow one to reason about when two terms of a type are considered equal. *Intensional* type theory is relatively strict, allowing equality only when two terms are ‘exactly’ identical (for example: 4 is identical to 4, but  $3 + 1$  is not necessarily considered identical to 4 inside this theory), whereas *extensional* type theory allows for more flexible equality (so:  $3 + 1$  is considered identical to 4). Inductive types allow familiar mathematical objects

---

<sup>4</sup>Dependent function types roughly correspond to universal quantification,  $\forall$ , dependent pair types roughly correspond to existential quantification,  $\exists$ .

<sup>5</sup>Indeed, this way of considering functions – as terms themselves – forms the basis of functional programming languages, in which functions can be passed around as terms.

#### 4. Third movement: Adagio

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to be generated by constructors: natural numbers from zero and successor, finite types from explicit labels, lists from empty lists and cons, and so on. The more formal definition of these can be found in Definitions A.1.12 and A.1.13.

In order to develop Martin-Löf-style type theory into a modal logic in the next movement, the extension by Gratzner and collaborators is employed [Gra+21; Gra23; GSB19]. For this reason, my design for Martin-Löf-style type theory is partly governed by the design of their theory. These choices are made explicit in Definitions A.1.6, A.1.8 and A.1.12 to A.1.14.

This concludes the philosophical and mathematical setting for the constructive part of the thesis. The Martin-Löf-style type theory used here is intuitionistic in its logic, epistemological in its interpretation, finite enough to avoid infinite paradoxing, and computational enough to connect with proof checking. It therefore answers Jaynes' demand for explicit constructions without inheriting his overly broad universality claim: the framework tracks what can be formed, proved and computed from a given context, while leaving ontology deliberately open.

## CHAPTER 5

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### **Fourth movement: Allegro maestoso**

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*“Scientists have calculated that the chance of anything so  
patently absurd actually existing are millions to one.  
But magicians have calculated that million-to-one chances crop  
up nine times out of ten”*

Terry Pratchett in *Mort*

## 5. Fourth movement: Allegro maestoso

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The previous movement left three issues to address:

1. **Subjectivity**: has to become part of the formal language rather than a slogan about priors.
2. **Probability**: has to be reconstructed over finite, decidable objects rather than arbitrary classical propositions.
3. **Causality**: has to enter as intervention and model change, not merely as statistical dependence.

The fourth movement takes these in order: *contexts and modalities* first, *probability types* next, and *finite structural causal models* last.

### 5.1 Context manipulation in Martin-Löf-style type theory

Contexts are the quiet machinery of subjectivity. A context records what is currently available: variables, assumptions, dependencies and previously constructed terms. In ordinary prose one says “given what we know”, in type theory one writes a context. The formal definitions can be found in Definitions A.1.2 and A.1.3.

Several operations on contexts have immediate epistemic readings. *Extending a context by a new variable* corresponds to learning or introducing a new assumption. *Allowing a later variable to depend on an earlier one* records directed dependence. *Deleting an unused variable* corresponds to forgetting or ignoring something that nothing else depends on. Furthermore, *if a question depends on a variable absent from the current context*, the appropriate answer is not false but unavailable – not in this context, not from here, not with this information. What is in scope can be used. What is out of scope cannot even be formed as a judgement. These operations give a first, austere account of subjectivity.

This also gives a first account of intervention. *To intervene* is not merely to observe a value; it is to set a variable and sever or alter the dependencies that normally determine it. In a context, that means changing the dependency structure. *To hypothesise* is similar but more theatrical: one extends the current context with alternative variables, changes something there, and asks what follows under that altered arrangement.

Context manipulation alone is not yet the full, subjective theory. It gives the raw operations, but not the classification of epistemic states as such. For that, the thesis treats certain context changes as transitions between modes. Strictly speaking, ordinary type theory could already perform the operations. However, subjectivity becomes clearer when specific operations are marked as movements between states of knowing. A **mode theory** supplies that marking, which will be discussed in the following section.

### 5.2 Modalities inside Martin-Löf-style type theory

A mode theory is the smallest extra structure that records an epistemic location: it declares a small collection of epistemic states (*modes*), together with the admissible transitions between them (*modalities*), and indexes every judgement by the mode at which it is made. A mode can be read, for present purposes,

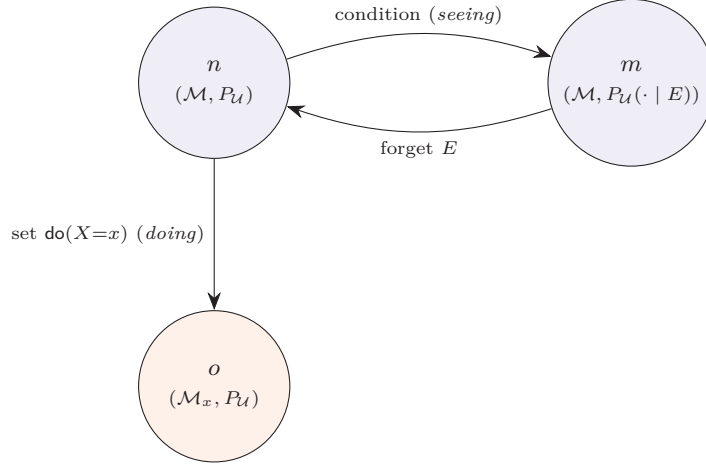


Figure 5.1: Modes as epistemic states; modalities as admissible transitions. Conditioning updates the exogenous distribution (abduction); setting replaces a mechanism (intervention). The fragment is introduction-only: modalities *mark* admissible moves rather than inspect the context (Definition A.2.4 and Remark A.2.8).

as an epistemic state. A modality can be read as a structured way of moving from one epistemic state to another, see Figure 5.1. The price is one extra index on every judgement; the benefit is that learning, forgetting, hypothesising and intervening become (a combination of) formal moves rather than informal annotations.

**Gratzer’s theory** provides an extension of Martin-Löf-style type theory, where one works not only with contexts and types, but with modes and modalities between modes [Gra+21; Gra23; GSB19]. Gratzer’s theory is rather extensive and provides many additional proofs and formal definitions. For the purposes of this thesis, the technical ambition is rather limited, as I only need *specific* modes and modalities to function properly, rather than a generalised case. The formal fragment used here is therefore rather modest in scope and made explicit in Definitions A.2.1 to A.2.6. The finite implementation is stated in Definition B.5.8; its preservation, conditioning and cancellation facts are in Propositions B.5.9 to B.5.11.

With subjectivity explicitly formalised, the next step is to allow uncertainty within the epistemic states themselves; that is, uncertainty over the available propositions within a given context. For this, a new type needs to be introduced: **probability types**. These are discussed next.

### 5.3 Probability types

The next task is to allow for uncertainty within each context. The guiding idea is simple: a probability assignment should be an explicit object attached to a finite and, ideally, decidable domain. It should therefore be represented by a type whose inhabitants carry the required structure, as laid down in Definitions A.3.1, A.3.4 and A.3.5.

## 5. Fourth movement: Allegro maestoso

In Martin-Löf-style type theory, however, an arbitrary proposition is *not* decidable, because excluded middle is not available. Probability within this theory is therefore not defined on the propositions themselves, but on their resolution status: proof found, refutation found, or neither found. The resolution-status for every proposition inside Martin-Löf-style type theory *is* decidable, even when a proposition itself is not. This definition is made precise in Definitions A.3.2 and A.3.3.

Clayton and Waddington provide the bridge from Cox-style reasoning to a finite, constructive reconstruction [CW17]. The rules named in Section 3.2 are made formal in Definition A.3.6. The order of use is slightly different from the original Clayton–Waddington route. Furthermore, as already mentioned, universality, continuum regularity and arbitrary-proposition complementation are not used in the constructive proof, but are part of the original paper, as it operates on a different domain.

| Step | Ingredient or result            | Role in the argument                                                                                                                      | Where                                                                |
|------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1    | Finite resolution-status events | Replaces probability on arbitrary unresolved propositions by a decidable three-way event partition.                                       | Proposition B.4.1 and Corollary B.4.9                                |
| 2    | Finite urn principles           | Give the order-preserving rescaling $U_N(k) \mapsto k/N$ ; decidable urn complements then satisfy the finite complement rule.             | Proposition B.4.2                                                    |
| 3    | Product urn compatibility       | Gives multiplication for rectangular product events after the ratio rescaling, so the product rule is recovered after the sum/ratio step. | Proposition B.4.3                                                    |
| 4    | Record–urn equivalence          | Shows that urn presentations and common-denominator finite rational records agree on every decidable event.                               | Proposition B.4.5                                                    |
| 5    | Finite event calculus           | Proves non-negativity, normalisation and finite additivity, with monotonicity and inclusion–exclusion as corollaries.                     | Theorem B.4.6 and Corollaries B.4.7 and B.4.8                        |
| 6    | Conditional probability         | Proves Bayes’ theorem and Bayesian update for decidable events with positive conditioning probability.                                    | Definition A.3.7, Theorem B.4.10, and Propositions B.4.11 and B.4.12 |

Table 5.1: Route through the finite probability reconstruction.

The full proof route is summarised in Table 5.1 and carried out in Section B.4. Starting from the rules of Clayton and Waddington, the construction recovers

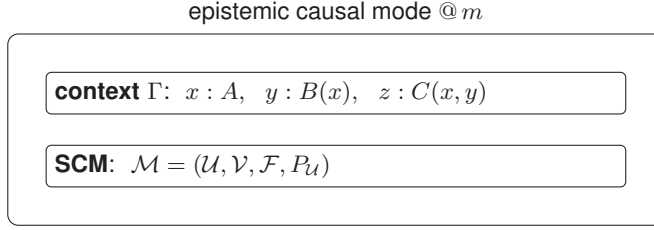


Figure 5.2: An epistemic causal mode bundles an ordered context with a finite structural causal model; moving between modes changes the model, the available information, or which interventions are in scope (Definition A.3.37).

both the finite Kolmogorov laws and Bayes’ theorem. Since these rules are appealing and intuitive, and since the final result appears in an epistemological and constructive setting, the construction captures many features desired by both Jaynes and his critics. The main victory is conceptual: probability is made into something fundamentally constructible and checkable.

For a complete formalism, causality still needs to be explicitly introduced. In the next section, **structural causal models** will serve both as an implementation of causality and as a vehicle for epistemic states.

## 5.4 Structural causal models in Martin-Löf-style type theory

Pearl-style structural causal models can now be rebuilt from the same basis. A finite structural causal model consists of variables, exogenous noise, structural functions and a finite rational probability distribution over the exogenous variables [Pea09; PM18]. The formal definition can be found in Definitions A.3.8, A.3.9 and A.3.25. In the constructive setting, each component can be represented explicitly: variables as finite types or indexed families, structural functions as terms, exogenous uncertainty as a probability type – defined in the previous section – and interventions as transformations of the structural functions.

The ordered shape of a type-theoretic context is doing real work here: later declarations may depend on earlier ones; earlier declarations cannot depend on later ones. Thus, acyclicity is not merely an external graph condition in the restricted fragment: it is reflected in what the structural term is allowed to mention. This is the ordered-telescope reading of finite recursive structural causal models.

The modal reading gives structural causal models their epistemic interpretation. A causal model is more than an abstract graph: it represents a genuine state of knowledge, namely what variables are distinguished, what mechanisms are assumed, what interventions are meaningful, and which counterfactual questions can be asked. Moving from one structural causal model to another is therefore a change of epistemic state; see Definition A.3.37 for a formal definition. Modes containing these structural causal models can represent these states; modalities can represent admissible transitions between them, see Figure 5.2.

Causal reasoning fits this modal reading compositionally. This means that an intervention or twin-network construction is not a single primitive move: to build a twin network, one learns the copied endogenous variables; relates

## 5. Fourth movement: Allegro maestoso

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the actual and copied sectors to the shared exogenous references; sets the intervention in the copied sector; and evaluates the resulting record. This is why I do not introduce a separate ‘do modality’ or ‘twin-network modality’. The point of the modal machinery introduced in Section 5.2 is precisely that smaller modalities can be composed into larger ones. In Gratzner’s theory, such composites are again legitimate modalities [Gra+21; Gra23]. It is therefore better to keep the primitive moves small, since larger causal operations can be assembled from them when needed.

Seen in this way, current version of Pearl’s framework suggests transitions beyond ordinary causal queries. In an ordinary query, a structural causal model is usually held fixed; in scientific reasoning, however, the model itself may change: a mechanism may be discovered; a variable may be split in two; an intervention may be reclassified; a hidden confounder may be introduced; or a previous assumption may be withdrawn. Treating causal models as epistemic modes makes such changes visible as transitions between states of knowledge. The present thesis does not develop a general calculus for these model-changing transitions; its more limited aim is to reconstruct Pearl’s finite framework inside a constructive setting, while leaving this wider modal-causal direction for future work.

It is worth running a concrete example before presenting more general results. The natural choice is to revisit Mira’s situation from Example 4.2.1, but now from the hiring committee’s side. Officially, the committee claims to select on research dossier strength alone; in practice, candidates from prestigious institutions tend to do better, fashionable topics tend to attract committee members with matching expertise, and funding lines have their own seasonality. The interesting question is not whether such a committee is biased – almost every committee is – but whether the bias visible at the output reflects a stable causal mechanism or merely a hidden common cause. The model below is small enough to inspect line by line and rich enough to exhibit the standard causal phenomena, see Figure 5.3. The type-level proposition  $\text{Offer}(\text{Mira})$  now becomes the Boolean endogenous variable  $\text{Off}$ , and the three resolution statuses *proved*, *refuted* and *unresolved* from Definition A.3.3 become decidable events on the joint exogenous assignment.

**Example 5.4.1** (Mira’s tenure-track search, as a structural causal model). Returning to Mira from Example 4.2.1, the running example is now a stylised tenure-track hiring decision viewed from the search committee’s side. The finite probabilities below should be read as a toy model for a candidate already under serious committee consideration, not as empirical base rates for all tenure-track applications. All endogenous variables are Boolean. Let  $\text{Pr}$  denote institutional prestige,  $\text{Qu}$  research dossier strength,  $\text{Top}$  topic fashionability,  $\text{Com}$  committee expertise,  $\text{Fit}$  committee fit,  $\text{Short}$  being shortlisted,  $\text{Fund}$  funding-line availability and  $\text{Off}$  receiving the tenure-track offer; the previous proposition  $\text{Offer}(\text{Mira})$  is the event  $\text{Off} = 1$  on Mira’s joint exogenous assignment. One finite structural causal model is given by independent Boolean exogenous variables

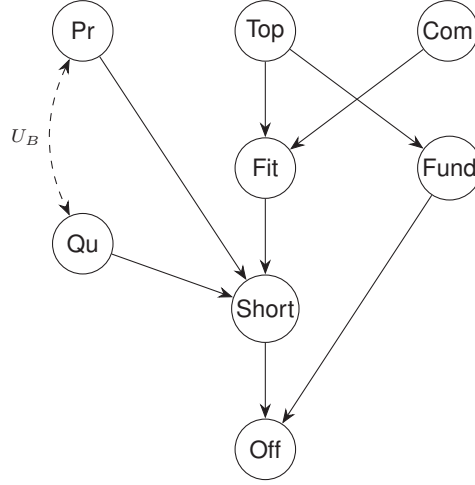


Figure 5.3: Mira's tenure-track structural causal model. The triples  $\text{Top} \rightarrow \text{Fit} \leftarrow \text{Com}$ ,  $\text{Pr} \rightarrow \text{Short} \leftarrow \text{Fit}$  and  $\text{Short} \rightarrow \text{Off} \leftarrow \text{Fund}$  are colliders; the dashed edge is the latent common cause  $U_B$  inducing  $\text{Pr} \leftrightarrow \text{Qu}$  (Example 5.4.1).

$U_B, U_{\text{Pr}}, U_{\text{Qu}}, U_{\text{Top}}, U_{\text{Com}}, U_{\text{Fund}}$  and structural equations

$$\begin{aligned} \text{Pr} &:= U_B \vee U_{\text{Pr}}, & \text{Qu} &:= U_B \vee U_{\text{Qu}}, \\ \text{Top} &:= U_{\text{Top}}, & \text{Com} &:= U_{\text{Com}}, \\ \text{Fit} &:= \text{Top} \wedge \text{Com}, & \text{Short} &:= \text{Qu} \wedge (\text{Pr} \vee \text{Fit}), \\ \text{Fund} &:= \text{Top} \vee U_{\text{Fund}}, & \text{Off} &:= \text{Short} \wedge \text{Fund}. \end{aligned}$$

The directed structure contains

$$\begin{aligned} U_B &\rightarrow \text{Pr}, \quad U_B \rightarrow \text{Qu}, \quad \text{Top} \rightarrow \text{Fit} \leftarrow \text{Com}, \\ \text{Pr} &\rightarrow \text{Short} \leftarrow \text{Fit}, \quad \text{Qu} \rightarrow \text{Short}, \quad \text{Top} \rightarrow \text{Fund}, \\ \text{Short} &\rightarrow \text{Off} \leftarrow \text{Fund}. \end{aligned}$$

If  $U_B$  is not observed, the observed semi-Markovian projection (in the sense of Definition B.6.9) has the bidirected edge  $\text{Pr} \leftrightarrow \text{Qu}$ . Thus the example contains an unobserved common cause, colliders, ordinary directed chains and the graph predicates used later for d-separation and c-components. With the finite rational probabilities in Example A.4.1, one obtains, for instance,

$$P(\text{Qu} = 1 \mid \text{Pr} = 1) = \frac{3}{4}, \quad P(\text{Qu} = 1 \mid \text{do}(\text{Pr} = 1)) = \frac{1}{2}.$$

Seeing prestige gives evidence about the hidden background variable that also affects dossier strength. Setting prestige by intervention does not change that background variable and therefore does not by itself improve the dossier.

The same finite model also supports a representative counterfactual calculation. Suppose the actual evidence is that the candidate had no prestige, had a strong dossier, and received no offer:

$$E := \{\text{Pr} = 0, \text{Qu} = 1, \text{Off} = 0\}.$$

## 5. Fourth movement: Allegro maestoso

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After conditioning the exogenous distribution on  $E$ , replacing the equation for  $\text{Pr}$  by the constant value 1, and evaluating the offer variable in the intervened model, one obtains

$$P(\text{Off}_{\text{Pr}=1} = 1 \mid E) = \frac{1}{2}.$$

This is a genuine abduction–action–prediction calculation for the concrete finite model, checked again in Example A.4.1; it is not a general counterfactual-identification theorem from a graph alone.

In the tenure-track example the epistemic reading means that the model is not treated as background prose, but takes a more active role. Schematically, at the tenure-track mode there is a judgement

$$\Gamma_{\text{ten}} \vdash \mathcal{M}_{\text{ten}} : \text{FinSCM}@m_{\text{ten}},$$

where  $\text{FinSCM}$  is the restricted finite type-theoretic structural-causal-model fragment of Definition B.5.1. An intervention such as  $\text{do}(\text{Pr} = 1)$  moves to an epistemic causal mode whose structural-causal-model record is  $\mathcal{M}_{\text{ten}, \text{Pr}=1}$ , obtained by replacing the structural equation for  $\text{Pr}$  by the constant value 1 and leaving the other mechanisms intact. A calculation such as

$$\Gamma_{\text{ten}} \vdash P_{\mathcal{M}_{\text{ten}, \text{Pr}=1}}(\text{Qu} = 1) = \frac{1}{2}@m_{\text{ten}}$$

is therefore a judgement about a finite probability expression inside the current epistemic causal mode.

In this movement, I am not claiming to rebuild all of Pearl’s theory from scratch, but I am making a finite correspondence claim: in the finite, discrete and recursive case, the ordinary ingredients of a structural causal model can be represented inside the type-theoretic setting used here. Variables become finite types; structural equations become terms; exogenous uncertainty becomes a finite rational probability type; interventions become transformations of the relevant structural equations; and the graph tests used by do-calculus remain finite combinatorial checks.

Once this bridge is in place, the classical finite results do not need to be reinvented, as they can be transported. The appendix proves this in stages, moving from the restricted finite record fragment, through probability and graph transport, to the transferred identification and do-calculus results; see Table 5.2. The point is not to reprove Pearl–Tian–Shpitser completeness from scratch, but to show that the restricted type-theoretic fragment is an adequate target for current finite results [HV06; Pea12; SP06; TS09].

| Step | Ingredient or result             | Role in the argument                                                                                                                                                                 | Where                                               |
|------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1    | Restricted finite causal records | Fixes the type-theoretic fragment that is meant to correspond to ordinary finite recursive structural causal models.                                                                 | Definition B.5.1                                    |
| 2    | Model correspondence             | Shows that classical finite models and restricted type-theoretic records carry the same finite variables, values, structural equations, probabilities, interventions and graph data. | Theorem B.5.2                                       |
| 3    | Expression and graph transport   | Moves probability expressions and finite graph predicates across the correspondence, so that later causal arguments speak about the same finite objects on both sides.               | Propositions B.5.4 and B.5.5                        |
| 4    | Query and derivation transport   | Transfers interventional queries, identifiability claims and do-calculus derivations across the bridge.                                                                              | Corollary B.5.13 and Propositions B.5.14 and B.5.15 |
| 5    | Completeness transfer            | Applies the published finite Pearl–Tian–Shpitser results on the classical side and transports the result back to the restricted type-theoretic fragment.                             | Section B.6                                         |

Table 5.2: Route through the finite causal correspondence and completeness transfer.

The transport argument is mostly bookkeeping, but this is precisely why it matters, as the architecture is now explicit: probability types supply uncertainty; modal dependent type theory supplies epistemic movement; structural causal models supply intervention and counterfactual vocabulary; and Lean checks representative finite pieces of the construction. The Lean results are discussed in the following section.

## 5.5 Lean 4 implementation

The mathematical manifesto asks for explicitness, and in this work Lean 4 is used as a proof checker [MU21]. Three Lean modules accompany this thesis: `lean/Thesis/Probability.lean` for the finite probability core, `lean/Thesis/SCM.lean` for finite structural causal models, and `lean/Thesis/Modalities.lean` for the finite epistemic transition layer. The aim is not to mechanise every philosophical claim of the text, but to keep a small executable witness for the central finite constructions, so that hidden assumptions surface as type errors rather than as forgiving prose.

The boundary of the formalisation is just as important. Strictly speaking, a Lean proof is not a proof in raw intensional Martin-Löf type theory. Lean 4 is a proof assistant based on dependent type theory [MU21], but it has its own kernel, inductive types and conventions. This distinction does not undermine the present

## 5. Fourth movement: Allegro maestoso

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use of Lean, because the mechanised claims are modest, finite constructions: rational finite probability records; recursive finite structural causal models; and explicit finite transition records. The code at `lean/Thesis/Modalities.lean` therefore formalises a finite transition layer compatible with the Gratzner-style discipline described in Definitions A.2.1, A.2.4 and A.2.5, not Gratzner’s full modal dependent type theory [Gra+21; Gra23; GSB19]. Furthermore, it does not mechanise the full do-calculus completeness transfer of Theorem B.6.11 or the causal-completeness transfer of Theorem B.6.13. Those broader results are stated and argued in the appendix. Lean only checks the finite ingredients on which the current construction rests.

The flavour of the Lean code is best seen in an example, here the record–urn equivalence theorem:

Listing 5.1: Record–urn extensionality (`lean/Thesis/Probability.lean`).

---

```
theorem urn_to_record_extensional (m : UrnProb 0) :  
  forall E : Event 0, QProb.Equiv ((ofUrn m).probVal E) (m.probVal E) :=  
    ofUrn_agrees m  
  
theorem record_to_urn_extensional (R : FiniteProbRecord 0) :  
  forall E : Event 0, QProb.Equiv (R.probVal E) (R.toUrn.probVal E) :=  
    R.toUrn_agrees
```

---

Read informally, the two statements say that converting an urn to a finite record, or a finite record to an urn, does not change the probability assigned to any decidable event. The proof is not deep, and that is exactly the sort of bookkeeping that prose often hides: counts are preserved when atoms are expanded, unit-weight atoms count as one cell, and the resulting rational probabilities agree by cross multiplication. Lean is used here not for theorem-proving acrobatics, but to keep the finite construction honest.

The Lean implementation is best read as a test of explicitness. The probability module checks that the finite probability calculus really can be written as finite data and finite proofs. The causal module checks that recursive structural causal models, interventions and representative counterfactual calculations can be handled similarly. The modalities module checks that the finite epistemic transitions used above preserve the required well-formedness conditions. These correspondences are collected in Table 5.3.

| Module                      | What it checks                                                                                                                                                                                                                       | Where it connects                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Thesis/<br>Probability.lean | Finite urns, common-denominator weighted records, the record–urn equivalence, finite Kolmogorov laws, Bayes calculations, the Clayton–Waddington rescaling steps and Bayesian-update normalisation.                                  | Section 5.3; Table 5.1; Section B.4                          |
| Thesis/SCM.lean             | Recursive finite structural causal models, observational and interventional probabilities, the tenure-track calculations, the model-level counterfactual helper and the local binary conditional-probability-table row construction. | Section 5.4; Table 5.2; Section B.5                          |
| Thesis/<br>Modalities.lean  | The finite transition layer: learning, forgetting, conditioning, relating, unrelating, setting, unsetting, preservation of well-formedness, the basic cancellation checks and the modal version of the tenure-track counterfactual.  | Sections 5.1 and 5.2; Definition B.5.8 and Proposition B.5.9 |

Table 5.3: Lean modules and their checked finite constructions.

## 5.6 Results

The research questions postulated in the Prelude were:

1. Can Jaynes’ probabilistic framework be cast in the language of Martin-Löf-style type theory?
2. Can Jaynes’ probabilistic framework be extended so as to natively include subjectivity?
3. Can Jaynes’ probabilistic framework be extended so as to include Pearl-style causal reasoning?
4. Can Jaynes’ commentary on bad scientific practice be placed in a wider institutional context?

This movement answers the first three questions in the restricted finite setting of this thesis. First, Jaynes’ probabilistic framework can be cast in Martin-Löf-style type-theoretic language, provided probability is assigned to finite decidable event types rather than to arbitrary propositions. This reconstruction is developed in Section 5.3, while the proof route is summarised in Table 5.1.

Second, the framework can be extended with a native account of subjectivity, where subjectivity is no longer reduced to the choice of a prior. Contexts, modes and modalities make the epistemic position itself part of the formal language. This account is developed in Sections 5.1 and 5.2, while the corresponding finite transition layer is summarised in Table 5.3.

## 5. Fourth movement: Allegro maestoso

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Third, Pearl-style causal reasoning can be embedded in the same finite constructive setting. Structural causal models, interventions and representative counterfactual calculations are developed in Section 5.4, while the correspondence and completeness-transfer route is summarised in Table 5.2.

The fourth question is deliberately left open here. It belongs to Chapter 6, where the formal concern with hidden assumptions is connected to the institutional conditions under which scientific inference is produced.

## CHAPTER 6

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### **Fifth movement: Presto con fuoco**

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*“For far too long economists have sought to define themselves in terms of their supposedly scientific methods. In fact, those methods rely on an immoderate use of mathematical models, which are frequently no more than an excuse for occupying the terrain and masking the vacuity of the content. Too much energy has been and still is being wasted on pure theoretical speculation without a clear specification of the economic facts one is trying to explain or the social and political problems one is trying to resolve”*

Thomas Piketty in *Capital in the Twenty-First Century*



## 6.1 Questionable research practices on a large scale

Jaynes' criticism of bad scientific attitude can be read as a criticism of individual character, as was seen in Section 2.3:

1. **Gamesmanship:** the desire to display technical virtuosity, rather than to clarify a problem;
2. **Cutting corners:** the focus on polished results, without explaining the painstaking work that led to these outcomes;
3. **Hidden assumptions:** the silent transfer of what should have been stated into what is merely presumed;
4. **Hyperfocus:** the habit to only investigate an incredibly specialised subfield, whilst forgetting the wider context in which this research sits;
5. **Theoretical navel-gazing:** the production of mathematics that neither explains, predicts, nor helps anyone build or test anything practical.

In his text he then continues to blame Fisher for having all these vices, while Jeffreys is presented as a model scientist. Besides the fact that this is rather simplistic view of both quite complicated characters, the following observation by Jaynes is at least as interesting: Jaynes laments that although Jeffreys was correct in his view on probability theory<sup>1</sup>, he only trained one Ph.D. student during his whole career<sup>2</sup>. Meanwhile, Fisher, who according to Jaynes was utterly wrong on his views on probability theory, trained several tens of influential Ph.D. students [Jay03]. Although in itself completely anecdotal, this observation suggests that perhaps the mechanisms by which scientific ideas and scientific careers spread do not select primarily for truthfulness.

The business language that has become central to science matters here. Discovery is naturally highly nonlinear, but is nowadays often translated into a linear framework of targets, deliverables and measurable output. This also makes the risk more precise: when funding, promotion, job security, visas and prestige all depend directly or indirectly on publication rate, publication can become the object that is optimised, rather than any 'truthfulness' as such. The phrase 'publish or perish' is therefore not merely a complaint about stress, but it highlights a selection environment. As such, it is advisable to consider scientific inheritance, promotion and prestige as part of the machinery of inquiry as well. McElreath's reminder from the Prelude is therefore central here: inference is only one component of science [McE20]. A formally correct inference engine cannot rescue a pipeline in which claims are not selected for their truthfulness, but rather for fashionability, publishability and speed.

Surveys and meta-analyses of questionable research practices and misconduct already suggest that scientifically questionable behaviour is widespread, and even more so since the official record understates the problem, as (self-)reporting is naturally limited by stigma and personal risk [Fan09; JLP12; MAV05]. Retraction and paper-mill studies emphasise the same problem from another

<sup>1</sup>Jeffreys is widely known for having resisted the theory of plate tectonics throughout his life.

<sup>2</sup>Maybe not completely coincidentally, also Jaynes only had one Ph.D. student during his career, giving this observation a rather personal dimension.

## 6. Fifth movement: Presto con fuoco

angle [Agr+25a; FSC12; Ric+25]. More recent reports on the sheer scale of industrialised fraud, fake authorship markets, pseudo-scientific journals and batch retractions make the same point even more urgently [Bri25; Nad26; NDR18; OGr25].

| Study                              | Field                | What was checked                                                              | Reported signal                                                                                                                                                                         |
|------------------------------------|----------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Open Science Collaboration [Ope15] | Psychology           | 100 studies from three psychology journals                                    | 97% of original studies reported significant results; 36% of replications did so, 39% were subjectively judged to replicate, and replication effects were about half the original size. |
| Klein et al. [Kle+14]              | Social psychology    | 13 effects tested across 36 samples in the Many Labs project                  | 10 effects replicated consistently, one received weak support, and two did not replicate.                                                                                               |
| Camerer et al. [Cam+16]            | Economics            | 18 laboratory experiments from top economics journals                         | 11 of 18 replications, or 61.1%, found a significant effect in the same direction; replicated effect sizes averaged 66% of the originals.                                               |
| Begley and Ellis [BE12]            | Preclinical oncology | 53 landmark cancer findings checked in an industrial drug-development setting | Scientific findings were confirmed in only 6 cases, roughly 11%, well enough to continue the development programme.                                                                     |
| Button et al. [But+13]             | Neuroscience         | Statistical power and its consequences for neuroscience research              | The central result is not a direct replication rate: low power lowers the chance that a significant result reflects a true effect and inflates effect-size estimates.                   |

Table 6.1: Selected reproducibility signals across fields. The table is intentionally heterogeneous: significance rates, subjective replication judgements, confirmed findings and statistical power all expose different parts of the same institutional problem.

The scale of the consequences is difficult to explain by individual vice alone – contrary to what Jaynes posited. The studies in Table 6.1 all point to the same direction: when published claims are put under more explicit pressure, much of their apparent certainty disappears. Smaldino and McElreath’s point is precisely that bad methods can spread by selection even when most individuals are not trying to corrupt the record [SM16]. Bad science is therefore not only a matter of bad people; it is also a matter of bad selection pressures. The better conclusion to draw based on this data is not that science is full of vicious personalities, but that the current system actively rewards behaviour that is adjacent to fraud.

Explicit fraud cases are useful here, because they make the institutional pressures visible in extreme form. Stapel, Schön, Hwang, Obokata, Wakefield, the LaCour case involving Green’s coauthorship, Pandey, Raoult and Fujii are not typical in every detail, but they reveal familiar failure modes: fabricated or copied data, ethical shortcuts, failure to declare conflict of interest, salami-sliced output, hypothesising after the results are known (HARKing), reviewer manipulation, author manipulation, prestige dependence, citation games and highly inadequate correction mechanisms [BDK02; Cyr06; GSM11; Kon15;

Los11; MO15; OGr24; The16; The24]. Where the incentive of such behaviour stems from is the subject of the next sections.

## 6.2 Start of a research career

The pressure begins before a researcher has the power to resist it. A young researcher needs publications to obtain funding, funding to obtain positions, positions to obtain independence, and independence to obtain the time required for careful work. This is the catch-22 of grants and careers: one needs resources to do the work that justifies receiving resources [Ibá99; NI12]. The result favours rich institutions, already visible researchers and projects close enough to fashion that committees can recognise them. The inequality is not only personal: if ambitious research increasingly requires expensive equipment, time, grant-writing capacity and institutional prestige, then rich entities can attempt speculative projects while poorer entities either support the existing agenda or operate with much less room for risk. As a result, a conservative funding environment has an opportunity cost: low-quality but recognisable work is generally supported while high-quality but risky work is suppressed or delayed [Agr+25b; Lan+22; PLF23].

This structure makes positive findings unusually valuable. The incentive is not simply to do good work, but to produce something that can be written up as a finding. Nosek, Spies and Motyl describe the conflict directly: incentives for publishable results can pull against incentives for accurate results [NSM12]. Replication studies, negative findings and careful methodological work are all part of science, but they have often carried less immediate career value than mere novelty of results. Furthermore, the penalty for an irreproducible study is also often diffuse and delayed, whereas the reward for a publishable claim can be immediate.

The tragedy is that the system can then select against precisely the kind of slow work it later celebrates. Peter Higgs remarked that, under present productivity expectations, he would probably not have been considered productive enough to keep his post [Ait13; Al-14]. That is not an argument against funding or accountability, but it *is* an argument against confusing accountability with throughput.

As mentioned, the pressure to publish does not need bad intent in order to create bad science. Small samples, underpowered designs, flexible stopping rules and flexible analyses allow researchers to obtain publishable results without consciously committing fraud [But+13; SNS11]. Gelman and Loken's garden of forking paths emphasises that a study can be kneaded by many unnoticed analytical choices to obtain the desired results, even when nobody is deliberately 'fishing' [GL13; GL14]. A study can be made more publishable by post-hoc hypothesising, selective reporting, choosing the positive analysis among many alternatives, cutting power because more data takes more time, or replacing careful modelling by the quickest defensible model. These are not rare choices, but ordinary research decisions that can subconsciously support a publication economy that rewards significance, clarity and speed [Ioa05; NSM12; SM16; SNS11]. Gamesmanship, corner-cutting, not specifying hidden assumptions, hyperfocus and theoretical navel-gazing do not have to be taught, they can emerge as adaptive responses to the environment.

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Additionally, it should be noted that aforementioned pressure can be especially sharp for researchers whose employment or residence status depends on visible output. This should be stated carefully: visa pressure does not make a researcher fraudulent, but when the right to stay in a country or the ability to support a family is linked to publication metrics, fraudulent intermediaries obtain a ready market [OGr25; Ric+25]. It can only be concluded that with the advent of more protectionist research environments the coming years, together with intensifying competition, fraud cases and retracted studies related to visas, permits and direct livelihoods will most likely increase.

### 6.3 Reporting in journals

Firstly, the current ‘winner-takes-all’ economy within science actively damages collaboration, as everyone competes for the same (limited) publishing space. When the first team to publish the positive result wins the reward, independent verification becomes a less attractive use of time. This is especially dangerous in fields where the underlying process of discovery is slow, uncertain and nonlinear. Artificial intelligence can intensify the problem by increasing the speed at which manuscripts, variations and near-duplicates can be produced [Zha26]. The journal article then turns an uncertain process into a clean product, and the market rewards the product more directly than the process that produced it, or any corrections that come after.

Journals are not merely neutral containers for such products, as they are also competing for attention, subscriptions, impact and prestige. Randy Schekman, a Nobel laureate in biology, has openly criticised prestige journals for distorting incentives [Sch13]. The justified worry in a system where journals have a conflict of interest with the truth is that surprise, novelty and controversy can be rewarded before reliability is known. As such, correction can remain slow or absent even after reliability is doubted or proofs are provided of clear misconduct [Bik26a; FSC12]. In practice, investigations by journals can be pushed back onto the same institutions, laboratories and even the very same people who are under investigation [OGr24; OM23; The24]. As such, even when some of the behaviours leading to publication are formally prohibited, the consequences are often lighter, slower or less predictable than the original reward [Cam98; MAV05; Smi06].

Once publication itself becomes a currency, a market grows around that currency: salami slicing, extreme self-citation, exploit editorial and review roles for citation, citation brokers, ghostwritten papers, fake authorship advertisements, paper mills, predatory or pseudo-scientific journals, unusual review processes, identity manipulation and fake reviewer accounts all exploit the fact that careers, visas, rankings and institutional prestige depend on visible output [Agr+25a; Bri25; Cam98; MAV05; Nad26; NDR18; OGr25; Smi06]. Also here remains true that gamesmanship, corner-cutting, not specifying hidden assumptions, hyperfocus and theoretical navel-gazing do not have to be taught as principles, but they appear as responses to the environment. At this point the problem is no longer only a set of individual lapses, as it can be readily concluded there has been built an entire industry around the weaknesses of current publication system.

## 6.4 Peer review and editorial process

Peer review is supposed to prevent many of the described behaviours, but the task assigned to it is almost impossible. Reviewers are asked to judge novelty, correctness, significance, ethics, method, relevance and sometimes fraud, usually without enough time, sufficient training, necessary data or any reward to do so [Smi06]. Furthermore, without proper training and a good definition of the process, there are many problems directly identifiable: inconsistency between reviewers; rivalry between reviewers and authors; bias, conservatism and self-interest of reviewers; and lack of clear standards for what defines a good review [Cam98].

As a result, review actively stimulates conservatism and review systems can punish risk while actually claiming to value innovation [AS14; GB24; Lan+22]. Reviewers and researchers are trained to be critical, and any paper will reveal weaknesses if pressed hard enough. Since the process is also structurally conflicted – as reviewers and authors often compete for similar publication space, funding and prestige – referees can ask for extra citations, editors can choose reviewers with predictable objections, and anonymous review can leave authors with little recourse against poor or self-interested judgement [Cam98; SR14]. In a sense, the garden of forking paths that is present for authors, exists for referees and editors as well. The broader evidence that papers and patents have become less disruptive over time gives this observation a historical perspective [PLF23].

Conservatism also gives disproportionate power to senior researchers and prestigious institutes [AS14; Cam98; Lan+22]. Their names can help a paper survive review; their views can shape what is judged plausible; and their time, networks and authority make them more likely to be involved in evaluation. In extreme cases this allows for coercive citation, inappropriate authorship demands and reviewer manipulation, or even citation cartels on larger scales [Agr+25a; Ric+25]. This can slow the ordinary replacement of old ideas by new ones considerably; a process which is natural in society and one of the cornerstones of the scientific method. A radical paper may be rejected because the shared assumptions of the review community make it hard to read fairly. Gamesmanship, corner-cutting, not specifying hidden assumptions, hyperfocus and theoretical navel-gazing can then be reproduced by peer review rather than exposed by it.

The standards for reviewing are often also surprisingly thin. Authors are usually given detailed formal obligations on how and what to publish, but reviewers and editors often receive much less training and fewer explicit standards. Proper reviewing is therefore partly learned at the expense of authors: an anonymous referee can be careless, arbitrary or self-interested while the author carries the cost [Cam98; Smi06]. As reviewers are often anonymous, often competing researchers and often exceptionally time-pressured, the mentioned consequences for not having a clear review process are not a fringe phenomenon. Additionally, the remarkable fact that reviewers are not obliged to disclose their conflicts of interest is a source of concern, especially in fields where industry and academics heavily overlap. With such a system, it should not come as a surprise if both the false positive rate, as well as the false negative rate, would be considerable, even though both are extremely difficult to assess in practice [Ioa05; SNS11].

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Finally, the hopes are that peer review could function as a first defence against serious offences like fraud, however: fabricated data, plagiarism, image manipulation, conflicts of interest, salami slicing, p-hacking, hypothesising after the results are known (HARKing), paper-mill products, reviewer fraud, unusual editorial processes and lost original data are often invisible to (other) reviewers working with a submitted manuscript and a deadline. Therefore, this turns out to be an insufficient guardrail in practice. The very fact that platforms for post-publication scrutiny, such as: Retraction Watch, PubPeer, Science Integrity Digest, Clear Skies, Elisabeth Bik's blog and the Black Spatula Project exist, is precisely because pre-publication review can demonstrably not carry the whole burden [Bik26a; Bik26b; Cle26; Pub26; Ret26; The26]. As such, it can be concluded that the general expectations for the review system are unrealistically high, while the current state of affairs indicate that the result are in fact not as intended.

### 6.5 Career advancement

As was alluded to above, researchers are rewarded for publication counts, citation counts, grants, prestige and managerial expansion more directly than for careful reasoning. Salami slicing, excessive self-citation, strategic authorship, rushed review, exaggerated claims and selective reporting can then quietly turn into professional habits. In order to be successful in every successive step, mostly 'important' publications literally count, and thus quantity can quietly devour quality. In fact, when read carefully, the analysis of Martinson, Anderson and de Vries seems to imply that the occurrence of scientifically questionable behaviour increases between successive stages of a scientific career [MAV05]. If this extrapolation turns out to be correct, the system as a whole can be considered to actively support gamesmanship, corner-cutting, not specifying hidden assumptions, hyperfocus and theoretical navel-gazing.

For researchers the sheer growth of the scientific literature makes careful participation even harder. Keeping up with one's own field can in itself already become a full-time job, and reviewing every relevant paper with equal care is downright impossible [BHM21; Lan16]. Reports of sharply rising reviewing loads and batch retractions are therefore not side details, but a clear indication of an overstretched system [Bri25]. Artificial intelligence may help with search, drafting and checking, but it can also increase output, intensify competition and create a loop in which machine-generated manuscripts meet machine-assisted review [Zha26]. The expected result for the coming years is not more serious science, but simply more similar throughput.

Contrarily, researchers at poorer institutions, or outside wealthy national systems, can be blocked from reading the literature to which they could meaningfully contribute. This directly adds a severe form of inequality to career advancement, and actively pushes such researchers towards lower-quality publication venues and fraudulent intermediaries [Ric+25; Spa18]. This means that, fundamentally, the researchers situated at better-off institutions have a significant advantage over their peers situated at poorer institutions in terms of: publication venues, citation counts, prestige, grants and, ultimately, career advancement. Furthermore, a public research system that hides publicly funded knowledge behind tolls weakens its own corrective capacity, and actively hampers

potentially fruitful collaborations.

When focusing on the scientific career path itself, one sees that, rather remarkably, the role of the researcher fundamentally changes whilst advancing in successive stages. Researchers are – as the word gives away – trained to do research, but as their career progresses, they are silently turning into managers of human resources, finance, public relations, grant strategy and institutional survival. Some researchers genuinely flourish in their new role, but it is worth to emphasise two things:

1. This career path is not efficient, something that might matter as often substantial public funding is involved;
2. There is often no good alternative for researchers that genuinely wish to specialise – and remain specialised – in doing research.

As there is a clear mismatch between training and occupation, and there is often no link between funding and managerial behaviour, unintended or unprofessional behaviour could result from this mismatch. The mismatch matters, because badly supported operations can directly result into gamesmanship, shortcutting, delegation without oversight and prestige management. As such, it should be concluded that if academia would like to improve the quality of its output, it should also reflect on the career path of its researchers.

## 6.6 Reform proposals

Science can be imagined in at least two noble ways: as the sincere search for truth, and as the organised attempt to solve important societal problems. As can be concluded from previous analysis, a system increasingly governed by key performance indicators should not automatically inherit the name ‘science’, as it does not align with these noble goals [Gel14; NSM12]. That is why current set-up fully deserves the name *funded research* instead. As it happens, some funded research is excellent science, however, most of it is merely funded research.

For funded research to fundamentally become *science* once more, several structural changes are highly recommended. It is not claimed that following recommendations are exhaustive, but they can be considered a fruitful starting point. The focus in the following paragraphs will be about funding, peer reviewing, publication, correction and task division inside research groups.

**Funding should become more pluralistic.** An argument in favour of current construction can be that it maintains a relative autonomy for researchers. Research autonomy greatly matters, but it should be emphasised that this autonomy does not move in neutral space. It is pulled by whoever has the clearest agenda and the strongest purse, which can, directly and indirectly, simply be the industry within a country [Agr+25b; SM16; Spa18]. It is therefore worthwhile, if science is to be directed towards solving important societal problems, to consider other sources of funding too:

- **Citizen-directed funding** can support local and neglected problems;
- **Democratic funding priorities**, that is: parties or democratically elected bodies that decide on funding priorities;

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- **Specialist peer funding** can protect difficult work whose value is not obvious at first sight.

Such a system where funding can be obtained in more than one significant way rewards more kinds of contribution than only publishable results. Education, outreach, public problem-selection, good reviewing, replication, negative findings, careful methods, useful tools and well-curated data all create scientific and societal value. Funding and assessment should therefore be allowed to count these objects directly rather than pretending that their value only exists once it has been converted into a publication [BE12; DOR12; Mun+17; Nos+15].

**Publication should be rebuilt around public access and public scrutiny.** As an argument in favour of open access, it can be noted that now often considerable amounts of public spending are involved in creating publications that are publicly closed. With this argument in mind, the Council of the European Union has called for transparent, equitable and open access to scholarly publications [Cou23]. Indeed, it is justified that if public money pays for research, the public should not pay again to read it. Diamond open-access infrastructure, such as the programme described by DIAMAS, points in the right direction [DIA26]. A digital, public publication infrastructure should also make the older ‘limited space’ argument obsolete. Such a system can host well-defined, well-executed and correct work even when it is not necessarily novel.

Even more importantly, as space limitations are less stringent in the digital age, a finished paper should not be the only recognised scientific object shared. Indeed, such a platform can also host the parts of research that are usually hidden: protocols, raw data, analysis plans, code, materials, failed attempts, reviewer reports and later corrections. Public platforms such as the Open Science Framework, Wellcome Open Research and Open Research Europe show how protocols, data, methods, tools and results can all be made part of the public record [Cen26a; Eur26; Wel26]. If any specific aspect of a study, like data or a model, cannot be made fully public for ethical or proprietary reasons, the reason should itself be explicit. The repeated excuse that original data were lost should not be treated as a harmless administrative accident; in several fraud cases, missing raw data were central to the difficulty of correction [BDK02; BE12; NSM12; The16]. Registered reports, open materials, open data, better research assessment and explicit rewards for replication would make correction an integral part of the ordinary scientific cycle, rather than an embarrassing afterthought [Cen26b; DOR12; Mun+17; Nat26; Nos+15; NSM12].

Naturally, the same infrastructure would be most beneficial if linking were easy. A replication should be (visibly) connected to the original study; extensions should be connected to methods and datasets they reuse; comments and post-publication discussions should remain attached to the relevant research object; and reliability signals should be able to change as evidence accumulates. Standard researcher identifiers, such as ORCID-style identifiers, would strengthen accountability by making authors, reviewers and conflicts of interest of either harder to obscure. Templates, documentation and research-design tools could then help researchers find relevant methods, datasets and recent methodological improvements before embarking on a new study. Such an open system also changes the meaning of failure: a study that looked promising but is weakened by a later analysis, a failed replication or a better design need

not disappear, but can remain in the record as a corrected step in a growing process. That is closer to how knowledge actually develops, and it lowers the burden on pre-publication review to get everything right at once. In this way, a false start does not remain there without correction, linkage or context [Nat26; NSM12].

**Review should become more open, more cumulative and more clearly credited.** Anonymous review may still be useful in some settings, for example in swift double-blind review that the original author can request, to quickly assess the quality of the to-be published object. However, after its initial release, the system should also support signed reviews, public post-publication discussion and links between replications and originals, so that usability or credibility of scientific work can change over time. Researchers should, as such, be able to build reputations as solid reviewers and contributors. This would make review and replication itself a recognised scholarly contribution rather than invisible service work. In such a system a direct impetus would appear for institutions to offer serious training to reviewers, as good reviewers would become reputable members of the scientific community. For this to work properly, it is essential that, next to authors, also reviewers would be subjected to declare their conflict-of-interest with regards to the objects they are reviewing. Such a review system would considerably lower the burden on pre-publication review, while increasing the appreciation of review as a serious scholarly contribution.

**Research groups should be organised around a research-focused and an operational part.** The former concerns questions, methods and intellectual direction; the latter concerns people, budgets, compliance, project support, data stewardship and institutional logistics. It would therefore be commendable if different people within a research group would be considered for these different tasks. The head of research would be responsible for the research-focused part, while the head of operations would be responsible for the operational part. With a clear separation in place, two interrelated consequences would be:

1. Researchers would be able to focus on their own research, something which they *are* properly trained for and, as such, this set-up would arguably be more efficient than current situation;
2. As considerable public spending is involved in general research projects, it is important to ensure that the selected solution is efficient and transparent.

As a result, researchers would not have to become operations managers, something that often takes away a lot of time and attention and they did *not* receive proper training for. As budgets within research institutions are often tight, one could even imagine a set-up where the head of operations is shared between different groups or across an institute. Naturally, as a new role is defined, this could be a role that one specialises in during their tenure, but it should really be considered as such: a different specialisation, that requires a different set of skills and knowledge.

**It should be enforced by law that public research institutes make their income and expenses publicly inspectable.** As public research is publicly funded, serious financial interests, contribution claims and institutional

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dependencies should be visible enough that failure to disclose them becomes practically and legally meaningful rather than a polite administrative courtesy. There are several fraud cases that could have been revealed much earlier if the conflict of interest were immediately disclosed, and even in these cases the conflict of interest itself was often a mere afterthought rather than the main reason for the fraud [GSM11; OGr24; OM23; The24]. It should always be clearly remembered that these institutes are *public* and, as such, hold a serious obligation towards society to generate objects that serve the general interest. Sponsorship need not automatically imply a direct mingling in the end-results, however they increase the odds for serious interference. Cases in the past where researchers accepted funding to steer the scientific debate around, e.g.: the active support around harmful treatments or products, the suppression of helpful treatments, climate change causes or effects, or the effects of social media on psychological well-being, are pointers to the fact that researchers are mere human beings that can be influenced by power or money [Gol12; OC10; PM18; Zub19]. Rather than blindly trust institutions to self-correct when such processes take place, it is wiser and more efficient to be ahead of it.

**Synthesis.** Above proposals create an environment where more sources for funding will be made available to a researcher. Furthermore, the online environment where the research would be developed in, would actively stimulate collaboration and sharing of a much broader set of scientific objects. In this system, rather than all developing in prolific producers of publishable papers, researchers are allowed to specialise in different roles: Enthusiastic teachers, thorough reviewers, precise replicators, exceptional data-collectors, deep methodologists, gifted public speakers, creative out-of-the-box thinkers, and others. As research groups would have at least one dedicated employee focusing on operations and funding, the researcher can indeed focus on those aspects of science that the community and society needs most. By ensuring that any form of interference can be readily detected, society also ensures that the interests of the researchers stays aligned with them. As a result, it can be expected that the achieved results can be faithfully judged on their current reliability or truthfulness.

Circling back on the habits that Jaynes mentioned as bad scientific behaviours, the new system is expected to reduce these greatly:

1. **Gamesmanship:** would be counteracted by an incentive to increase linkage to the rest of the scientific web;
2. **Cutting corners:** would be placed under much heavier scrutiny by reviewers and replication studies;
3. **Hidden assumptions:** would be placed under much heavier scrutiny by reviewers and replication studies;
4. **Hyperfocus:** would be counteracted by an incentive to increase linkage to the rest of the scientific web;
5. **Theoretical navel-gazing:** would be discouraged in citizen- or democratically-directed projects.

By unlocking a wider view and a more society-driven approach among researchers, more society-focused results can be expected. It is likely that

as a direct result the mutual trust between researchers and society will increase, to the benefit of all. Whatever one's definition of science, an activity that serves both the pursuit of truthfulness and society at large, is entitled to receive this qualification. As such, above system would rightfully deserve the name *science* once again.

Of course it cannot be expected that above system eradicates fraud, vanity or ignorance. However, by changing what the system rewards, it makes replication, correction, negative results, careful review, public access, transparent data practices, operational competence and public problem-selection part of the normal scientific economy. It also reduces the value of a single spectacular publication by increasing the value of the surrounding process that checks, links and corrects it. That is the institutional version of the formal demand made throughout this thesis: if scientific bad habits are the enemy of good inference, then scientific institutions should not be stimulated to develop them.



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## Acknowledgements

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*“Teleutias truly sailed home under the happiest of circumstances. When he went down to the sea to begin his journey home, there was not a soldier who did not clasp his hand. One placed a wreath on his head, another put a fillet around him, and those who came too late, when he was already putting out to sea, still threw wreaths into the water and wished him every good thing.*

*I realize that I am not describing any great expenditure, any danger, or any strategy worth mentioning; but by Zeus, I do think it is important for a man to consider what kind of conduct enabled Teleutias to evoke such a response from his subordinates, for that is, beyond any amount of money or danger, a man’s most noteworthy achievement”*

Xenophon in *Hellenica*



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## **Appendices**

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## APPENDIX A

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### Definitions

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*“So if you are an expert and, as I said just now, have deceived me with your promise to give a demonstration about Homer, you are playing a trick on me. But if you are not an expert, but are possessed by Homer through a divine gift and, without knowledge, say many beautiful things about the poet, as I claimed is the case with you, then you are not playing a trick on me. So you need only choose how you wish us to regard you: as a man who plays tricks, or as a man governed by the gods”*

Socrates in *Ion*



## A.1 Type-theoretic definitions

The definitions in this appendix fix the fragment used in the main text. They are intentionally finite and judgemental. The type-theoretic notation follows the Martin-Löf tradition and the presentations used by Nordström–Petersson–Smith and Gratzner [Gra23; Mar84; NPS90].

**Definition A.1.1** (Type-theoretic judgements). The basic judgemental forms used here are

$$\Gamma \text{ ctx}, \quad \Gamma \vdash A \text{ type}, \quad \Gamma \vdash a : A, \quad \Gamma \vdash \delta : \Delta,$$

together with definitional equality judgements

$$\Gamma \vdash A \equiv B \text{ type}, \quad \Gamma \vdash a \equiv b : A, \quad \Gamma \vdash \delta \equiv \epsilon : \Delta.$$

These are read as follows:  $\Gamma$  is a well-formed context;  $A$  is a type in context  $\Gamma$ , that is, a domain of possible constructions available in that context;  $a$  is a term of type  $A$  in context  $\Gamma$  (under Definition A.1.17, evidence for the proposition represented by  $A$ ); and  $\delta$  is a substitution from  $\Gamma$  to  $\Delta$ . The symbol  $\Gamma$  is always a context;  $\delta$  and  $\epsilon$  are substitutions.

**Definition A.1.2** (Context). A context is a finite list of typed variable declarations

$$\Gamma := x_1 : A_1, x_2 : A_2(x_1), \dots, x_n : A_n(x_1, \dots, x_{n-1}),$$

in which each type  $A_{i+1}$  may depend on the earlier variables  $x_1, \dots, x_i$ . Such a dependently-typed sequence is called a *telescope* in the de Bruijn–Jacobs tradition. Contexts are generated from the empty context by context extension:

$$\frac{\Gamma \text{ ctx} \quad \Gamma \vdash A \text{ type}}{\Gamma, x : A \text{ ctx}}.$$

Later declarations may depend on earlier declarations; earlier declarations may not depend on later ones.

**Definition A.1.3** (Structural use of contexts). In the fragment used here, weakening, exchange and forgetting are admissible only when the dependency order permits them. Thus a fresh declaration  $x : A$  may be added to a context when  $\Gamma \vdash A \text{ type}$ ; two declarations may be exchanged only if neither dependency is broken; and a declaration  $x : A$  may be forgotten from  $\Gamma, x : A, \Delta$  only when neither  $\Delta$  nor the target judgement contains a free occurrence of  $x$ .

**Definition A.1.4** (Substitution). If  $\Gamma, x : A, \Delta \vdash J$  is any judgement and  $\Gamma \vdash a : A$ , substitution replaces  $x$  by  $a$ :

$$\Gamma, \Delta[a/x] \vdash J[a/x].$$

In explicit-substitution presentations this is internalised as a judgement  $\Gamma \vdash \delta : \Delta$  and an operation sending  $\Delta \vdash b : B$  to  $\Gamma \vdash b[\delta] : B[\delta]$ .

**Definition A.1.5** (Definitional equality). Definitional equality, written  $\equiv$ , is equality by computation, unfolding or stipulated judgemental equation. It is part of the judgemental structure of the theory, not a proposition internal to a type.

## A. Definitions

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**Definition A.1.6** (Formation, introduction, elimination and computation). The rules for a type former are organised as follows: formation rules state when the type is well formed; introduction rules construct canonical inhabitants; elimination rules specify how inhabitants may be used; and computation rules state how eliminations applied to introductions reduce by definitional equality.

*Remark A.1.7.* This meta-schema is used as a structuring device for presenting individual type formers below; no proof in this thesis quantifies over rule-schemas as objects.

**Definition A.1.8** (Universe). A universe level  $\mathcal{U}_i$  is a type of codes for small types. In the Coquand-style notation used by Gratzer, the relevant operations are

$$\Gamma \vdash M : \mathcal{U}_i \quad \Rightarrow \quad \Gamma \vdash \text{El}(M) \text{ type}_i,$$

and

$$\Gamma \vdash A \text{ type}_i \quad \Rightarrow \quad \Gamma \vdash \text{Code}(A) : \mathcal{U}_i,$$

with definitional equations  $\text{El}(\text{Code}(A)) \equiv A$  and  $\text{Code}(\text{El}(M)) \equiv M$ , within the chosen universe level.

**Definition A.1.9** (Dependent function type). A dependent type over  $A$  in context  $\Gamma$  is a family specified by a judgement  $\Gamma, x : A \vdash B(x) \text{ type}$ ; it assigns a type  $B(a)$  to each term  $\Gamma \vdash a : A$  by substitution. Given such an  $A$  and  $B$ , the dependent function type is formed by

$$\Gamma \vdash \Pi_{x:A} B(x) \text{ type}.$$

Its introduction, elimination and computation rules are

$$\frac{\Gamma, x : A \vdash b(x) : B(x)}{\Gamma \vdash \lambda x. b(x) : \Pi_{x:A} B(x)}, \quad \frac{\Gamma \vdash f : \Pi_{x:A} B(x) \quad \Gamma \vdash a : A}{\Gamma \vdash f(a) : B(a)},$$

and  $(\lambda x. b(x))(a) \equiv b(a)$ . When  $B$  does not depend on  $x$ , this is the ordinary non-dependent function type  $\Gamma \vdash A \rightarrow B \text{ type}$ , with the same introduction, elimination and computation rules and the usual beta reduction  $(\lambda x. b)(a) \equiv b[a/x]$ .

**Definition A.1.10** (Dependent pair type). Given  $\Gamma \vdash A \text{ type}$  and a dependent type  $\Gamma, x : A \vdash B(x) \text{ type}$ , the dependent pair type is formed by

$$\Gamma \vdash \Sigma_{x:A} B(x) \text{ type}.$$

Its canonical inhabitants are pairs  $(a, b)$  with  $\Gamma \vdash a : A$  and  $\Gamma \vdash b : B(a)$ . The projections satisfy

$$\text{pr}_1(a, b) \equiv a, \quad \text{pr}_2(a, b) \equiv b.$$

When  $B$  does not depend on  $x$ , this is the ordinary non-dependent product type  $\Gamma \vdash A \times B \text{ type}$ , with pair inhabitants  $(a, b)$  and the same projections.

**Definition A.1.11** (Sum type). If  $\Gamma \vdash A \text{ type}$  and  $\Gamma \vdash B \text{ type}$ , then  $\Gamma \vdash A+B \text{ type}$ . Its introductions are  $\text{inl}(a) : A+B$  for  $a : A$  and  $\text{inr}(b) : A+B$  for  $b : B$ . To define a term out of  $A+B$ , one must give a case for the left injection and a case for the right injection.

**Definition A.1.12** (Identity type). If  $\Gamma \vdash A$  type and  $\Gamma \vdash a, b : A$ , then

$$\Gamma \vdash \text{Id}_A(a, b) \text{ type.}$$

Its canonical introduction is reflexivity:

$$\frac{\Gamma \vdash a : A}{\Gamma \vdash \text{refl}_a : \text{Id}_A(a, a)}.$$

In intensional type theory, an inhabitant of  $\text{Id}_A(a, b)$  is evidence of equality inside the theory; it is not automatically the same as external or definitional equality.

**Definition A.1.13** (Inductive type). An inductive type is specified by constructors and by an eliminator. If  $I$  is generated by constructors  $c_k : C_k \rightarrow I$ , then to define a dependent function

$$f : \prod_{x:I} P(x)$$

it suffices to give the corresponding constructor cases demanded by the eliminator for  $I$ . Natural numbers, lists and finite types are standard examples. The natural-number type  $\mathbb{N}$  is the inductive type generated by  $0 : \mathbb{N}$  and  $\text{succ} : \mathbb{N} \rightarrow \mathbb{N}$ . In particular, the type  $\text{List}(A)$  of finite lists over a type  $A$  is the inductive type with constructors  $\text{nil} : \text{List}(A)$  and  $\text{cons} : A \rightarrow \text{List}(A) \rightarrow \text{List}(A)$ ; its eliminator yields ordinary list recursion. This is the `List` used in the weighted finite-support form of Definition A.3.5.

**Definition A.1.14** (Finite type). For  $n : \mathbb{N}$ , the finite type  $\text{Fin}(n)$  has exactly the canonical inhabitants  $0, \dots, n-1$ . Its eliminator says that to define  $f : \prod_{i:\text{Fin}(n)} P(i)$ , one gives terms  $p_i : P(i)$  for each  $i < n$ .

**Definition A.1.15** (Empty type). The empty type  $\perp$  is the inductive type (Definition A.1.13) with no constructors; equivalently,  $\perp \equiv \text{Fin}(0)$ . Its eliminator says that to define

$$\text{absurd}_P : \perp \rightarrow P$$

for any type  $P$ , no constructor cases are needed: the empty type has no canonical inhabitants to dispatch on.

**Definition A.1.16** (Boolean type). The Boolean type `Bool` is the primitive inductive type with two constructors `true, false : Bool`. Its eliminator gives Boolean case analysis: to define  $f : \text{Bool} \rightarrow P$ , one provides values  $p_{\text{true}}, p_{\text{false}} : P$  with computation rules  $f(\text{true}) \equiv p_{\text{true}}$  and  $f(\text{false}) \equiv p_{\text{false}}$ . Equality on `Bool` is decidable, and `true`  $\neq$  `false`.

**Definition A.1.17** (Propositions as types). The propositions-as-types reading identifies a proposition with a type and a proof with a term of that type:

$$A \text{ is proved in } \Gamma \iff \Gamma \vdash a : A \text{ for some term } a.$$

*Remark A.1.18.* This reading is used informally throughout the thesis but is not invoked as a rule inside the formal transfer of Theorem B.6.13; it serves as a background interpretation of the constructive judgemental discipline.

**Definition A.1.19** (Negation). Negation is the type-theoretic function into the empty type:

$$\neg A := A \rightarrow \perp.$$

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From  $a : A$  and  $n : \neg A$  one obtains  $n(a) : \perp$ . Double negation  $A \rightarrow \neg\neg A$  is always derivable by  $\lambda a. \lambda n. n(a)$ ; the converse  $\neg\neg A \rightarrow A$  is a classical principle and is not available uniformly (cf. Proposition B.3.1).

**Definition A.1.20** (Logical connectives under propositions-as-types). Under Definition A.1.17, the standard logical connectives correspond to type formers:

$$\begin{aligned} A \wedge B &:= A \times B, & A \vee B &:= A + B, \\ A \Rightarrow B &:= A \rightarrow B, & \neg A &:= A \rightarrow \perp, \\ \top &:= \text{Fin}(1). \end{aligned}$$

The constructive content of a logical statement is then a term of the corresponding type. In particular,  $A \wedge B$  requires both a witness of  $A$  and a witness of  $B$ ;  $A \vee B$  requires a witness of one side together with evidence of which side; and  $\neg A$  requires a procedure converting any witness of  $A$  into an inhabitant of the empty type.

**Definition A.1.21** (Decidable type and decidable predicate). A type  $A$  is *decidable* when the sum type  $A + \neg A$  is inhabited, using Definitions A.1.11 and A.1.19; a term  $d : A + \neg A$  is a *decision procedure* for  $A$ , returning either a witness of  $A$  or a refutation of  $A$ . A predicate  $P : X \rightarrow \mathcal{U}$  is *decidable* when  $P(x)$  is decidable for every  $x : X$ , equivalently when there is a function  $X \rightarrow \text{Bool}$  in the sense of Definition A.1.16 together with a soundness proof. A type  $A$  has *decidable equality* when the predicate  $\text{Id}_A(a, b)$  of Definition A.1.12 is decidable for every  $a, b : A$ . Finite types (Definition A.1.14) and the Boolean type have decidable equality by their eliminators, and finite sums and finite products of types with decidable equality have decidable equality.

*Remark A.1.22* (Decidability conventions). Throughout this thesis, “decidable event”, “decidable predicate” and “decidable equality” are used in the sense of Definition A.1.21. The three role-tagged finite-type definitions used later — Definition A.3.1 for sample spaces, Definition A.3.8 for variable values, and Definition A.1.14 for the inductive families  $\text{Fin}(n)$  — are all finite types with decidable equality; the distinct names mark distinct roles, not distinct constructions. “Decidable event” means the predicate form  $E : \Omega \rightarrow \text{Bool}$  of Definition A.3.1, equivalently a finite subset of  $\Omega$ .

**Definition A.1.23** (Indicator). For a decidable predicate  $E : X \rightarrow \text{Bool}$ , the indicator function  $\mathbb{1}_E : X \rightarrow \mathbb{N}$  is defined by Boolean case analysis:  $\mathbb{1}_E(x) := 1$  when  $E(x) = \text{true}$  and  $\mathbb{1}_E(x) := 0$  when  $E(x) = \text{false}$ .

## A.2 Modal definitions

The modal notation follows Gratzer’s multimodal dependent type theory: a mode theory is a small strict 2-category; objects are modes; morphisms are modalities; and 2-cells record transformations between modalities [Gra+21; Gra23; GSB19]. I use this notation locally; it is meant to track the relevant fragment, not to reproduce every rule of Gratzer’s full systems.

**Definition A.2.1** (Mode theory). A mode theory is a small strict 2-category  $\mathcal{M}$ . Its objects  $m, n, o \in \mathcal{M}$  are modes, its morphisms  $\mu : n \rightarrow m$  are modalities, and its 2-cells  $\alpha : \mu \Rightarrow \nu$  are transformations between modalities. No rule in

this thesis exercises 2-cells; they are declared here to match the mode-theory definition of Gratzer and collaborators [Gra+21; Gra23].

**Definition A.2.2** (Mode). A mode is an object  $m$  of a mode theory  $\mathcal{M}$ . In this thesis, modes are interpreted as epistemic states: structured positions from which certain variables, assumptions, questions and causal models are available.

**Definition A.2.3** (Modal judgement). For each mode  $m$ , judgements are indexed by that mode:

$$\Gamma \text{ ctx } @m, \quad \Gamma \vdash A \text{ type } @m, \quad \Gamma \vdash a : A @m, \quad \Gamma \vdash \delta : \Delta @m.$$

The annotation  $@m$  belongs to the judgement, not to the type expression  $A$ . Thus a judgement is not made in an unlocated universe of discourse; it is made at a mode. The full modal-substitution discipline of Gratzer's multimodal type theory [Gra+21; Gra23] is not reproduced here. For the purposes of this thesis, substitutions are used at a single mode at a time; inter-mode movement is mediated by Definition A.2.5 and Definition A.2.6 rather than by inter-mode substitutions.

**Definition A.2.4** (Modality). A modality is a morphism  $\mu : n \rightarrow m$  in a mode theory. The displayed arrow direction follows the Gratzer-style lock notation of Definition A.2.5: data judged at mode  $n$  may be marked by  $\mu$  and used from mode  $m$ . In the intended interpretation, a modality is an admissible transition between epistemic states: learning, forgetting, changing attention, hypothesising or intervening.

**Definition A.2.5** (Locked context). Let  $\mu : n \rightarrow m$  be a modality in a mode theory  $\mathcal{M}$ . The locked context is the context extension

$$\frac{\Gamma \text{ ctx } @m \quad \mu : n \rightarrow m}{\Gamma.\{\mu\} \text{ ctx } @n}.$$

The lock entry  $\{\mu\}$  is a context entry, not a postfix operator on contexts; it records that subsequent declarations are read at mode  $n$  rather than  $m$ . Composition of locks follows the modality 2-category: for  $\mu : n \rightarrow m$  and  $\nu : o \rightarrow n$ ,

$$\Gamma.\{\mu\}.\{\nu\} \equiv \Gamma.\{\mu \circ \nu\}, \quad \Gamma.\{1_m\} \equiv \Gamma,$$

where  $\mu \circ \nu : o \rightarrow m$  is the composite modality and  $1_m$  is the identity modality on  $m$  [Gra+21; Gra23]. An ordinary context extension under a lock takes the form  $\Gamma.\{\mu\}, x : A$ , adding the hypothesis  $x : A$  in the locked context at mode  $n$ ; no separate "modal-hypothesis" entry is needed.

**Definition A.2.6** (Modal type former). Let  $\mu : n \rightarrow m$  be a modality. Given a context  $\Gamma \text{ ctx } @m$ , form the locked context  $\Gamma.\{\mu\} \text{ ctx } @n$  of Definition A.2.5. If  $\Gamma.\{\mu\} \vdash A \text{ type } @n$ , then the modal type former gives

$$\Gamma \vdash \langle \mu \mid A \rangle \text{ type } @m.$$

If  $\Gamma.\{\mu\} \vdash a : A @n$ , then the modal introduction form gives

$$\Gamma \vdash \text{mod}_\mu(a) : \langle \mu \mid A \rangle @m.$$

This is the fragment needed here: a modality lets a type or term from mode  $n$  appear, under modal marking, at mode  $m$ .

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*Remark A.2.7* (Modal fragment is introduction-only). Definition A.2.6 states only formation and introduction for  $\langle \mu \mid A \rangle$ . The corresponding modal elimination rule (the **letmod** or modal pattern-matching rule) of multimodal type theory [Gra+21; Gra23], the action of 2-cells  $\alpha : \mu \Rightarrow \nu$  on modal terms, and the full inter-mode substitution discipline are deliberately not reproduced. No definition or proof in this thesis destructs a  $\text{mod}_\mu(a)$ , applies a 2-cell to a term, or transports a substitution across modes. The modal vocabulary used here is therefore the introduction-only fragment of multimodal type theory: enough to record that data lives at a mode and that an admissible modality lets it surface, under modal marking, at another mode. Results that genuinely depend on modal elimination, on 2-cell action or on inter-mode substitution lie outside the scope of this fragment and are deferred to the cited works.

*Remark A.2.8* (Epistemic modalities mark record transitions). The intended epistemic transition names — learning, forgetting, conditioning, relating, unrelating, setting and unsetting — are not context-dependent modal operators. In a strict Gratzner-style reading, the mode theory first supplies modalities  $\mu : n \rightarrow m$  between already specified modes. The finite context or structural-causal-model records at those endpoint modes may then differ: a fresh unused variable may have been added, an unused variable may have been forgotten, the exogenous distribution may have been conditioned on evidence, a structural equation may have been replaced by a constant for an intervention, or a relation may have been inserted or removed. Those changes are ordinary record or context transformations attached to the endpoint modes; the modality itself only witnesses that the transition is admissible in the mode theory.

Conditioning is deliberately separate from setting. Setting is the Pearl-style action step: it changes an endogenous mechanism and keeps the exogenous uncertainty fixed. Conditioning is the abduction or evidence-update step: it replaces the exogenous distribution by a posterior distribution and keeps the structural equations fixed. If the evidence determines the exogenous state, the posterior may be a point mass, but this is still conditioning rather than an intervention on an exogenous structural equation. Keeping the two labels separate prevents the formal layer from conflating seeing with doing.

This separation is important. If the modality inspected the current context and performed the transformation by itself, the system would move towards context-dependent modalities, which are substantially stronger than the fragment used here and would require new metatheory. In the present fragment, forward-only epistemic change is harmless: one declares two endpoint modes and a modality between them. Inverse-looking operations, such as adding a fresh variable and then forgetting exactly that unused variable, are not automatic. They require a stated equation in the mode theory or a propositional theorem saying that the composite record transformation returns a context equivalent to the starting one.

*Remark A.2.9* (Narrative-only terms: subjectivity, interest, epistemic position). Several umbrella terms in the main narrative — *subjectivity*, *interest*, *epistemic position* — are not given formal definitions of their own. They are deliberately narrative-only: they name the conceptual target that the modal apparatus is built to serve, not separate objects of the calculus. Inside this fragment they are captured informally by the structural shape of the current judgement: *subjectivity* is captured by the dependence of every judgement on a mode and

a context (Definitions A.1.2 and A.2.3); *epistemic position* is captured by the current mode together with the current context (Definitions A.1.2 and A.2.2); and *interest* is captured by which variables, locks and modalities are present in  $\Gamma$  at the current mode — a question only enters scope when its variables are declared, an interest is forgotten when the corresponding declaration is removed, and a change of interest is a context manipulation or a modal transition. No proof in this thesis quantifies over *subjectivity*, *interest* or *epistemic position* as objects; they are names for the discipline that contexts, modes and modalities together impose.

### A.3 Probabilistic and causal definitions

Three notation conventions are used throughout this section and the proofs that depend on it.

**Independence.** The symbol  $\perp\!\!\!\perp$  has two readings. Probabilistic independence is always subscripted by the probability assignment, for example  $A \perp\!\!\!\perp_{\mu} B$ ,  $\mathcal{X} \perp\!\!\!\perp_{\mu} \mathcal{Y} \mid \mathcal{Z}$  or  $\perp\!\!\!\perp_{P_{\mathcal{M}}}$ . Graphical d-separation always carries a graph subscript on the surrounding expression, for example  $(X \perp\!\!\!\perp Y \mid Z)_{\mathcal{G}}$ . The full convention is fixed in Definition A.3.10.

**Decidable events.** Decidable events on a finite sample type  $\Omega$  are predicates  $E : \Omega \rightarrow \text{Bool}$ , in the sense of Definitions A.1.16, A.1.21 and A.3.1. The notation  $E \subseteq \Omega$  is used interchangeably with the predicate notation, with  $\cap$ ,  $\cup$ ,  $\setminus$  and complement read pointwise on Booleans.

**Intervention.** The expression  $\text{do}(X = x)$  always denotes Pearl’s intervention operator (Definition A.3.28). Its model-theoretic content is the intervened structural causal model  $\mathcal{M}_x$  of Definition A.3.26; its graph-level content is the mutilated graph  $\mathcal{G}_{\overline{X}}$  of Definition A.3.27.

**Definition A.3.1** (Finite event type). A finite event type is a finite type  $\Omega$  with decidable equality. An event is a decidable predicate  $E : \Omega \rightarrow \text{Bool}$  in the sense of Definition A.1.16, equivalently a finite subset  $E \subseteq \Omega$ .

**Definition A.3.2** (Resolution status). Let  $A(s)$  be a proposition or question available in scenario  $s : S$ , where  $S$  is a finite event type of *scenarios* available to a finite inspection procedure. The word “scenario” is reserved here for the input type of the inspection procedure, in order to avoid confusion with Definition A.2.2, where the phrase “epistemic state” refers to a mode. The resolution-status type for  $A$  is the three-constructor finite type

$\text{Status}_A$  generated by `proved`, `refuted`, `unresolved`.

A finite inspection procedure for  $A$  is a total finite procedure

$$\iota_A : S \rightarrow \text{Status}_A.$$

To read the tags as genuine proof status, the procedure must also be sound:

$$\text{sound}_{\text{proved}} : \Pi_{s:S} (\iota_A(s) = \text{proved}) \rightarrow A(s),$$

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and

$$\text{sound}_{\text{refuted}} : \Pi_{s:S} (\iota_A(s) = \text{refuted}) \rightarrow \neg A(s),$$

where  $\neg A(s) := A(s) \rightarrow \perp$  as in Definition A.1.19. The value `unresolved` means only that neither kind of witness is found by the chosen finite procedure. It does not mean that no proof or refutation exists in principle.

**Definition A.3.3** (Resolution-status events). Suppose  $S$  is a finite event type of scenarios in the sense of Definition A.3.2, and  $\iota_A : S \rightarrow \text{Status}_A$  is a finite inspection procedure. The three resolution-status events associated with  $A$  are

$$\text{Prov}_A(s) := \begin{cases} \text{true}, & \iota_A(s) = \text{proved}, \\ \text{false}, & \text{otherwise}, \end{cases} \quad \text{Ref}_A(s) := \begin{cases} \text{true}, & \iota_A(s) = \text{refuted}, \\ \text{false}, & \text{otherwise}, \end{cases}$$

and

$$\text{Unr}_A(s) := \begin{cases} \text{true}, & \iota_A(s) = \text{unresolved}, \\ \text{false}, & \text{otherwise}. \end{cases}$$

Formally these are Boolean predicates  $S \rightarrow \text{Bool}$  obtained by case analysis on the finite resolution-status type, with equality tests justified by decidable equality of  $\text{Status}_A$ . They form a partition of  $S$ : every  $s : S$  lies in exactly one of them.

**Definition A.3.4** (Urn-generated probability type). Let  $X$  be any type. An urn-generated finite probability assignment into  $X$  is

$$\text{Prob}_{\mathbb{Q}}^{\text{urn}}(X) := \Sigma_{N:\mathbb{N}} \Sigma_{h:N>0} (\text{Fin}(N) \rightarrow X).$$

If  $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$  and  $E : X \rightarrow \text{Bool}$  is a decidable predicate, then

$$P_{\mu}(E) := \frac{1}{N} \sum_{i:\text{Fin}(N)} \mathbf{1}_{E(f(i))},$$

using the indicator of Definition A.1.23. The finite type  $\text{Fin}(N)$  is the urn of  $N$  equipossible cells. The map  $f$  pushes that urn forward to  $X$ . No finiteness of  $X$  is required for this definition: the finiteness lies in the source urn. Thus  $\mu$  is a finite-supported assignment into  $X$ , and only explicitly decidable predicates on  $X$  are evaluated.

**Definition A.3.5** (Finite probability type). Let  $\Omega$  be a finite event type. In this thesis the coefficient field is fixed to the rational numbers  $\mathbb{Q}$ ;  $\mathbb{Q}_{\geq 0}$  is the subtype of rational numbers equipped with a proof of non-negativity. Finite sums over a finite event type are computed by the eliminator or by a chosen finite enumeration, and are invariant under finite reindexing. The record form of a probability type over  $\Omega$  is

$$\text{Prob}_{\mathbb{Q}}(\Omega) := \Sigma_{p:\Omega \rightarrow \mathbb{Q}_{\geq 0}} \left( \sum_{\omega:\Omega} p(\omega) = 1 \right).$$

If  $\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$  and  $E \subseteq \Omega$ , then

$$P_{\mu}(E) := \sum_{\omega:\Omega} \mathbf{1}_E(\omega) p(\omega),$$

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using the indicator of Definition A.1.23. If  $E \cap F = \emptyset$ , then  $P_\mu(E \cup F) = P_\mu(E) + P_\mu(F)$  by finite summation. When only one probability assignment is in scope, the subscript is sometimes dropped:  $P(E) \equiv P_\mu(E)$ .

After a common denominator has been chosen, the same finite rational record can also be written in weighted finite-support form:

$$\begin{aligned} R &= (a, D, h, d), \\ a &: \text{List}(\Omega \times \mathbb{N}), \\ D &: \mathbb{N}, \\ h &: \sum_{(\omega, m) \in a} m = D, \\ d &: D > 0. \end{aligned}$$

Its value on a decidable event  $E : \Omega \rightarrow \text{Bool}$  is

$$P_R(E) := \frac{\sum_{(\omega, m) \in a, E(\omega) = \text{true}} m}{D}.$$

This is the common-denominator form formalised in Lean.

When the codomain is itself a finite event type  $\Omega$ , every urn-generated probability assignment  $\nu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(\Omega)$  induces such a record by

$$p_\nu(\omega) := \frac{\#\{i : \text{Fin}(N) \mid f(i) = \omega\}}{N}.$$

Equivalently, in weighted finite-support form,  $\nu$  is represented by the list of its urn cells with weight 1. Conversely, every weighted common-denominator record can be represented by an urn by replacing each pair  $(\omega, m)$  by  $m$  urn cells labelled  $\omega$ . Thus the urn and common-denominator record presentations are extensionally equivalent: they assign the same value to every decidable event. For ordinary finite rational records on finite event types, this applies after multiplying by a common denominator. This equivalence is not asserted for arbitrary rational-valued functions on arbitrary codomains  $X$  in Definition A.3.4; it uses finite support, and the usual record form over  $\Omega$  additionally uses the finite enumeration of  $\Omega$ .

**Definition A.3.6** (Constructive Clayton–Waddington urn assumptions). For  $N > 0$  and  $0 \leq k \leq N$ , let  $U_N(k)$  denote the plausibility assigned to any  $k$ -cell decidable event in a symmetric  $N$ -cell urn. The constructive Clayton–Waddington assumptions used here are:

A.3.6.1. finite urn existence: for every positive  $N$ , the symmetric urn  $\text{Fin}(N)$  is available;

A.3.6.2. qualitative coherence: equality of plausibility is an equivalence relation, strict plausibility comparison respects this equivalence, and equivalent plausibilities are not strictly ordered;

A.3.6.3. logical equivalence: equivalent decidable events receive the same plausibility;

A.3.6.4. relabelling invariance: permuting  $\text{Fin}(N)$  does not change plausibility;

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A.3.6.5. strict monotonicity: if  $k < \ell \leq N$ , then  $U_N(k)$  is strictly less plausible than  $U_N(\ell)$ ;

A.3.6.6. refinement invariance: replacing every cell by  $c > 0$  indistinguishable subcells preserves plausibility, so  $U_N(k) = U_{cN}(ck)$ .

These assumptions concern finite decidable urn events. They do not include a complement rule for arbitrary propositions. Product urn compatibility is used only after this qualitative rescaling step: if an  $M$ -cell urn and an  $N$ -cell urn are combined, their rectangular product is an  $MN$ -cell urn.

**Definition A.3.7** (Conditional probability and Bayesian update). Let  $\mu : \text{Prob}_{\mathbb{Q}}(\Omega)$  and let  $A, B \subseteq \Omega$ . If  $P_{\mu}(B) > 0$ , then

$$P_{\mu}(A \mid B) := \frac{P_{\mu}(A \cap B)}{P_{\mu}(B)}.$$

Bayesian update is the same operation read as a transformation on probability assignments: given  $\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$  and an event  $E \subseteq \Omega$  with  $P_{\mu}(E) > 0$ , Bayesian update produces  $\mu_E : \text{Prob}_{\mathbb{Q}}(\Omega)$  with mass function

$$p_E(\omega) := \frac{\mathbb{1}_E(\omega)p(\omega)}{P_{\mu}(E)}.$$

Both conditional probability and Bayesian update are partial: they are defined only when the conditioning event has positive probability under  $\mu$ . Bayes' theorem is not part of the definition; it is the finite-event identity proved in Theorem B.4.10.

**Definition A.3.8** (Value type). A value type for a variable  $X$  is a finite type  $\text{Val}(X)$  whose inhabitants are the possible values of  $X$ . If  $x : \text{Val}(X)$ , then  $x$  is an admissible value assignment for  $X$ .

**Definition A.3.9** (Finite variable). A finite variable is a pair consisting of a variable name  $X$  and a value type  $\text{Val}(X)$ . A finite family of variables  $\mathcal{W}$  means a finite index type together with one finite variable at each index. For such a family, write

$$\text{Val}(\mathcal{W}) := \prod_{W \in \mathcal{W}} \text{Val}(W)$$

for the type of joint assignments to all variables in  $\mathcal{W}$ . A joint event on  $\mathcal{W}$  is a decidable predicate on  $\text{Val}(\mathcal{W})$ .

**Definition A.3.10** (Independence and conditional independence of decidable events). Let  $\mu : \text{Prob}_{\mathbb{Q}}(\Omega)$  and let  $A, B, C \subseteq \Omega$  be decidable events.

*Unconditional independence.* The events  $A$  and  $B$  are *independent under  $\mu$* , written  $A \perp_{\mu} B$ , when

$$P_{\mu}(A \cap B) = P_{\mu}(A) \cdot P_{\mu}(B).$$

Equivalently, when  $P_{\mu}(B) > 0$ ,  $P_{\mu}(A \mid B) = P_{\mu}(A)$ ; and when  $P_{\mu}(A) > 0$ ,  $P_{\mu}(B \mid A) = P_{\mu}(B)$ .

*Conditional independence.* The events  $A$  and  $B$  are *conditionally independent given  $C$  under  $\mu$* , written  $A \perp_{\mu} B \mid C$ , when

$$P_{\mu}(A \cap B \cap C) \cdot P_{\mu}(C) = P_{\mu}(A \cap C) \cdot P_{\mu}(B \cap C).$$

Equivalently, when  $P_\mu(C) > 0$ ,  $P_\mu(A \cap B \mid C) = P_\mu(A \mid C) \cdot P_\mu(B \mid C)$ ; and when additionally  $P_\mu(B \cap C) > 0$ ,  $P_\mu(A \mid B \cap C) = P_\mu(A \mid C)$ . The product form is the primary definition because it does not require a positivity side condition; the conditional forms are recorded as equivalent characterisations when the relevant denominators are realised.

*Variable families.* Suppose  $\Omega = \text{Val}(\mathcal{V})$  for a finite family of variables  $\mathcal{V}$ , and  $\mathcal{X}, \mathcal{Y}, \mathcal{Z} \subseteq \mathcal{V}$ . A joint event on  $\mathcal{X}$  is read under  $\mu$  as the corresponding cylinder event on  $\Omega$ , obtained by precomposition with the projection  $\text{Val}(\mathcal{V}) \rightarrow \text{Val}(\mathcal{X})$ . The families  $\mathcal{X}$  and  $\mathcal{Y}$  are independent under  $\mu$  when every joint event on  $\mathcal{X}$  is independent of every joint event on  $\mathcal{Y}$  in the product sense. They are conditionally independent given  $\mathcal{Z}$ , written  $\mathcal{X} \perp_\mu \mathcal{Y} \mid \mathcal{Z}$ , when, for every joint value assignment  $z$  to  $\mathcal{Z}$ , every joint event on  $\mathcal{X}$  is conditionally independent of every joint event on  $\mathcal{Y}$  given the cylinder event  $\mathcal{Z} = z$  in the product sense.

*Notation conventions.* The symbol  $\perp$  is used in two readings throughout this thesis: *probabilistic* independence, always subscripted by the probability assignment (e.g.  $A \perp_\mu B$ ,  $\mathcal{X} \perp_\mu \mathcal{Y} \mid \mathcal{Z}$ , and  $\perp_{P_\mathcal{M}}$  when the assignment is the observational distribution induced by a structural causal model  $\mathcal{M}$  in the sense of Definition A.3.25); and *graphical* d-separation, always carrying a graph subscript on the surrounding expression, e.g.  $(X \perp Y \mid Z)_\mathcal{G}$ , as in Definitions A.3.18 and A.3.33. Every occurrence of  $\perp$  in this thesis carries one of these two subscripts; the probabilistic subscript is the distribution under discussion at that point, as in Definition A.3.20.

*Remark A.3.11.* The equivalence between the product and conditional forms is the one-line calculation  $P_\mu(A \mid B) \cdot P_\mu(B) = P_\mu(A \cap B)$  for the unconditional case, and the analogous  $P_\mu(A \cap B \mid C) \cdot P_\mu(C) = P_\mu(A \cap B \cap C)$  for the conditional case, valid when the relevant denominator is positive by Definition A.3.7. Because the product form imposes no positivity side condition, it is the form used inside the SCM proofs of Sections B.5 and B.6, including Lemma B.6.1. In those proofs, conditioning events on observed variables can have zero rational mass when the structural functions only realise a proper subset of the declared value type; the conditional form  $P_\mu(A \mid B) = P_\mu(A)$  is then undefined, while the product form  $P_\mu(A \cap B) = P_\mu(A) \cdot P_\mu(B)$  remains a well-defined equality, with both sides equal to zero whenever  $B$  is unreachable.

**Definition A.3.12** (Marginalisation). *Event form.* Let  $\mu : \text{Prob}_\mathbb{Q}(\Omega)$ , and let  $\{F_y\}_{y:Y}$  be a finite family of decidable events on  $\Omega$  indexed by a finite type  $Y$  that is pairwise disjoint and jointly exhaustive (every  $\omega : \Omega$  lies in exactly one  $F_y$ ). For any decidable event  $A \subseteq \Omega$ , the *marginal of  $A$  over  $Y$*  is the finite sum

$$P_\mu(A) = \sum_{y:Y} P_\mu(A \cap F_y).$$

This is finite additivity (Theorem B.4.6) applied to the disjoint decomposition  $A = \bigcup_{y:Y} (A \cap F_y)$ .

*Variable form.* If  $\Omega = \text{Val}(\mathcal{V})$  for a finite family of variables  $\mathcal{V}$  in the sense of Definition A.3.9, and  $\mathcal{Y} \subseteq \mathcal{V}$  is a subset of variables, the marginal of  $\mu$  over the complement  $\mathcal{V} \setminus \mathcal{Y}$  is the probability assignment on  $\text{Val}(\mathcal{Y})$  obtained by finite summation over the remaining variables:

$$P_\mu(\mathcal{Y} = y) := \sum_{z:\text{Val}(\mathcal{V} \setminus \mathcal{Y})} P_\mu(\mathcal{Y} = y, \mathcal{V} \setminus \mathcal{Y} = z).$$

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The summation is over a *finite fibre*: the fibre of the projection  $\text{Val}(\mathcal{V}) \rightarrow \text{Val}(\mathcal{Y})$  over  $y$  is the finite type  $\text{Val}(\mathcal{V} \setminus \mathcal{Y})$ . Both forms are special cases of finite additivity; the variable form is the version used inside the do-calculus derivation steps of Definition A.3.34 and in the SCM proofs of Sections B.5 and B.6.

**Definition A.3.13** (Directed acyclic graph). A directed graph is a pair  $\mathcal{G} = (V, E)$  with finite node type  $V$  and decidable edge relation  $E : V \times V \rightarrow \text{Bool}$ . Write  $X \rightarrow Y$  when  $E(X, Y) = \text{true}$ . A directed path is a sequence  $X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_k$ . A graph path, used for d-separation, is a sequence of adjacent nodes in which each consecutive pair is connected by an edge in either direction. The directed graph is acyclic when there is no directed path with  $k > 0$  and  $X_0 = X_k$ ; a directed acyclic graph is such an acyclic directed graph.

**Definition A.3.14** (Causal graph). A causal graph is a graph whose nodes are variables and whose edges encode the direct causal structure assumed by a model. Two cases are distinguished:

- *Markovian (finite) case*:  $\mathcal{G}$  is a directed acyclic graph with no bidirected edges. All causal dependence between observed variables is mediated by declared parents in  $\mathcal{G}$ .
- *Semi-Markovian case*:  $\mathcal{G}$  is a finite directed-and-bidirected graph over observed variables, with a decidable directed edge relation and a decidable symmetric bidirected edge relation. Directed arrows represent direct causal dependence between observed variables; a bidirected edge  $X \leftrightarrow Y$  represents an unobserved common cause of  $X$  and  $Y$ . In this setting, a graph path may use directed edges in either orientation and bidirected edges as adjacencies. The directed part of the graph is acyclic. This is the usual graphical setting for c-components and hedges in finite identification theory.

**Definition A.3.15** (Parents, ancestors and descendants). In a directed graph  $\mathcal{G}$ , the parents, ancestors and descendants of  $Y$  are

$$\text{Pa}_{\mathcal{G}}(Y) := \{X \mid X \rightarrow Y\},$$

$$\text{An}_{\mathcal{G}}(Y) := \{X \mid X = Y \text{ or there is a directed path } X \rightarrow \dots \rightarrow Y\},$$

and

$$\text{De}_{\mathcal{G}}(Y) := \{X \mid X = Y \text{ or there is a directed path } Y \rightarrow \dots \rightarrow X\}.$$

For a semi-Markovian directed-and-bidirected graph, these three notions are computed in the directed part of the graph; bidirected edges are used for c-components, hedges and the latent-variable reading of d-separation, but they do not themselves create directed parents, ancestors or descendants.

**Definition A.3.16** (Chains, forks and colliders). On a path through three variables, a chain has the form  $A \rightarrow B \rightarrow C$  or  $A \leftarrow B \leftarrow C$ , a fork has the form  $A \leftarrow B \rightarrow C$ , and a collider has the form  $A \rightarrow B \leftarrow C$ . Chains and forks are non-collider configurations; colliders behave differently under conditioning.

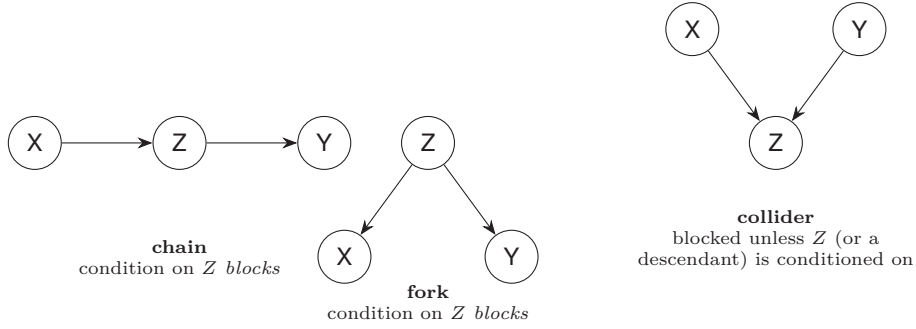


Figure A.1: The three path configurations behind d-separation (Definitions A.3.16 and A.3.18): chains and forks are non-colliders (conditioning on the middle node blocks the path); a collider blocks the path by default and is *opened* by conditioning on it or any of its descendants.

*Remark A.3.17.* This vocabulary is used to read d-separation (Definition A.3.18) informally; the formal d-separation predicate quantifies over paths and conditioning sets directly, so these three configurations appear inside its non-collider/collider clauses rather than as standalone objects.

**Definition A.3.18** (d-separation). d-separation is the graphical criterion for reading conditional independence from a directed acyclic graph. Here a path means a graph path in the sense of Definition A.3.13, not necessarily a directed path. A path is blocked by a conditioning set  $Z$  if it contains a non-collider on the path (in the sense of Definition A.3.16) that is in  $Z$ , or if it contains a collider such that neither the collider nor any of its descendants is in  $Z$ . If every path between variables in  $X$  and variables in  $Y$  is blocked by  $Z$ , then  $X$  and  $Y$  are d-separated by  $Z$ , written  $(X \perp\!\!\!\perp Y \mid Z)_{\mathcal{G}}$  in the graph [FK23; Pea09]. In a semi-Markovian graph with bidirected edges, this notation is read through the canonical finite latent-variable directed acyclic graph obtained by replacing each bidirected edge  $A \leftrightarrow B$  by a finite latent root with arrows into  $A$  and  $B$ .

**Definition A.3.19** (Causal Bayesian network). A causal Bayesian network is a directed acyclic graph  $\mathcal{G}$  over variables  $V_1, \dots, V_n$  together with a finite rational probability distribution  $P$  that factorises as

$$P(v_1, \dots, v_n) = \prod_{i=1}^n P(v_i \mid \mathbf{pa}_i),$$

where  $\mathbf{pa}_i$  is the tuple of values of  $\mathbf{Pa}_{\mathcal{G}}(V_i)$ . The intended causal reading is that each variable is independent of its non-descendants conditional on its parents.

**Definition A.3.20** (Causal Markov condition). Let  $\mathcal{V}$  be the full variable set of a directed acyclic graph  $\mathcal{G}$ . A finite rational probability distribution  $P$  satisfies the causal Markov condition with respect to  $\mathcal{G}$  when, for every variable  $X \in \mathcal{V}$ ,

$$X \perp\!\!\!\perp_P (\mathcal{V} \setminus (\text{De}_{\mathcal{G}}(X) \cup \mathbf{Pa}_{\mathcal{G}}(X))) \mid \mathbf{Pa}_{\mathcal{G}}(X).$$

This is the bridge from the graph to ordinary conditional independence claims [Pea09].

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*Remark A.3.21.* This condition is recorded for completeness and for comparison with classical presentations; the formal correspondence theorem of Theorem B.5.2 works directly with the factorisation of Definition A.3.19 rather than with this independence form.

**Definition A.3.22** (Faithfulness). A distribution  $P$  is faithful to a causal graph  $\mathcal{G}$  when every conditional independence in  $P$  is entailed by d-separation in  $\mathcal{G}$  and every d-separation in  $\mathcal{G}$  corresponds to a conditional independence in  $P$ . Markovness says that the graph licenses certain independencies; faithfulness says that the distribution contains no extra independencies produced by accidental parameter cancellation.

*Remark A.3.23.* Faithfulness is not assumed by any result in this thesis. It is recorded as background context for readers familiar with the causal-discovery literature, where faithfulness is often used to infer graph structure from observed independencies. The identification and do-calculus completeness results transferred in Section B.6 do not rely on faithfulness; they work with the graph and model class as given.

**Definition A.3.24** (Endogenous and exogenous variables). In a structural causal model, endogenous variables are the variables whose values are determined inside the model by structural functions. Exogenous variables are background or noise variables whose values are not explained inside the model but are assigned a finite rational probability distribution. The split between observed and unobserved variables is orthogonal: endogenous variables are observed by convention here, while exogenous variables may be either observed or unobserved. Exogenous variables that are unobserved are called *latent*; the projection onto the observed variables that records latent common causes as bidirected edges is the finite latent projection of Definition B.6.9.

**Definition A.3.25** (Structural causal model, solution map and observational distribution). A finite structural causal model is a tuple

$$\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$$

where  $\mathcal{U}$  is a finite family of exogenous finite variables,  $\mathcal{V}$  is a finite family of endogenous finite variables,  $P_{\mathcal{U}}$  is a finite rational probability distribution over  $\text{Val}(\mathcal{U})$ , and  $\mathcal{F}$  is the structural package consisting of a decidable declared endogenous-parent relation  $\text{Pa}(V) \subseteq \mathcal{V}$ , finite declared exogenous-input families  $\mathcal{U}_V \subseteq \mathcal{U}$ , and structural functions

$$f_V : \text{Val}(\text{Pa}(V)) \times \text{Val}(\mathcal{U}_V) \rightarrow \text{Val}(V) \quad (V \in \mathcal{V}).$$

Thus each endogenous variable is assigned from its declared endogenous parents and relevant exogenous inputs. The exogenous-input declaration is the exogenous-incidence relation used in Definition B.5.1. The *induced causal graph*  $\mathcal{G}$  (in the sense of Definition A.3.14) is read off the declared parent relation and this incidence relation: its nodes are the endogenous variables  $\mathcal{V}$ ; there is a directed edge  $X \rightarrow V$  exactly when  $X$  is a declared endogenous parent of  $V$ , i.e.  $X \in \text{Pa}(V)$ ; and, in the semi-Markovian reading obtained after the finite latent projection of Definition B.6.9, there is a bidirected edge  $A \leftrightarrow B$  exactly when some unobserved exogenous variable lies in both  $\mathcal{U}_A$  and  $\mathcal{U}_B$ .

All structural causal models in this thesis are *recursive*: the directed part of the induced causal graph  $\mathcal{G}$  is acyclic, and the endogenous variables admit a topological order in which each structural equation depends only on earlier endogenous variables and on its exogenous inputs. In this case the structural equations have a unique solution for each exogenous assignment  $u : \text{Val}(\mathcal{U})$  [Pea09].

The *solution map* of  $\mathcal{M}$  is the total function

$$\text{sol}_{\mathcal{M}} : \text{Val}(\mathcal{U}) \rightarrow \text{Val}(\mathcal{V})$$

defined by recursion over the topological order: for each endogenous variable  $V_i$ ,

$$\text{sol}_{\mathcal{M}}(u)_i := f_{V_i}(\text{sol}_{\mathcal{M}}(u)_{<i} \upharpoonright_{\text{Pa}(V_i)}, u \upharpoonright_{\mathcal{U}_{V_i}}),$$

where  $\text{sol}_{\mathcal{M}}(u)_{<i}$  denotes the already-computed values of the earlier endogenous variables and the restriction symbols pick out the declared parents and exogenous inputs of  $V_i$ . The recursion terminates after  $|\mathcal{V}|$  steps by recursiveness.

The *observational distribution induced by  $\mathcal{M}$*  is the pushforward of  $P_{\mathcal{U}}$  along  $\text{sol}_{\mathcal{M}}$ . For any decidable event  $E \subseteq \text{Val}(\mathcal{V})$ ,

$$P_{\mathcal{M}}(E) := \sum_{u : \text{Val}(\mathcal{U})} \mathbb{1}_E(\text{sol}_{\mathcal{M}}(u)) P_{\mathcal{U}}(u),$$

using the indicator of Definition A.1.23 and the finite probability assignment of Definition A.3.5. Marginal and conditional probabilities derived from  $P_{\mathcal{M}}$  are taken in the sense of Definitions A.3.7 and A.3.12; (conditional) independence under  $P_{\mathcal{M}}$  is written  $\perp\!\!\!\perp_{P_{\mathcal{M}}}$  in the notation convention of Definition A.3.10.

**Definition A.3.26** (Intervention and interventional distribution). For a structural causal model  $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$  in the sense of Definition A.3.25 and a joint assignment  $x : \text{Val}(X)$  to a finite set of endogenous variables  $X \subseteq \mathcal{V}$ , the *intervened model* is

$$\mathcal{M}_x := (\mathcal{U}, \mathcal{V}, \mathcal{F}_x, P_{\mathcal{U}}),$$

where  $\mathcal{F}_x$  replaces, for each  $V \in X$ , the declared endogenous-parent and exogenous-input families of  $V$  by empty families and replaces  $f_V$  by the constant value  $x_V : \text{Val}(V)$ ; all other declarations and structural functions are left unchanged. The intervened model is again recursive over the same topological order; in particular, the solution map  $\text{sol}_{\mathcal{M}_x}$  and the observational distribution  $P_{\mathcal{M}_x}$  are defined as in Definition A.3.25. The *interventional distribution* associated with  $\text{do}(X = x)$  is then

$$P_{\mathcal{M}_x}(E) = \sum_{u : \text{Val}(\mathcal{U})} \mathbb{1}_E(\text{sol}_{\mathcal{M}_x}(u)) P_{\mathcal{U}}(u)$$

for any decidable event  $E \subseteq \text{Val}(\mathcal{V})$ . The interventional distribution  $P_{\mathcal{M}_x}$  is the model-theoretic content of the intervention  $\text{do}(X = x)$ ; its graph-level content is the mutilated graph  $\mathcal{G}_{\overline{X}}$  of Definition A.3.27.

**Definition A.3.27** (Graph mutilation). Graph mutilation is Pearl's graphical representation of intervention. For a graph  $\mathcal{G}$  and a set of variables  $X$ , the mutilated graph  $\mathcal{G}_{\overline{X}}$  is obtained by deleting all directed arrows into  $X$ , because an intervention on  $X$  replaces its usual causes. The graph  $\mathcal{G}_{\underline{X}}$  is obtained

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by deleting all directed arrows out of  $X$ . In the semi-Markovian reading, these operations are applied after passing to the canonical finite latent-variable directed acyclic graph of Definition A.3.18; equivalently, a bidirected edge incident with an intervened variable no longer contributes a latent arrowhead into that variable in  $\mathcal{G}_{\overline{X}}$ . These operations are combined when several marks occur together.

**Definition A.3.28** (Do-operator). The expression  $\text{do}(X = x)$  denotes an intervention that sets  $X$  to  $x$ . It differs from the observation  $X = x$  because it changes the causal mechanism for  $X$  rather than only conditioning on the event that  $X$  has value  $x$ . The model-theoretic content of  $\text{do}(X = x)$  is the intervened structural causal model of Definition A.3.26; its graph-level content is the mutilated graph of Definition A.3.27.

**Definition A.3.29** (Compatibility and identification). Let  $\mathcal{G}$  be a finite causal graph in the Markovian or semi-Markovian sense of Definition A.3.14, and let  $\mathcal{M}$  be a finite recursive structural causal model in the sense of Definition A.3.25. The model  $\mathcal{M}$  is *compatible with*  $\mathcal{G}$  when the induced causal graph of  $\mathcal{M}$  agrees with  $\mathcal{G}$  on its directed edges and, in the semi-Markovian case, on its bidirected edges as induced by the exogenous-incidence relation, and when  $\mathcal{M}$  satisfies the finite recursive structure required by the fragment. Write  $\text{Mod}(\mathcal{G})$  for the class of finite rational recursive models compatible with  $\mathcal{G}$ .

A finite interventional query is an expression  $Q = P_x(E)$ , often written  $P_x(y)$  for the event  $Y = y$ , where  $P_x(E)$  abbreviates  $P_{\mathcal{M}_x}(E)$  for the intervened model of Definition A.3.26. Such a query is *identifiable from  $\mathcal{G}$  and a finite rational observational distribution  $P^{\text{obs}}$*  when, for every  $\mathcal{M} \in \text{Mod}(\mathcal{G})$  whose observational distribution  $P_{\mathcal{M}}$  in the sense of Definition A.3.25 equals  $P^{\text{obs}}$ , the corresponding interventional distribution  $P_{\mathcal{M}_x}$  returns the same value on the target event  $E$ . Equivalently, any two compatible models in  $\text{Mod}(\mathcal{G})$  with the same observational distribution give the same value to the target interventional query. In that case the effect can, in principle, be written as a formula involving only observational quantities [Pea09; SP06; TS09].

**Definition A.3.30** (c-component). In a semi-Markovian causal graph, a c-component is a maximal set of observed variables connected by bidirected paths, where a bidirected path is a graph path using only bidirected edges. The bidirected edges represent shared unobserved causes, so c-components mark the parts of the graph tied together by latent confounding [TP02; TS09].

**Definition A.3.31** (c-forest). A c-forest is a directed-and-bidirected subgraph whose observed variables form one c-component and in which each observed variable has at most one directed child inside the subgraph. The roots of the c-forest are the variables with no directed children inside it.

**Definition A.3.32** (Hedge). A hedge for the effect of  $X$  on  $Y$  is a pair of c-forests  $F$  and  $F'$  with the same roots such that  $F' \subset F$ ,  $F$  contains at least one variable in  $X$ ,  $F'$  contains no variable in  $X$ , and the roots lie among the ancestors of  $Y$  in the graph where arrows into  $X$  have been deleted. The existence of a hedge witnesses that the causal effect is not identifiable from the available finite rational observational distribution [TS09].

**Definition A.3.33** (Do-calculus rules). Let  $\mathcal{G}$  be a causal graph. All probability expressions below are finite conditional expressions in the sense of Defini-

tion A.3.7, so any displayed conditioning event carries the required positive-denominator side condition. Pearl's do-calculus consists of three transformation rules [Pea09; Pea12]:

A.3.33.1. insertion and deletion of observations:

$$P(y \mid \text{do}(x), z, w) = P(y \mid \text{do}(x), w)$$

whenever  $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}}}$ ;

A.3.33.2. action and observation exchange:

$$P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), z, w)$$

whenever  $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}, \underline{Z}}}$ ;

A.3.33.3. insertion and deletion of actions:

$$P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), w)$$

whenever  $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}, \overline{Z(W)}}$ , where  $Z(W)$  contains the variables in  $Z$  that are not ancestors of any variable in  $W$  in  $\mathcal{G}_{\overline{X}}$ .

**Definition A.3.34** (Do-calculus derivation). Let  $\mathcal{G}$  be a finite causal graph in the Markovian or semi-Markovian sense of Definition A.3.14. A *do-calculus derivation* of an equality  $E_0 = E_n$  between finite probability expressions over  $\mathcal{G}$  is a finite sequence

$$E_0 \longrightarrow E_1 \longrightarrow \cdots \longrightarrow E_n$$

in which each step  $E_i \longrightarrow E_{i+1}$  is one of the following finite rule applications:

A.3.34.1. *Reflexivity step*:  $E_i = E_{i+1}$  as a definitional equality of finite rational expressions.

A.3.34.2. *Model-evaluation step*:  $E_i$  contains a sub-expression  $P_{\mathcal{M}_x}(y)$  for an intervened structural causal model  $\mathcal{M}_x$  (Definition A.3.26) whose recursive evaluation determines that probability as a finite rational expression  $r$  in the exogenous distribution and structural functions of  $\mathcal{M}_x$ ;  $E_{i+1}$  replaces  $P_{\mathcal{M}_x}(y)$  by  $r$  in  $E_i$ . This step is used in finite countermodel arguments (such as the proof of Lemma B.6.2) where a concrete numerical value of  $P_{\mathcal{M}_x}(y)$  is needed. Each such step is licensed by Theorem B.5.2.

A.3.34.3. *Probability step*:  $E_{i+1}$  is obtained from  $E_i$  by a finite rewrite using one of finite additivity (Theorem B.4.6), the definition of conditional probability (Definition A.3.7) with explicit positive denominator, or marginalisation in the sense of Definition A.3.12.

A.3.34.4. *Do-calculus step*:  $E_{i+1}$  is obtained from  $E_i$  by one application of the insertion-and-deletion-of-observations rule, the action-and-observation-exchange rule, or the insertion-and-deletion-of-actions rule of Definition A.3.33, together with a finite proof that the licensing d-separation condition holds in the appropriate mutilated graph.

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A derivation is therefore a finite sequence of finite rule applications, each carrying its own finite side conditions. The set of do-calculus derivations is closed under composition: if  $E_0 \rightarrow \dots \rightarrow E_n$  and  $E_n \rightarrow \dots \rightarrow E_m$  are derivations, their concatenation is a derivation of  $E_0 = E_m$ . This is the formal object on which the induction in the proof of Proposition B.5.15 runs.

**Definition A.3.35** (Counterfactual query). Given a structural causal model  $\mathcal{M}$  in the sense of Definition A.3.25, an exogenous assignment  $u$ , and an intervention  $X = x$  in the sense of Definition A.3.26, the counterfactual  $Y_x(u)$  is the value of  $Y$  in the solution of the intervened model  $\mathcal{M}_x$  under  $u$ , that is,  $\text{sol}_{\mathcal{M}_x}(u)$  projected to  $Y$ . With observed evidence, Pearl’s evaluation procedure is abduction, action and prediction: update the finite rational probability distribution over  $u$ , replace the relevant equations, and compute the resulting value or finite rational probability distribution.

Concretely, if  $E \subseteq \text{Val}(\mathcal{V})$  is a decidable observed event with positive probability, write

$$p_E(u) := \frac{\mathbb{1}_E(\text{sol}_{\mathcal{M}}(u)) P_{\mathcal{U}}(u)}{P_{\mathcal{M}}(E)}.$$

Then the model-level counterfactual probability of a decidable target event  $F \subseteq \text{Val}(\mathcal{V})$  under the action  $X = x$  is

$$P(F_x \mid E) := \sum_{u: \text{Val}(\mathcal{U})} \mathbb{1}_F(\text{sol}_{\mathcal{M}_x}(u)) p_E(u).$$

This is a calculation inside the given structural causal model. It is not, by itself, an identifiability claim saying that the value is determined by an observational distribution or a graph alone.

*Remark A.3.36.* Counterfactual queries are placed here, after the identification machinery of Definitions A.3.29 to A.3.32, because they sit on Pearl’s third level of inquiry: the unconditional and conditional interventional fragment (Pearl’s second level) is closed by the identification definitions above, and counterfactuals add an extra step (abduction) on top of intervention. They are recorded for orientation; the formal completeness transfer of Theorem B.6.13 is restricted to ordinary and conditional interventional distributions, and does not cover counterfactual identification.

**Definition A.3.37** (Epistemic causal mode). In this thesis, an epistemic causal mode is a mode whose content includes a finite structural causal model. This is a thesis-specific construction: it uses Gratzner-style modes to organise Pearl-style causal models as epistemic states. Moving between such modes represents changing the causal model, changing the available information, or changing which interventions and counterfactuals are in scope.

## A.4 Examples

The example collected here is the worked-out finite calculation supporting the running tenure-track example of Example 5.4.1. It is placed in this appendix rather than alongside the definitions because it is a numeric illustration, not a definitional commitment of the theory. The narrative version in the fourth movement carries the conceptual weight; the calculations below carry the rational arithmetic.

**Example A.4.1** (Mira's tenure-track search, finite calculation). This is the fully finite version of the tenure-track model used in Example 5.4.1. All endogenous value types are the two-element Boolean type. The endogenous variables are

$$\mathcal{V} = \{\text{Pr}, \text{Qu}, \text{Top}, \text{Com}, \text{Fit}, \text{Short}, \text{Fund}, \text{Off}\}.$$

The exogenous variables are independent Boolean variables

$$U_B, U_{\text{Pr}}, U_{\text{Qu}}, U_{\text{Top}}, U_{\text{Com}}, U_{\text{Fund}},$$

with finite rational masses

$$P(U_B = 1) = \frac{1}{3}, \quad P(U_{\text{Pr}} = 1) = P(U_{\text{Qu}} = 1) = P(U_{\text{Fund}} = 1) = \frac{1}{4},$$

and

$$P(U_{\text{Top}} = 1) = P(U_{\text{Com}} = 1) = \frac{1}{2}.$$

The structural functions are

$$\begin{aligned} \text{Pr} &:= U_B \vee U_{\text{Pr}}, & \text{Qu} &:= U_B \vee U_{\text{Qu}}, \\ \text{Top} &:= U_{\text{Top}}, & \text{Com} &:= U_{\text{Com}}, \\ \text{Fit} &:= \text{Top} \wedge \text{Com}, & \text{Short} &:= \text{Qu} \wedge (\text{Pr} \vee \text{Fit}), \\ \text{Fund} &:= \text{Top} \vee U_{\text{Fund}}, & \text{Off} &:= \text{Short} \wedge \text{Fund}. \end{aligned}$$

In the full finite directed acyclic graph,  $U_B$  is a common cause of  $\text{Pr}$  and  $\text{Qu}$ . If  $U_B$  is hidden, the observed semi-Markovian graph has the bidirected edge  $\text{Pr} \leftrightarrow \text{Qu}$ , so  $\{\text{Pr}, \text{Qu}\}$  is a c-component. The graph also contains the colliders

$$\text{Top} \rightarrow \text{Fit} \leftarrow \text{Com}, \quad \text{Pr} \rightarrow \text{Short} \leftarrow \text{Fit}, \quad \text{Short} \rightarrow \text{Off} \leftarrow \text{Fund}.$$

Thus  $\text{Top}$  and  $\text{Com}$  are d-separated by the empty conditioning set, but the path through  $\text{Fit}$  is opened when one conditions on  $\text{Fit}$  or one of its descendants. Numerically, the deterministic equation  $\text{Fit} := \text{Top} \wedge \text{Com}$  gives

$$P(\text{Top} = 1 \mid \text{Com} = 1) = \frac{1}{2} = P(\text{Top} = 1),$$

while conditioning on the collider event  $\text{Fit} = 0$  changes the relation:

$$P(\text{Top} = 1 \mid \text{Fit} = 0) = \frac{1}{3}, \quad P(\text{Top} = 1 \mid \text{Com} = 1, \text{Fit} = 0) = 0.$$

This is the collider phenomenon in a finite deterministic form.

The distinction between observation and intervention is also visible by finite summation. First,

$$P(\text{Pr} = 1) = 1 - P(U_B = 0, U_{\text{Pr}} = 0) = 1 - \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}.$$

Similarly  $P(\text{Qu} = 1) = 1/2$ , and

$$P(\text{Pr} = 1, \text{Qu} = 1) = 1 - \frac{1}{2} - \frac{1}{2} + \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{3}{8}.$$

## A. Definitions

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Hence

$$P(\mathbf{Qu} = 1 \mid \mathbf{Pr} = 1) = \frac{P(\mathbf{Pr} = 1, \mathbf{Qu} = 1)}{P(\mathbf{Pr} = 1)} = \frac{3/8}{1/2} = \frac{3}{4}.$$

But the intervention  $\text{do}(\mathbf{Pr} = 1)$  replaces the equation for  $\mathbf{Pr}$  by the constant value 1 and leaves the mechanism for  $\mathbf{Qu}$  unchanged, so

$$P(\mathbf{Qu} = 1 \mid \text{do}(\mathbf{Pr} = 1)) = P(\mathbf{Qu} = 1) = \frac{1}{2}.$$

A second calculation shows why observing committee fit is not the same as forcing committee fit. Since  $\mathbf{Fit} = 1$  implies  $\mathbf{Top} = 1$  and  $\mathbf{Com} = 1$ , it also implies  $\mathbf{Fund} = 1$ . Under  $\mathbf{Fit} = 1$ , the shortlist equation reduces to  $\mathbf{Short} = \mathbf{Qu}$ , and therefore

$$P(\mathbf{Off} = 1 \mid \mathbf{Fit} = 1) = P(\mathbf{Qu} = 1) = \frac{1}{2}.$$

By contrast, in the intervened model  $\text{do}(\mathbf{Fit} = 1)$ , the variable  $\mathbf{Top}$  is not forced. The offer requires both  $\mathbf{Qu} = 1$  and  $\mathbf{Fund} = 1$ , and these are independent because they depend on disjoint exogenous variables. Since

$$P(\mathbf{Fund} = 1) = 1 - P(\mathbf{Top} = 0, U_{\mathbf{Fund}} = 0) = 1 - \frac{1}{2} \cdot \frac{3}{4} = \frac{5}{8},$$

one gets

$$P(\mathbf{Off} = 1 \mid \text{do}(\mathbf{Fit} = 1)) = \frac{1}{2} \cdot \frac{5}{8} = \frac{5}{16}.$$

Finally, the same finite model permits a model-level counterfactual calculation. Let

$$E := \{\mathbf{Pr} = 0, \mathbf{Qu} = 1, \mathbf{Off} = 0\}.$$

Under the structural equations,  $\mathbf{Pr} = 0$  forces  $U_B = 0$  and  $U_{\mathbf{Pr}} = 0$ . Together with  $\mathbf{Qu} = 1$ , this forces  $U_{\mathbf{Qu}} = 1$ . Moreover, when  $\mathbf{Pr} = 0$  and  $\mathbf{Qu} = 1$ , the equation for  $\mathbf{Short}$  reduces to  $\mathbf{Short} = \mathbf{Fit}$ . Since  $\mathbf{Fit} = 1$  implies  $\mathbf{Top} = 1$ , hence  $\mathbf{Fund} = 1$ , the additional evidence  $\mathbf{Off} = 0$  forces  $\mathbf{Fit} = 0$ . Thus

$$P(E) = \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32},$$

so the abduction step is well defined.

Condition on  $E$ , then perform the action  $\text{do}(\mathbf{Pr} = 1)$ . In the intervened model,  $\mathbf{Short} = 1$ , because  $\mathbf{Qu} = 1$  is retained from the posterior evidence and  $\mathbf{Pr}$  has been set to 1. Therefore the counterfactual offer depends only on

$$\mathbf{Fund} = \mathbf{Top} \vee U_{\mathbf{Fund}}.$$

The evidence about  $\mathbf{Pr}$  and  $\mathbf{Qu}$  is independent of  $U_{\mathbf{Top}}, U_{\mathbf{Com}}, U_{\mathbf{Fund}}$ , and the remaining relevant information is  $\mathbf{Fit} = 0$ , i.e. not both  $\mathbf{Top}$  and  $\mathbf{Com}$ . Hence

$$P(\mathbf{Fund} = 1 \mid E) = P(\mathbf{Fund} = 1 \mid \mathbf{Fit} = 0) = \frac{1/4 + 1/8}{3/4} = \frac{1}{2}.$$

Consequently

$$P(\mathbf{Off}_{\mathbf{Pr}=1} = 1 \mid E) = \frac{1}{2}.$$

The same posterior also gives

$$P(\text{Off}_{\text{Fit}=1} = 1 \mid E) = \frac{1}{2},$$

because setting  $\text{Fit} = 1$  likewise makes  $\text{Short} = 1$  while leaving the same posterior uncertainty about  $\text{Fund}$ . This is Pearl’s abduction–action–prediction procedure in the finite model: first condition the exogenous distribution, then change the structural equation, then evaluate the target. It is not a proof that arbitrary counterfactuals are identifiable from the observational distribution alone.

Internally to the thesis framework, this example is a finite structural-causal-model record at an epistemic causal mode:

$$\Gamma_{\text{ten}} \vdash \mathcal{M}_{\text{ten}} : \text{FinSCM}@m_{\text{ten}},$$

where  $\text{FinSCM}$  abbreviates the restricted finite type-theoretic structural-causal-model fragment of Definition B.5.1. The structural equations are terms in the Boolean value types; the exogenous distribution is a finite rational probability type; and the displayed probability calculations are equalities of finite sums and quotients with positive denominators. In the restricted Lean core of `lean/Thesis/SCM.lean`, the same shape is represented by a recursive finite record whose evaluation map pushes the exogenous distribution forward to observational and interventional probabilities.<sup>1</sup>

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<sup>1</sup>The Lean module is called `Thesis.SCM` for backward compatibility with the existing formalisation.



## APPENDIX B

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### Proofs and arguments

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*“The important thing is life — life and nothing else! What is any ‘discovery’ whatever compared with the incessant, eternal discovery of life?”*

Hippolyte in *Idiot*

## B. Proofs and arguments

The appendix contains several different kinds of support: philosophical motivation, metatheoretic limitation, finite probability reconstruction, causal-model correspondence and completeness transfer. The route through these arguments is summarised in Table B.1.

| Proof or argument                         | Purpose                                                                                                                                           | Depends on                                                                                                                              | Used by                                                        |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Causality argument                        | Explains why mathematical inference for science should include causal reasoning, not only probabilistic prediction.                               | Scientific explanation and Pearl’s seeing–doing–imagining distinction.                                                                  | Motivates Section 5.4 and the causal transfer arguments.       |
| Physics argument                          | Defends the constructive and epistemological stance: finite measurements should not force a commitment to completed infinities or fixed ontology. | Gisin-style finite information and Spekkens-style epistemic readings.                                                                   | Supports the constructive stance used in Sections 5.3 and 5.4. |
| Excluded middle proof                     | Shows why global excluded middle is not available in the base constructive theory.                                                                | Martin-Löf-style type theory and Kripke/presheaf semantics.                                                                             | Justifies restricting probability to finite decidable events.  |
| Clayton–Waddington probability proof      | Reconstructs finite probability from urn principles, then derives finite Kolmogorov laws, Bayes’ theorem and Bayesian update.                     | Definition A.3.6 and finite decidable event types.                                                                                      | Supports Section 5.3 and the Lean probability checks.          |
| Structural-causal-model equivalence proof | Builds the bridge between ordinary finite recursive structural causal models and the restricted type-theoretic record fragment.                   | Finite probability records, finite variables and recursive structural causal models.                                                    | Supports Section 5.4 and the completeness transfer.            |
| Causality-completeness proof              | Transfers published finite do-calculus and identification completeness results into the restricted type-theoretic fragment.                       | Model correspondence, graph/probability-expression transport and the external Shpitser–Pearl, Huang–Valtorta and Tian–Shpitser results. | Supports the final causal completeness claim, Theorem B.6.13.  |

Table B.1: Route through the proofs and arguments appendix.

### B.1 Causality argument

Human beings reason causally almost before they reason formally. We ask what made the glass fall, why the patient recovered, what caused the model to fail, which policy changed the outcome, and what would have happened had the intervention been different. This is not an exotic philosophical habit. It is one of the ordinary grammars of human orientation.

Science can be described, at least in part, as the disciplined extension of this causal grammar. It discovers, tests, rejects and stabilises causal relations in the material world, the biological world and the social world. A science without causality would not simply be incomplete in a technical sense. It would fail to speak the language in which many scientific questions are actually asked.

None of which is to say that any causal story will do. Scientific inquiry is communal, historical and bounded. With Huizinga, one may say that it contains a playful and creative element: scientists construct models, concepts and experimental situations together [Hui49]. With Erasmus, one may add that such constructions retain an ironic or provisional character: they are humanly made, and therefore should not be confused with reality itself [Era93]. With Kierkegaard, one may add that irony is not limitless: it is bounded by what has already been inherited [Kie80; Kie83; Kie89]. Fleck sharpens this social point: causal judgement is stabilised inside communities, inherited styles of thought, shared techniques and historically formed expectations [Fle79]. Hacking sharpens the practical point: science is not only representational, but interventionist, because scientists isolate, manipulate, produce, measure and stabilise phenomena [Hac83]. To this I add that scientific construction is bounded by intersubjective accountability: claims must pass through other people, practices, instruments, institutions and traditions of criticism.

The result is a middle position. Against a crude Feyerabendian reading, not anything goes [Fey75]. Against a crude Humean scepticism, causal talk need not be metaphysically guaranteed in order to be scientifically indispensable [Hum75; Hum78]. Against a purely Kantian framing, the question whether causality originates in the world or in human cognition becomes secondary once science is understood as a human practice of shared orientation, intervention and explanation [Kan98].

Pearl's framework is therefore justified pragmatically before it is justified metaphysically [Pea09; PM18]. It gives a rigorous language for something science already needs to do: distinguish seeing from doing, association from intervention, and prediction from counterfactual explanation. This thesis uses Pearl not because Pearl solves every philosophical problem about causality, but because any formal theory of mathematical inference for science must give causal reasoning a home.

## B.2 Physics argument

The physics argument begins with a modest observation: measurements contain finite information. A measured value is not an actually completed real number handed down from the heavens. It is a finite construction, with precision, procedure, instrument and context. Gisin's work is valuable because it presses this point against the habitual use of completed real numbers in physics [Bar26; BDG25; DG19; Gis20].

On a finite-information picture, measurement can actualise new information and the future can be genuinely open. One can recover analogues of quantum-style indeterminism, measurement-problem-like tensions, Wigner's-friend-like situations and no-cloning phenomena. This is a helpful direction. Yet it does not recover the whole quantum core: incompatible observables, Bell violations,

## B. Proofs and arguments

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contextuality, the precise Born-rule structure and full quantum entanglement remain outside the classical indeterministic reconstruction.

The difficulty is that Gisin’s proposed numbers still risk importing infinity through the back door. If an observed number is treated as an infinite sequence of propensities, then the number may look constructive while depending on an indefinitely extended object. That is not the constructivist instinct taken seriously enough. A genuinely constructive reading should say that measurement constructs a value up to the precision of the measurement, not that it reveals part of a hidden infinite propensity sequence.

Spekkens’ toy theory supports the epistemological half of the argument [HS10; Spe07]. Many features often advertised as distinctively quantum can arise from restrictions on knowledge: interference-like behaviour, noncommuting measurements, no-cloning, no-broadcasting, teleportation-like structure and epistemic update. Yet Spekkens also does not recover full quantum mechanics. Pusey, Barrett and Rudolph then sharpen the warning: one cannot keep a completely definite underlying ontic state and expect the epistemic view to do all the work [PBR12].

The natural conclusion is agnostic and constructive. Theories should be treated as epistemological vehicles for finite observations. There is no need, for the purposes of this thesis, to decide what the ultimate ontology of physics is. Measurements are finite; formal reasoning about them should therefore begin constructively. The epistemological reading gives room for Spekkens-style insight; the constructive reading gives room for Gisin-style finite information; and the refusal to fix a hidden ontology leaves space for the quantum phenomena that neither classical finite information nor purely epistemic toy theory fully captures.

### B.3 Excluded middle is not derivable

The claim used in the main text is modest. It is not that Martin-Löf-style type theory proves the negation of excluded middle as an internal theorem. The claim is metatheoretic: global excluded middle is not derivable in the bare constructive theory. If one wants to use it, one must add it as an additional classical principle.

**Proposition B.3.1** (Global excluded middle is not derivable). *In the Martin-Löf-style fragment used in this thesis, the uniform principle, read over small propositions or proposition-like types in the universe,*

$$\prod_{A:\mathcal{U}}(A + \neg A)$$

*is not derivable without adding a classical axiom or restricting  $A$  to a decidable class of types.*

*Proof.* The argument is by soundness with respect to standard Kripke/presheaf semantics for dependent type theory [AR11; Gra23]. The model used here validates the underlying non-modal fragment needed in this argument: contexts and substitution, dependent function types, finite product and sum types, the empty type, negation as function into the empty type and a small universe containing the proposition-like type used below. It is not a Boolean model, and it does not validate uniform excluded middle.

## B.4. Clayton–Waddington probability proof

The countermodel is used only for the propositional fragment of the argument, so the intensional/extensional distinction for identity types does not affect the failure of uniform excluded middle.

Consider the two-stage Kripke frame

$$w_0 \leq w_1,$$

and let  $P$  be an atomic proposition which is not forced at  $w_0$  but is forced at  $w_1$ . Forcing is persistent along extensions: if a proposition is forced at an earlier stage, it remains forced at later stages. Thus information may grow from  $w_0$  to  $w_1$ , so it is coherent for  $P$  not to be forced at  $w_0$  while being forced at  $w_1$ .

At  $w_0$ ,  $P$  is not forced by construction. Moreover,  $\neg P$  is not forced at  $w_0$ , because forcing  $\neg P$  would mean that no extension of  $w_0$  could force  $P$ ; but  $w_1$  is an extension of  $w_0$  where  $P$  is forced. Hence neither disjunct of

$$P + \neg P$$

is forced at  $w_0$ . Thus excluded middle is not valid in this sound model of the constructive propositional fragment.

If the displayed uniform excluded-middle term were derivable in bare Martin-Löf-style type theory for a universe containing this small propositional type  $P$ , its interpretation would give, at every stage and for every such proposition, either a witness of the proposition or a witness of its negation. Instantiating it with  $P$  at  $w_0$  would force  $P + \neg P$ , contradicting the model above. Therefore no such term is derivable in the base theory. ■

The same warning applies to double-negation elimination as a global principle. Constructively one always has

$$A \rightarrow \neg\neg A,$$

by sending  $a : A$  and  $n : A \rightarrow \perp$  to  $n(a) : \perp$ . The converse

$$\neg\neg A \rightarrow A$$

is a classical principle and is not available uniformly in the base theory.

This does not conflict with the finite probability constructions used in this thesis. For particular finite types, equality, membership and Boolean predicates are decidable by construction. Thus one may use  $E + \neg E$  for a finite decidable event  $E$ , and one may take complements inside finite event algebras. What is rejected is the global move from an arbitrary proposition  $A$  to  $A + \neg A$  without a decision procedure or an explicit classical assumption.

## B.4 Clayton–Waddington probability proof

This proof is deliberately finite. It recovers the ordinary finite probability calculus from urn-generated probability types, not the full measure-theoretic theory of countably additive probability. The constructive point is also important: no global law of excluded middle is used. Complements appear only inside finite decidable event algebras. The urn rescaling below is a representation result for the stated Clayton–Waddington-style assumptions; it does not claim that every possible plausibility calculus must satisfy those assumptions. The corresponding Lean 4 formalisation is in the module `Thesis.Probability`, in `lean/Thesis/Probability.lean`.

## B. Proofs and arguments

| Classical Cox / Van Horn route                                                                                       | This thesis (finite, constructive)                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| desiderata: real-valued, common sense, consistency<br>associativity equation $\Rightarrow$ <b>product rule first</b> | finite urns + relabelling, refinement, strict monotonicity<br>urn rescaling $U_N(k) \mapsto k/N \Rightarrow$ <b>ratio and sum rule first</b> |
| negation function $S \Rightarrow$ <b>sum rule second</b><br><i>universality</i> (R4) density assumed/derived         | product urns $\Rightarrow$ <b>product rule second</b><br><i>universality eliminated</i> : values are $k/N$ by construction                   |
| continuity on $[0, 1]$                                                                                               | no continuum; complements only on decidable events                                                                                           |
| construct rescaling $p$                                                                                              | record–urn equivalence $\rightarrow$ Kolmogorov laws<br>$\rightarrow$ Bayes                                                                  |

Figure B.1: Where this thesis’s finite derivation departs from the Cox–Jaynes–Van Horn argument refined by Clayton and Waddington [CW17]: the sum and product rules are derived in the opposite order, and the universality assumption is dropped rather than patched.

**Proposition B.4.1** (Resolution-status partition). *Let  $S$  be a finite event type and let  $\iota_A : S \rightarrow \text{Status}_A$  be a finite inspection procedure as in Definition A.3.2. Then the events  $\text{Prov}_A$ ,  $\text{Ref}_A$  and  $\text{Unr}_A$  of Definition A.3.3 are decidable, pairwise disjoint and jointly exhaustive.*

*Proof.* Decidability follows because  $\text{Status}_A$  is the finite three-element type

$$\{\text{proved}, \text{refuted}, \text{unresolved}\}$$

and equality in a finite type is decidable. For any  $s : S$ , the value  $\iota_A(s)$  is exactly one of these three constructors. If it is **proved**, then  $s$  lies in  $\text{Prov}_A$  and in neither of the other two events. The cases **refuted** and **unresolved** are identical. Thus the three events are pairwise disjoint and their union is  $S$ . ■

**Proposition B.4.2** (Qualitative urn rescaling). *Assume the constructive Clayton–Waddington urn assumptions of Definition A.3.6. Then the plausibility scale for finite symmetric urn events can be rescaled so that*

$$p(U_N(k)) = \frac{k}{N}$$

*for every  $N > 0$  and  $0 \leq k \leq N$ . This rescaling is well-defined and order-preserving: equal ratios give the same qualitative plausibility class, and strictly smaller ratios give strictly smaller plausibility. Under this rescaling, decidable complements in an  $N$ -cell urn satisfy*

$$p(E^{\complement}) = 1 - p(E).$$

*Proof.* Relabelling invariance implies that the plausibility of a  $k$ -cell event in  $\text{Fin}(N)$  depends only on  $k$  and  $N$ , not on which particular  $k$  cells were chosen. Thus  $U_N(k)$  is well-defined.

Refinement invariance gives

$$U_N(k) = U_{cN}(ck)$$

#### B.4. Clayton–Waddington probability proof

for every  $c > 0$ . If two fractions agree,  $k/N = k'/N'$ , then refinement to the common urn of size  $NN'$  gives

$$U_N(k) = U_{NN'}(kN') = U_{N'N}(k'N) = U_{N'}(k').$$

Hence the rational number  $k/N$  determines a single plausibility class. If  $k/N < \ell/M$ , refinement to the common urn of size  $NM$  gives  $kM < \ell N$ , and strict monotonicity makes the first refined event strictly less plausible than the second. Therefore the assignment

$$U_N(k) \mapsto \frac{k}{N}$$

is a well-defined order-preserving rescaling of the finite urn plausibilities, using qualitative coherence to transport strict comparison across plausibility equivalence.

In a decidable  $N$ -cell urn, the complement of a  $k$ -cell event has  $N - k$  cells. Therefore

$$p(E^c) = \frac{N - k}{N} = 1 - \frac{k}{N} = 1 - p(E).$$

This is a statement about decidable urn complements, not about arbitrary propositions in Martin-Löf-style type theory. ■

**Proposition B.4.3** (Product urns after rescaling). *After the rescaling of Proposition B.4.2, product urn compatibility gives the expected multiplication law for rectangular decidable events. If  $E$  has  $k$  cells in an  $M$ -cell urn and  $F$  has  $\ell$  cells in an  $N$ -cell urn, then*

$$p(E \times F) = p(E)p(F).$$

*Proof.* The product event  $E \times F$  has  $k\ell$  cells in the product urn of size  $MN$ . Therefore the rescaling of Proposition B.4.2 gives

$$p(E \times F) = \frac{k\ell}{MN} = \frac{k}{M} \frac{\ell}{N} = p(E)p(F).$$

■

**Proposition B.4.4** (Finite rational records admit a common denominator). *Let  $\Omega$  be a finite event type and let*

$$\mu = (p, h) : \text{Prob}_{\mathbb{Q}}(\Omega)$$

*be a finite rational probability record in the sense of Definition A.3.5. Then  $\mu$  has an extensionally equal weighted common-denominator presentation: there are a positive natural number  $D$  and natural numbers  $m_{\omega}$  indexed by  $\omega : \Omega$  such that*

$$p(\omega) = \frac{m_{\omega}}{D} \quad \text{for every } \omega : \Omega,$$

*and*

$$\sum_{\omega : \Omega} m_{\omega} = D.$$

## B. Proofs and arguments

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*Proof.* Choose a finite enumeration  $\omega_1, \dots, \omega_r$  of  $\Omega$ . Since each  $p(\omega_i)$  is a non-negative rational number, write it as

$$p(\omega_i) = \frac{a_i}{b_i}$$

with  $a_i, b_i : \mathbb{N}$  and  $b_i > 0$ . Let

$$D := \prod_{i=1}^r b_i,$$

taking the empty product to be 1. For each  $i$ , define

$$m_i := a_i \prod_{j \neq i} b_j.$$

Then  $m_i : \mathbb{N}$  and

$$\frac{m_i}{D} = \frac{a_i}{b_i} = p(\omega_i).$$

Summing over the finite enumeration and using the normalisation proof  $h : \sum_{\omega \in \Omega} p(\omega) = 1$  gives

$$\frac{1}{D} \sum_{i=1}^r m_i = \sum_{i=1}^r \frac{m_i}{D} = \sum_{i=1}^r p(\omega_i) = 1,$$

hence  $\sum_i m_i = D$ . This construction uses only the finite enumeration of  $\Omega$  and the numerator-denominator data of the finitely many rational masses. It is therefore constructive in the finite rational fragment.  $\blacksquare$

**Proposition B.4.5** (Common-denominator record–urn equivalence). *Let  $X$  be any type. For weighted finite-support records in the common-denominator form of Definition A.3.5, urn presentations and record presentations are extensionally equivalent. More precisely:*

B.4.5.1. *every urn-generated assignment  $\mu : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$  determines a weighted record  $R_\mu$  such that*

$$P_{R_\mu}(E) = P_\mu(E)$$

*for every decidable event  $E : X \rightarrow \text{Bool}$ ;*

B.4.5.2. *every weighted common-denominator record  $R$  determines an urn-generated assignment  $\mu_R$  such that*

$$P_R(E) = P_{\mu_R}(E)$$

*for every decidable event  $E : X \rightarrow \text{Bool}$ .*

*Consequently, after choosing a common denominator, finite rational records on finite event types may be treated as urn presentations without changing any event probabilities.*

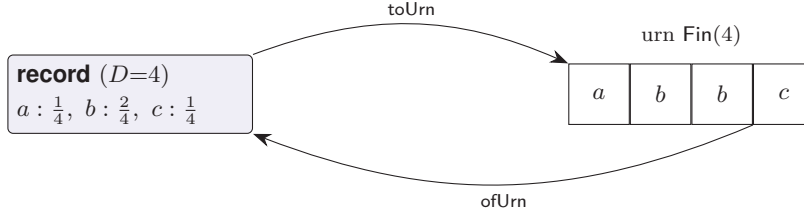


Figure B.2: Record and urn presentations of a finite rational distribution agree on every decidable event (Proposition B.4.5); the maps `toUrn/ofUrn` are the computable witnesses checked in Lean.

*Proof.* For the first direction, let  $\mu = (N, h, f)$  be an urn-generated assignment. Form the weighted record whose denominator is  $N$  and whose weighted list is

$$[(f(i), 1) \mid i : \text{Fin}(N)].$$

For a decidable event  $E : X \rightarrow \text{Bool}$ , the record numerator is the number of urn cells  $i$  for which  $E(f(i)) = \text{true}$ . This is exactly the numerator in the definition of  $P_\mu(E)$ .

For the second direction, let  $R = (a, D, h, d)$  be a weighted common-denominator record, where

$$a = [(x_1, m_1), \dots, (x_r, m_r)] \quad \text{and} \quad D : \mathbb{N} \quad \text{with} \quad \sum_j m_j = D.$$

The proof  $d : D > 0$  supplies the positivity side condition for the urn denominator. Construct an urn with  $D$  cells by expanding each weighted atom  $(x_j, m_j)$  into  $m_j$  cells labelled  $x_j$ . For any decidable event  $E$ , the number of expanded cells satisfying  $E$  is

$$\sum_{j, E(x_j)=\text{true}} m_j,$$

which is precisely the numerator of  $P_R(E)$ . The denominators are both  $D$ , so the probabilities agree for every decidable event.

The equality is extensional rather than literal equality of data structures: permuting cells, splitting cells into repeated equal labels or choosing a different but equivalent support list may produce a different urn or record with the same event probabilities. The Lean formalisation proves these two directions as `FiniteProbRecord.urn_to_record_extensional` and `FiniteProbRecord.record_to_urn_extensional`. ■

**Theorem B.4.6** (Finite Kolmogorov laws). *Let  $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$ , and let  $E, F : X \rightarrow \text{Bool}$  be decidable events. Then:*

B.4.6.1.  $P_\mu(E) \geq 0$ ;

B.4.6.2.  $P_\mu(\top_X) = 1$ , where  $\top_X(x) := \text{true}$  is the total event;

B.4.6.3. if  $E$  and  $F$  are disjoint, then

$$P_\mu(E \cup F) = P_\mu(E) + P_\mu(F).$$

## B. Proofs and arguments

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*More generally, every finite pairwise disjoint family of decidable events is finitely additive.*

*Proof.* For a decidable event  $E$ , write

$$\chi_E(i) := \mathbb{1}_{E(f(i))}$$

for the indicator of  $E$  after pulling it back along  $f : \text{Fin}(N) \rightarrow X$ . By Definition A.3.4,

$$P_\mu(E) = \frac{1}{N} \sum_{i:\text{Fin}(N)} \chi_E(i).$$

Each  $\chi_E(i)$  is either 0 or 1, so the finite sum is non-negative. Since  $N > 0$ , division by  $N$  preserves non-negativity. This proves the first law.

For the total event  $\top_X$ , every indicator is 1. Hence

$$P_\mu(\top_X) = \frac{1}{N} \sum_{i:\text{Fin}(N)} 1 = \frac{N}{N} = 1.$$

This proves normalisation.

If  $E$  and  $F$  are disjoint, then for each  $i : \text{Fin}(N)$  one has

$$\chi_{E \cup F}(i) = \chi_E(i) + \chi_F(i),$$

because at most one of  $E(f(i))$  and  $F(f(i))$  can hold. Finite summation distributes over addition, so

$$\begin{aligned} P_\mu(E \cup F) &= \frac{1}{N} \sum_i (\chi_E(i) + \chi_F(i)) \\ &= \frac{1}{N} \sum_i \chi_E(i) + \frac{1}{N} \sum_i \chi_F(i) \\ &= P_\mu(E) + P_\mu(F). \end{aligned}$$

The finite-family version follows by induction on the length of the family. This is the finite form of Kolmogorov additivity; countable additivity has not been assumed and is not derived here.  $\blacksquare$

**Corollary B.4.7** (Probability monotonicity). *Let  $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$ , and let  $E, F : X \rightarrow \text{Bool}$  be decidable events with  $E \subseteq F$ , that is,  $E(x) = \text{true}$  implies  $F(x) = \text{true}$  for every  $x : X$ . Then*

$$P_\mu(E) \leq P_\mu(F).$$

*Proof.* Define the decidable event  $G(x) := F(x) \wedge \neg E(x)$ , which is decidable because  $E$  and  $F$  are. By construction,  $E$  and  $G$  are disjoint, and the union  $E \cup G$  coincides with  $F$  pointwise: any  $x$  with  $F(x) = \text{true}$  either satisfies  $E(x) = \text{true}$ , lying in  $E$ , or satisfies  $E(x) = \text{false}$ , lying in  $G$ ; in either case it lies in  $E \cup G$ . Conversely, any  $x$  in  $E \cup G$  satisfies  $F(x) = \text{true}$ , since  $E \subseteq F$  and  $G \subseteq F$ . Disjoint additivity from Theorem B.4.6 therefore gives

$$P_\mu(F) = P_\mu(E) + P_\mu(G).$$

Non-negativity from Theorem B.4.6 gives  $P_\mu(G) \geq 0$ , so  $P_\mu(F) \geq P_\mu(E)$ .  $\blacksquare$

**Corollary B.4.8** (Two-event inclusion–exclusion). *Let  $\mu = (N, h, f) : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$ , and let  $E, F : X \rightarrow \text{Bool}$  be decidable events. Then*

$$P_{\mu}(E \cup F) + P_{\mu}(E \cap F) = P_{\mu}(E) + P_{\mu}(F).$$

*Proof.* Decompose  $E$  and  $F$  into pieces relative to  $E \cap F$ . Define decidable events

$$E^{\text{only}}(x) := E(x) \wedge \neg F(x), \quad F^{\text{only}}(x) := F(x) \wedge \neg E(x).$$

Then  $E^{\text{only}}$ ,  $F^{\text{only}}$  and  $E \cap F$  are pairwise disjoint, and

$$E \equiv E^{\text{only}} \cup (E \cap F), \quad F \equiv F^{\text{only}} \cup (E \cap F), \quad E \cup F \equiv E^{\text{only}} \cup F^{\text{only}} \cup (E \cap F),$$

where  $\equiv$  is pointwise equality of decidable predicates. Disjoint additivity from Theorem B.4.6 therefore gives

$$P_{\mu}(E) + P_{\mu}(F) = P_{\mu}(E^{\text{only}}) + P_{\mu}(F^{\text{only}}) + 2 P_{\mu}(E \cap F),$$

and

$$P_{\mu}(E \cup F) + P_{\mu}(E \cap F) = P_{\mu}(E^{\text{only}}) + P_{\mu}(F^{\text{only}}) + 2 P_{\mu}(E \cap F).$$

The two right-hand sides agree, so the two left-hand sides agree. The corresponding Lean equality is `UrnProb.inclusion_exclusion` in `lean/Thesis/Probability.lean`. ■

**Corollary B.4.9** (Resolution-status probabilities). *Let  $S$  be a finite event type, let  $\mu : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(S)$ , and let  $\iota_A : S \rightarrow \text{Status}_A$  be a finite inspection procedure. Then*

$$P_{\mu}(\text{Prov}_A) + P_{\mu}(\text{Ref}_A) + P_{\mu}(\text{Unr}_A) = 1.$$

*Proof.* By Proposition B.4.1, the three resolution-status events are pairwise disjoint and jointly exhaustive. Finite additivity from Theorem B.4.6 gives

$$P_{\mu}(\text{Prov}_A \cup \text{Ref}_A \cup \text{Unr}_A) = P_{\mu}(\text{Prov}_A) + P_{\mu}(\text{Ref}_A) + P_{\mu}(\text{Unr}_A).$$

The union is the total event  $\top_S$ , whose probability is 1 by normalisation. ■

**Theorem B.4.10** (Bayes’ theorem for decidable finite events). *Let  $\mu : \text{Prob}_{\mathbb{Q}}^{\text{urn}}(X)$ , and let  $E, F : X \rightarrow \text{Bool}$  be decidable events with  $P_{\mu}(E) > 0$  and  $P_{\mu}(F) > 0$ . With conditional probability defined by*

$$P_{\mu}(E \mid F) := \frac{P_{\mu}(E \cap F)}{P_{\mu}(F)},$$

*one has*

$$P_{\mu}(E \mid F) = \frac{P_{\mu}(F \mid E) P_{\mu}(E)}{P_{\mu}(F)}.$$

*Proof.* By the definition of conditional probability,

$$P_{\mu}(E \mid F) P_{\mu}(F) = P_{\mu}(E \cap F).$$

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Since intersection is commutative,

$$P_\mu(E \cap F) = P_\mu(F \cap E).$$

Using the definition of conditional probability again, and using  $P_\mu(E) > 0$ , gives

$$P_\mu(F \cap E) = P_\mu(F \mid E)P_\mu(E).$$

Combining these equalities yields

$$P_\mu(E \mid F)P_\mu(F) = P_\mu(F \mid E)P_\mu(E).$$

Since  $P_\mu(F) > 0$ , division by  $P_\mu(F)$  gives the desired formula.  $\blacksquare$

**Proposition B.4.11** (Bayesian update is conditional probability). *Let  $\mu = (p, h) : \text{Prob}_\mathbb{Q}(\Omega)$  and let  $E \subseteq \Omega$  be a decidable event with  $P_\mu(E) > 0$ . Let  $\mu_E$  be the Bayesian-updated probability assignment of Definition A.3.7, with mass function  $p_E(\omega) := \mathbb{1}_E(\omega)p(\omega)/P_\mu(E)$ . Then for every decidable event  $F \subseteq \Omega$ ,*

$$P_{\mu_E}(F) = P_\mu(F \mid E).$$

*Proof.* Unfolding the definition of  $p_E$  and using finite linearity of summation,

$$P_{\mu_E}(F) = \sum_{\omega:\Omega} \mathbb{1}_F(\omega)p_E(\omega) = \sum_{\omega:\Omega} \mathbb{1}_F(\omega) \frac{\mathbb{1}_E(\omega)p(\omega)}{P_\mu(E)} = \frac{1}{P_\mu(E)} \sum_{\omega:\Omega} \mathbb{1}_F(\omega)\mathbb{1}_E(\omega)p(\omega).$$

For each  $\omega : \Omega$ ,  $\mathbb{1}_F(\omega)\mathbb{1}_E(\omega) = \mathbb{1}_{F \cap E}(\omega)$  because both indicators are Boolean and their product is the indicator of the conjunction. Hence

$$P_{\mu_E}(F) = \frac{1}{P_\mu(E)} \sum_{\omega:\Omega} \mathbb{1}_{F \cap E}(\omega)p(\omega) = \frac{P_\mu(F \cap E)}{P_\mu(E)} = P_\mu(F \mid E),$$

the last equality being the definition of conditional probability from Definition A.3.7. The corresponding Lean equality is `UrnProb.bayesian_update_consistency` in `lean/Thesis/Probability.lean`, which checks the equality at the urn level via cross multiplication of `QProb` records.  $\blacksquare$

**Proposition B.4.12** (Bayesian update is a probability assignment). *Let  $\mu = (p, h) : \text{Prob}_\mathbb{Q}(\Omega)$  and let  $E \subseteq \Omega$  be a decidable event with  $P_\mu(E) > 0$ . The updated mass function*

$$p_E(\omega) := \frac{\mathbb{1}_E(\omega)p(\omega)}{P_\mu(E)}$$

*is non-negative and normalised:*

$$\sum_{\omega:\Omega} p_E(\omega) = 1.$$

*Hence  $\mu_E$  is again an inhabitant of  $\text{Prob}_\mathbb{Q}(\Omega)$ .*

*Proof.* Non-negativity follows pointwise:  $\mathbb{1}_E(\omega)$  is either 0 or 1,  $p(\omega) \geq 0$  by the definition of  $\text{Prob}_\mathbb{Q}(\Omega)$ , and the denominator  $P_\mu(E)$  is positive by assumption.

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For normalisation, compute by finite linearity:

$$\sum_{\omega:\Omega} p_E(\omega) = \sum_{\omega:\Omega} \frac{\mathbb{1}_E(\omega)p(\omega)}{P_\mu(E)} = \frac{1}{P_\mu(E)} \sum_{\omega:\Omega} \mathbb{1}_E(\omega)p(\omega).$$

The final sum is exactly  $P_\mu(E)$  by definition of event probability. Since  $P_\mu(E) > 0$ ,

$$\frac{1}{P_\mu(E)} P_\mu(E) = 1.$$

The corresponding urn-level normalisation check is formalised as `UrnProb.bayesian_update_normalization` in `lean/Thesis/Probability.lean`. ■

The theorem applies immediately to the resolution-status events  $\text{Prov}_A$ ,  $\text{Ref}_A$  and  $\text{Unr}_A$ , provided the conditioning event has positive probability. It does not turn an arbitrary proposition  $A$  into a decidable proposition. In particular, there is no general constructive theorem

$$P(\neg A \mid I) = 1 - P(A \mid I).$$

What is available is the finite resolution-status decomposition:

$$P(\text{Prov}_A \mid B) + P(\text{Ref}_A \mid B) + P(\text{Unr}_A \mid B) = 1$$

whenever  $B$  is a decidable conditioning event of positive probability. Thus the probability of proof and refutation may sum to less than 1; the remaining mass is exactly  $P(\text{Unr}_A \mid B)$ , the probability assigned to unresolved cases under the chosen finite inspection procedure.

## B.5 Structural-causal-model equivalence proof

The purpose of this appendix is not to reprove Pearl’s identification theory. It is to prove the bridge needed for importing that theory into the constructive setting used in this thesis. The fragment is deliberately narrow: finite, discrete, recursive structural causal models with finite rational probability distributions. Recursive means acyclic: the endogenous variables can be ordered so that each structural equation depends only on earlier endogenous variables and on exogenous noise.

For the completeness transfer below, the relevant classical fragment is the standard finite recursive semi-Markovian fragment: observed endogenous variables are finite, the directed part of the graph is acyclic, latent exogenous variables are finite roots with finite rational product distribution, and the observed graph records latent common causes by bidirected edges. The Markovian case is the special case with no bidirected edges. The semi-Markovian representation may be taken in its usual canonical edge-latent form, where bidirected structure records which observed variables share an unobserved finite cause.

I write  $\llbracket - \rrbracket$  for the translation from the classical finite structural-causal-model notation to the Martin-Löf-style encoding. The translation is not meant to change the mathematics; it repackages finite sets as finite types, functions

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as terms, products of value spaces as dependent products, and finite rational distributions as probability types.

Conversely, not every arbitrary type-theoretic object counts as a type-theoretic structural causal model for the correspondence theorem. The type-theoretic side is restricted to the record fragment generated by the same data: finite variable indices, finite value types, finite exogenous value types, finite rational probability records, structural terms typed over the declared parents, a recursive order, and, when latent confounding is represented, a finite incidence relation saying which exogenous components feed which observed variables. This restriction is what makes decoding back to an ordinary finite structural causal model possible.

**Definition B.5.1** (Restricted finite type-theoretic causal-model fragment). The fragment  $\text{SCM}_{\text{fin}}^{\text{tt}}$ , also abbreviated  $\text{FinSCM}$  when the type-theoretic side is clear from context, consists of records with the following data:

B.5.1.1. finite endogenous and exogenous index types, with chosen finite enumerations and decidable equality;

B.5.1.2. for each index, a finite value type, so that endogenous and exogenous assignments are finite dependent products;

B.5.1.3. a rational probability record on exogenous assignments, consisting of a non-negative rational mass function whose finite sum is 1;

B.5.1.4. a decidable declared parent relation on endogenous variables, a recursive order, and a proof that every declared endogenous parent of a variable occurs earlier in that order;

B.5.1.5. for each endogenous variable, a structural term whose context contains exactly the declared endogenous parents and the declared exogenous inputs of that variable;

B.5.1.6. in the semi-Markovian case, a finite decidable incidence relation recording which exogenous components feed which observed structural mechanisms.

The fourth and fifth clauses are the formal version of the ordered-telescope reading: the recursive order is a topological order, and the type-theoretic context for the structural term of  $V_i$  exposes only the permitted earlier endogenous variables, together with the declared exogenous inputs. Cyclic dependencies and dependencies on later endogenous variables are therefore excluded by the typing discipline of this restricted record. Only records of this form are quantified over in the correspondence and completeness transfer. General Martin-Löf-style terms that do not decode to such finite recursive structural-causal-model data are outside the fragment.

**Theorem B.5.2** (Finite recursive causal-model correspondence). *Let  $\text{SCM}_{\text{fin}}^{\text{cl}}$  be the classical finite recursive structural-causal-model fragment just described, and let  $\text{SCM}_{\text{fin}}^{\text{tt}}$  be the corresponding type-theoretic record fragment. There are translations*

$$\llbracket - \rrbracket : \text{SCM}_{\text{fin}}^{\text{cl}} \rightarrow \text{SCM}_{\text{fin}}^{\text{tt}}, \quad \text{cl}(-) : \text{SCM}_{\text{fin}}^{\text{tt}} \rightarrow \text{SCM}_{\text{fin}}^{\text{cl}}$$

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such that  $\text{cl}(\llbracket \mathcal{M} \rrbracket)$  is the same classical structural causal model as  $\mathcal{M}$  up to finite renaming of values and variables, and  $\llbracket \text{cl}(\mathcal{N}) \rrbracket$  is the same type-theoretic record as  $\mathcal{N}$  up to the corresponding finite equivalences. Under these equivalences, the following data and operations are preserved and reflected:

B.5.2.1. *finite variables and their value spaces;*

B.5.2.2. *exogenous assignments and their finite rational probability distribution;*

B.5.2.3. *structural functions and their recursive evaluation order;*

B.5.2.4. *observational distributions induced by the model;*

B.5.2.5. *interventions and the resulting interventional distributions;*

B.5.2.6. *the induced directed graph and the semi-Markovian bidirected structure.*

*Proof.* The proof has two parts: construction of the two translations, and verification that the operations commute with them.

Let

$$\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$$

be a finite recursive classical structural causal model. Since the model is recursive, fix a topological ordering

$$V_1, \dots, V_n$$

of the endogenous variables. For each exogenous variable  $U$  and endogenous variable  $V_i$ , the classical finite value set is represented by a finite type together with an enumeration. Thus

$$\llbracket \text{Val}(U) \rrbracket, \quad \llbracket \text{Val}(V_i) \rrbracket.$$

Joint assignments are represented by finite dependent products:

$$\llbracket \text{Val}(\mathcal{U}) \rrbracket \simeq \prod_{U \in \mathcal{U}} \llbracket \text{Val}(U) \rrbracket, \quad \llbracket \text{Val}(\mathcal{V}) \rrbracket \simeq \prod_{i=1}^n \llbracket \text{Val}(V_i) \rrbracket.$$

Because all domains are finite, these products are again finite types with decidable equality.

The exogenous probability distribution is translated directly:

$$P_{\mathcal{U}} : \text{Val}(\mathcal{U}) \rightarrow \mathbb{Q}_{\geq 0} \quad \mapsto \quad \llbracket P_{\mathcal{U}} \rrbracket : \text{Prob}_{\mathbb{Q}}(\llbracket \text{Val}(\mathcal{U}) \rrbracket).$$

Non-negativity and normalisation are finite equalities transported along the chosen finite equivalence. Hence this is preservation and reflection rather than a new probabilistic construction.

For the structural functions, write the classical equation for  $V_i$  as

$$f_i : \text{Val}(\text{Pa}_{\mathcal{G}}(V_i)) \times \text{Val}(\mathcal{U}_i) \rightarrow \text{Val}(V_i).$$

Under the chosen topological order, all endogenous parents of  $V_i$  occur among  $V_1, \dots, V_{i-1}$ . Hence its translation is a term in the context containing the exogenous assignment and the earlier endogenous values:

$$\begin{aligned} u : \llbracket \text{Val}(\mathcal{U}) \rrbracket, v_1 : \llbracket \text{Val}(V_1) \rrbracket, \dots, v_{i-1} : \llbracket \text{Val}(V_{i-1}) \rrbracket \\ \vdash \llbracket f_i \rrbracket(u, v_1, \dots, v_{i-1}) : \llbracket \text{Val}(V_i) \rrbracket. \end{aligned}$$

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This is exactly the type-theoretic form of recursive evaluation. Given  $u$ , the values  $v_1, \dots, v_n$  are constructed by induction over the topological order. The induction step computes  $v_i$  from  $u$  and the already constructed values  $v_1, \dots, v_{i-1}$ . Therefore, for every exogenous assignment  $u$  and every endogenous variable  $V_i$ , there is a propositional equality between the transported values,

$$\text{sol}_{\llbracket \mathcal{M} \rrbracket}(\llbracket u \rrbracket)_i = \llbracket \text{sol}_{\mathcal{M}}(u)_i \rrbracket.$$

This proves preservation and reflection of pointwise solutions.

The decoding map  $\text{cl}(-)$  is the same construction read in the other direction. Given a type-theoretic structural-causal-model record  $\mathcal{N}$ , its finite types are decoded to ordinary finite value sets by their finite enumerations, its probability record is decoded to a rational mass function, and its structural terms are decoded to total functions on the corresponding finite products. The declared parent relation, recursive order and exogenous-incidence relation are finite decidable relations, so they decode to the usual graph data. Applying  $\text{cl}$  after  $\llbracket - \rrbracket$  returns the original classical model up to the chosen finite names. Applying  $\llbracket - \rrbracket$  after  $\text{cl}$  returns the original type-theoretic record up to the corresponding finite type equivalences. These are the expected isomorphisms rather than judgemental equalities, because finite sets and finite types may use different names and enumerations for the same values.

Observational distributions are pushforwards of the exogenous distribution along the solution map. For any event  $E \subseteq \text{Val}(\mathcal{V})$ ,

$$P_{\mathcal{M}}(E) = \sum_{u: \text{Val}(\mathcal{U})} \mathbb{1}_E(\text{sol}_{\mathcal{M}}(u)) P_{\mathcal{U}}(u).$$

The translated model computes

$$P_{\llbracket \mathcal{M} \rrbracket}(\llbracket E \rrbracket) = \sum_{\llbracket u \rrbracket: \llbracket \text{Val}(\mathcal{U}) \rrbracket} \mathbb{1}_{\llbracket E \rrbracket}(\text{sol}_{\llbracket \mathcal{M} \rrbracket}(\llbracket u \rrbracket)) \llbracket P_{\mathcal{U}}(\llbracket u \rrbracket) \rrbracket.$$

The index type, mass function and solution map are the translations of the classical ones. The equality of the two displayed sums is therefore an instance of Lemma B.5.3. Thus observational distributions are preserved and reflected.

Now let  $\text{do}(X = x)$  be an intervention. Classically,  $\mathcal{M}_x$  is obtained by replacing the structural function for each intervened variable  $X$  by the constant function  $x$ , while leaving all other structural functions unchanged. The type-theoretic intervention performs the same substitution on the corresponding terms. Hence

$$\llbracket \mathcal{M}_x \rrbracket \simeq \llbracket \mathcal{M} \rrbracket_x.$$

Applying the previous pushforward argument to the intervened model gives, for every finite event  $E$ ,

$$P_{\mathcal{M}_x}(E) = P_{\llbracket \mathcal{M} \rrbracket_x}(\llbracket E \rrbracket).$$

Therefore interventional distributions are also preserved and reflected.

Finally, consider the graph structure. Directed edges are part of the declared endogenous parent relation in the model record, and the structural terms are typed over those declared parents. Since the translation changes finite sets into finite types but does not change the declared parent arguments of each structural function, directed edges are preserved and reflected. In the semi-Markovian case, the bidirected edge  $V_i \leftrightarrow V_j$  is present exactly when the finite

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exogenous-incidence relation says that some latent exogenous component feeds both structural mechanisms. This is again a finite decidable relation and is unchanged by the translation. Thus the directed part and the bidirected latent projection commute with encoding and decoding.

The remaining graph operations used by do-calculus and identification are finite combinatorial predicates built from the same directed and bidirected edge relations. Their predicate-by-predicate preservation is derived separately in Proposition B.5.5. This completes the two-sided correspondence proof for the stated finite graph and structural-causal-model data. ■

**Lemma B.5.3** (Finite reindexing for equivalent finite types). *Let  $A$  and  $B$  be finite types with decidable equality, let  $e : A \simeq B$  be a finite equivalence, and let*

$$f : A \rightarrow \mathbb{Q}, \quad g : B \rightarrow \mathbb{Q}$$

*be rational summands such that  $g(e(a)) = f(a)$  for every  $a : A$ . Then*

$$\sum_{a:A} f(a) = \sum_{b:B} g(b).$$

*Proof.* Choose a finite enumeration  $a_1, \dots, a_n$  of  $A$  without repetition. Since  $e$  is an equivalence,  $e(a_1), \dots, e(a_n)$  is a finite enumeration of  $B$  without repetition. Replacing each summand  $g(e(a_i))$  by  $f(a_i)$  gives equality of the two finite sums. This is a propositional equality proved by induction over the finite enumeration, not a judgemental equality; different enumerations give the same result because finite sums over decidable finite types are invariant under permutation. ■

**Proposition B.5.4** (Finite probability-expression transport). *Let  $E$  be an expression built from finite observational or interventional probabilities in the fragment using finite sums, finite products, marginalisation, substitution of interventions, and quotients whose denominators are positive. Then  $E$  is defined in a classical finite structural causal model if and only if  $\llbracket E \rrbracket$  is defined in its type-theoretic encoding, and the two expressions have the same rational value after transport along the finite equivalence of assignments.*

*Proof.* The proof is by induction on the construction of  $E$ . The base cases are observational and interventional probabilities, already covered by Theorem B.5.2. Finite sums and products are preserved by Lemma B.5.3. Marginalisation is a finite sum over a finite fibre, so it is another instance of the same lemma. Substitution of interventions is preserved because  $\llbracket \mathcal{M}_x \rrbracket \simeq \llbracket \mathcal{M} \rrbracket_x$ . For quotients, the denominator is a rational number preserved by the induction hypothesis; positivity is therefore preserved and reflected, so the quotient is defined on one side exactly when it is defined on the other. ■

**Proposition B.5.5** (Finite graph-predicate transport). *Every graph predicate used by identification, conditional identification, hedges and do-calculus in the finite semi-Markovian fragment is preserved and reflected by the encoding. This includes membership, inclusion, parents, ancestors, descendants, graph mutilation, d-separation, c-components, c-forests, root sets and hedge conditions.*

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*Proof.* The encoding preserves and reflects the directed parent relation, the bidirected latent-incidence relation, and the finite node equivalence. The verification then proceeds predicate by predicate.

For membership and inclusion, a finite subset is represented by a decidable predicate or finite list of nodes. Transporting along the node equivalence sends each node to its encoded node and preserves decidable equality. Hence  $X \in S$  and  $S \subseteq T$  hold classically if and only if their encoded versions hold type-theoretically.

For parents, the predicate  $X \in \text{Pa}_{\mathcal{G}}(Y)$  is exactly the directed edge predicate  $X \rightarrow Y$ . Directed edges are declared graph data in the model record and are unchanged by encoding and decoding, so parenthood is preserved and reflected.

For ancestors and descendants,  $X \in \text{An}_{\mathcal{G}}(Y)$  and  $X \in \text{De}_{\mathcal{G}}(Y)$  mean that a finite directed path exists. Encoding maps a directed path

$$X = X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_k = Y$$

to the corresponding encoded directed path, and decoding maps it back. Since the graphs are finite and acyclic, the relevant path search is finite. Thus directed reachability is preserved and reflected.

For graph mutilation,  $\mathcal{G}_{\overline{X}}$  deletes directed arrows into nodes in  $X$ , and  $\mathcal{G}_{\underline{X}}$  deletes directed arrows out of nodes in  $X$ ; in the semi-Markovian case this is read in the canonical finite latent-variable directed graph of Definition A.3.27. Deletion is defined by a finite edge predicate together with finite membership in  $X$ . Both ingredients have already been shown invariant, so mutilation commutes with encoding:

$$\llbracket \mathcal{G}_{\overline{X}} \rrbracket = \llbracket \mathcal{G} \rrbracket_{\llbracket \overline{X} \rrbracket}, \quad \llbracket \mathcal{G}_{\underline{X}} \rrbracket = \llbracket \mathcal{G} \rrbracket_{\llbracket \underline{X} \rrbracket}.$$

For d-separation, first replace a semi-Markovian graph by its canonical finite latent-variable directed acyclic graph as in Definition A.3.18; in the Markovian case this step is trivial. The canonical replacement is finite and is determined by the same bidirected edge relation, so it is preserved by encoding. A path in the canonical directed acyclic graph is mapped to a corresponding encoded path; collider status, non-collider status, membership in the conditioning set and descendant membership are all preserved by the previous clauses. Therefore a path is blocked by  $Z$  if and only if its encoded path is blocked by  $\llbracket Z \rrbracket$ . To turn this into a finite universal check, it suffices to quantify only over *simple* paths — paths in which no node repeats — rather than over all walks. The standard reduction lemma for d-separation [Pea09, §1.2.3] states that, in a finite directed graph, two nodes are d-connected by  $Z$  along a walk if and only if they are d-connected by  $Z$  along a simple path; the proof proceeds by induction on walk length, repeatedly contracting a repeated-vertex segment and using a standard case analysis on the collider status of the join node to argue that each contraction preserves d-connection. Since the directed-and-bidirected graph is finite, the set of simple paths between any two nodes is finite, and the universal statement “every simple path between  $X$  and  $Y$  is blocked by  $Z$ ” is a finite decidable check. Encoding preserves and reflects this finite check, so d-separation is preserved and reflected.

For c-components, two observed nodes are in the same c-component exactly when they are connected by a finite bidirected path. Bidirected paths are preserved by the latent-incidence relation. Maximality of a component is a finite

inclusion condition, already handled above. Thus c-components are preserved and reflected.

For c-forests, the defining predicates are: the observed nodes form one c-component; each observed node has at most one directed child inside the subgraph; and the roots are precisely the nodes with no directed child inside the subgraph. The first clause is c-component preservation, the second is a finite bounded child-count predicate over preserved directed edges, and the third is a finite negated-existence predicate over the same child relation. Hence c-forests and their root sets are preserved and reflected.

Finally, a hedge for  $P_x(y)$  is the conjunction of finite predicates:  $F$  and  $F'$  are c-forests, they have the same roots,  $F' \subset F$ ,  $F$  meets the intervention set,  $F'$  does not meet it, and the common roots are ancestors of  $Y$  in the graph with arrows into  $X$  deleted. Each conjunct has already been shown invariant: c-forests, roots, inclusion, finite intersection, graph mutilation and ancestry. Therefore hedge witnesses are preserved and reflected. ■

**Proposition B.5.6** (Compatible-model class equivalence). *For every finite recursive Markovian or semi-Markovian graph  $G$ , the translations restrict to an equivalence between classical finite structural causal models compatible with  $G$  and restricted type-theoretic structural-causal-model records compatible with  $\llbracket G \rrbracket$ . This equivalence preserves and reflects observational distributions and all ordinary finite interventional queries in the fragment.*

*Proof.* Compatibility with  $G$  in the sense of Definition A.3.29 is preserved and reflected by Theorem B.5.2: the declared directed and bidirected graph data and the finite recursive structure are exactly the data the correspondence preserves. The two composites  $\text{cl}(\llbracket - \rrbracket)$  and  $\llbracket \text{cl}(-) \rrbracket$  return the original compatible model up to finite renaming and finite type equivalence. Observational and interventional distributions are preserved by the same theorem. ■

**Proposition B.5.7** (Modal locality). *If an epistemic causal mode  $m$  contains a restricted finite type-theoretic structural-causal-model record, then the correspondence above holds internally at that mode. Modal transitions between epistemic causal modes do not change the transferred Pearl-style completeness result unless they are equipped with additional structure specifying how model records are transformed.*

*Proof.* The structural-causal-model correspondence is local to the record available at a mode: all variables, value types, structural terms, probability records, graph relations and finite proofs are judged in the context of that mode. The identification, conditional identification and do-calculus operations used here act on that local finite record. A modality can move information between modes only according to the modal structure supplied for that transition. Therefore the ordinary completeness transfer applies at each mode containing a restricted finite model record, while claims about changing from one causal model to another require a separate theorem about the chosen modality. ■

**Definition B.5.8** (Finite epistemic transition layer). The finite transition layer used in the mechanisation consists of the following data. An *epistemic record* is a tuple

$$R = (\mathcal{M}, p, e, \rho, b)$$

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where  $\mathcal{M}$  is a finite recursive structural-causal-model record,  $p$  proves that its exogenous finite distribution is a probability distribution, and  $e$  is a finite bound for exogenous references. A variable reference is either an endogenous reference  $\text{endo}(i)$  with  $i < |\mathcal{M}|$ , or an exogenous reference  $\text{exo}(j)$  with  $j < e$ . The list  $\rho$  is a finite list of relation edges between such references, and  $b$  proves that every edge in  $\rho$  is bounded by the corresponding endogenous and exogenous counts. An *epistemic mode* is a name paired with such a record. A *finite modal transition* is a label

learn, forget, condition, relate, unrelate, set, unset

together with a source mode and a target mode.

The labels are fixed transition names; they do not inspect the source context. The actual work is in the source–target record transformation. Learning appends a fresh last endogenous variable, with all old structural equations unchanged. Forgetting removes the last endogenous variable and requires a proof that all remaining relation edges are still bounded after the removal. Conditioning replaces the exogenous distribution by a posterior finite probability distribution and leaves the structural equations unchanged. The posterior may be supplied explicitly with a proof of normalisation, or it may be constructed from a decidable evidence event by finite Bayesian update. Setting replaces one endogenous structural equation by a constant value. Unsetting restores a stored old structural equation. Relating inserts a bounded relation edge between any two variable references, endogenous or exogenous. Unrelating erases the inserted edge.

**Proposition B.5.9** (Finite modal transition preservation). *Each generating transition of Definition B.5.8 sends a well-formed epistemic record to a well-formed epistemic record, provided the explicit side conditions of the transition are supplied: a learned variable is appended at the end, a forgotten variable is unused by the remaining relation data, a conditioning step is given either a posterior finite probability distribution or a positive-probability evidence event from which such a posterior is constructed, a related edge is bounded by the current endogenous and exogenous reference counts, and an unset operation is given the structural equation it restores.*

*Proof.* Learning, forgetting, relating, unrelating, setting and unsetting preserve the exogenous distribution, so the old probability proof  $p$  is reused for those cases. Conditioning is the one exception: it replaces the exogenous distribution by the supplied posterior and uses the supplied proof that this posterior is normalised. Learning increases the endogenous variable count from  $n$  to  $n + 1$  and leaves the exogenous reference count  $e$  unchanged, hence every previously bounded relation edge remains bounded. Forgetting has the boundedness proof as an explicit premise: all edges must already be bounded by  $n - 1$  and  $e$ , which says that the removed last endogenous variable is not mentioned in the remaining relation data. Setting, unsetting and conditioning leave the endogenous count, exogenous reference count and relation list unchanged. Relating requires a proof that the inserted edge is bounded by both counts; all old edges retain their existing proof. Unrelating removes an edge from a finite list, so any edge still present was already present before removal and keeps its previous boundedness proof. These

are exactly the preservation lemmas in `lean/Thesis/Modalities.lean`:

```
learnFresh_preserves_wellFormed,
forgetLast_preserves_wellFormed,
conditionNoise_preserves_wellFormed,
setVariable_preserves_wellFormed,
unsetVariable_preserves_wellFormed,
relate_preserves_wellFormed,
unrelate_preserves_wellFormed.
```

■

**Proposition B.5.10** (Bayesian conditioning as a modal transition). *Let  $R = (\mathcal{M}, p, e, \rho, b)$  be a well-formed epistemic record, and let  $E$  be a decidable event on exogenous assignments with  $P_{\mathcal{M}}(E) > 0$ . Define the posterior exogenous distribution by*

$$p_E(u) := \begin{cases} p(u)/P_{\mathcal{M}}(E), & E(u), \\ 0, & \text{otherwise.} \end{cases}$$

*Then  $p_E$  is a probability distribution, and the conditioning transition that replaces  $p$  by  $p_E$  is well formed. Moreover, for every decidable event  $F$  on exogenous assignments,*

$$P_{p_E}(F) = \frac{P_{\mathcal{M}}(E \cap F)}{P_{\mathcal{M}}(E)}.$$

*If  $O$  is a decidable event on evaluated endogenous assignments, the same construction applies to the lifted exogenous event  $E_O(u) := O(\mathcal{M}(u))$  whenever  $P_{\mathcal{M}}(E_O) > 0$ .*

*Proof.* Normalisation is the finite Bayesian-update calculation of Proposition B.4.12, applied to the finite exogenous distribution. Non-negativity is preserved because the original masses are non-negative and the denominator  $P_{\mathcal{M}}(E)$  is positive. The displayed equation is finite conditional probability: summing  $p_E$  over  $F$  sums exactly the original mass over  $E \cap F$ , divided by  $P_{\mathcal{M}}(E)$ . In Lean, the checked posterior facts are

```
FinDist.bayesPosterior,
FinDist.bayesPosterior_isProbability,
FinDist.bayesPosterior_probOf
```

in `lean/Thesis/SCM.lean`. The corresponding modal record operations and facts in `lean/Thesis/Modalities.lean` are

```
conditionOnExogenousEvidence,
conditionOnObservation,
conditionOnExogenousEvidence_preserves_wellFormed,
conditionOnObservation_preserves_wellFormed,
conditionOnExogenousEvidence_probOf.
```

■

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**Proposition B.5.11** (Finite modal cancellation checks). *The basic inverse-looking transition pairs cancel at the finite record level:*

$$\text{forget}(\text{learn}(R)) = R, \quad \text{unset}(\text{set}(R)) = R, \quad \text{unrelate}(\text{relate}(R)) = R,$$

*with the intended side conditions: learning appends a fresh last variable, forgetting removes that same last variable, unsetting is supplied with the stored old structural equation, and unrelating erases the inserted relation edge.*

*Proof.* For learning followed by forgetting, the learned variable is appended as the last variable. In a recursive finite SCM, earlier structural equations cannot depend on a later variable; the Lean lemma `learnFreshModel_old_fn` records that the old equations are unchanged when viewed inside the extended model. Removing the last variable therefore recovers exactly the original model, and the relation boundedness proof reduces to the original boundedness proof. This is formalised as `forgetLast_learnFresh_cancel`.

For setting followed by unsetting, setting changes only the selected structural equation. If the old equation has been stored as finite data, unsetting puts that equation back and leaves all other equations untouched. This is `unset_set_cancel`. For relating followed by unrelating, the relation list is extended by one edge and then that inserted occurrence is erased, giving the previous list back. This is `unrelate_relate_cancel`. Conditioning has no such cancellation law in general: replacing a prior by a posterior loses information unless the prior and update evidence are stored separately. These cancellation facts are not consequences of Gratzner’s modal rules alone; they are the record-level equations that justify treating the corresponding finite transitions as inverse-looking sanity checks. ■

*Remark B.5.12* (Twin networks as finite composites). The transition layer therefore does not contain a primitive modality for “adding a twin network”. A twin network is too specific: it is a particular construction using ordinary record changes. In the finite recursive fragment it can be represented as a composite of existing transitions. First, when there is evidence about the actual unit, a `condition` transition replaces the exogenous distribution by the posterior distribution required by the abduction step. Second, repeated `learn` transitions append the copied endogenous variables and copied structural equations in topological order. Third, `relate` records the intended sharing: copied and actual variables may be related to the same exogenous references, and actual/counterfactual endogenous counterparts may also be related. Fourth, a `set` transition imposes the counterfactual intervention in the copied sector. Prediction is then ordinary evaluation of the resulting finite record.

This is enough to express the construction at the record level, but it is not yet a full counterfactual completeness theorem. Such a theorem would additionally have to prove that the composite just described is equivalent to Pearl’s twin-network or multi-world semantics for the class of counterfactual queries under discussion. That correspondence is deliberately left outside the transfer theorem of Theorem B.6.13.

A representative Lean 4 formalisation of the computational core of this correspondence is included in the modules `Thesis.SCM` and `Thesis.Modalities`, in `lean/Thesis/SCM.lean` and `lean/Thesis/Modalities.lean`. Their role

is deliberately narrow. They check the shared finite record, the encode/decode skeleton, recursive evaluation, intervention, observational and interventional probability, the model-level counterfactual probability helper, the finite modal transition layer above, and an abstract completeness-transport lemma. The modal module also checks the same tenure-track counterfactual through the explicit record composite `conditionOnObservationThenSet`, and proves that this composite agrees with the model-level counterfactual helper for a single set action. The SCM module also checks the local binary conditional-probability-table row functionalisation used by the finite causal-Bayesian-network argument: a finite uniform tagged seed type with prescribed true and false cell counts realises the corresponding rational row.

It does not mechanise Pearl’s graph theory. D-separation, c-components, c-forests, hedges, hedge countermodels, identification, conditional identification, arbitrary finite causal-Bayesian-network functionalisation and semi-Markovian projection remain ordinary mathematical arguments in this appendix. Put plainly, this is where the Lean check stops and the appendix proof takes over. In particular, the encode and decode maps in `Thesis.SCM` are the identity on the shared finite record, so the mechanised correspondence lemmas check that the classical and type-theoretic presentations carry the same finite data by construction; they are not an independent mechanical verification of Pearl’s identification theory, which remains the cited external result.

**Corollary B.5.13** (Interventional-query adequacy). *For every finite ordinary interventional query  $Q = P_x(y)$  in the recursive structural-causal-model fragment, where  $P_x(y)$  abbreviates  $P(Y = y \mid \text{do}(X = x))$  or the corresponding finite joint event,*

$$P_{\mathcal{M}_x}(Y = y) = q \iff P_{\llbracket \mathcal{M} \rrbracket_x}(\llbracket Y = y \rrbracket) = q.$$

*Moreover, two classical models in the fragment agree on their finite observational distribution if and only if their type-theoretic encodings agree on the encoded observational distribution.*

*Proof.* The first claim is the interventional-distribution part of Theorem B.5.2 applied to the singleton event or finite event that defines  $y$ . The second claim follows from the observational-distribution part of the same theorem, since observational distributions are finite pushforwards of the same exogenous distribution through corresponding solution maps. ■

**Proposition B.5.14** (Identifiability is transported). *Let  $G$  be a finite recursive causal graph in the Markovian or semi-Markovian representation used above, and let  $Q = P_x(y)$  be an ordinary finite interventional query. Then  $Q$  is identifiable in the classical finite structural-causal-model fragment over  $G$  if and only if  $\llbracket Q \rrbracket$  is identifiable in the type-theoretic record fragment over  $\llbracket G \rrbracket$ .*

*Proof.* Identifiability says that any two compatible models with the same observational distribution give the same value to the target interventional query. By Proposition B.5.6, the classical compatible-model class over  $G$  and the restricted type-theoretic compatible-record class over  $\llbracket G \rrbracket$  are equivalent. Observational equality and interventional-query values are preserved and reflected by Theorem B.5.2. Thus a classical counterexample to identifiability translates into a type-theoretic counterexample. Conversely,

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because the type-theoretic quantification is restricted to the record fragment decoded by  $\text{cl}(-)$ , a type-theoretic counterexample decodes into a classical counterexample. Equivalently, absence of such counterexamples is preserved in both directions. ■

**Proposition B.5.15** (Do-calculus derivations are transported). *Any finite classical do-calculus derivation whose expressions, graph operations and probability manipulations lie in the fragment covered by Theorem B.5.2 and Proposition B.5.5 translates into a type-theoretic derivation of the encoded equality of expressions.*

*Proof.* By Definition A.3.34, a derivation is a finite sequence of reflexivity, model-evaluation, probability and do-calculus steps. Reflexivity steps are preserved by definitional equality of encoded expressions. Each model-evaluation step rewrites  $P_{\mathcal{M}_x}(y)$  to a finite rational expression in the exogenous distribution and structural functions of  $\mathcal{M}_x$ ; this is licensed by Theorem B.5.2 and so is preserved by encoding. Each probability step is an equality of finite sums, finite products, marginalisations or conditionings with non-zero denominator; these are transported by Proposition B.5.4. Each do-calculus step is licensed by a d-separation claim in a mutilated graph; these graph conditions are transported by Proposition B.5.5. Therefore each step in the classical derivation has an encoded type-theoretic step. Induction on the length of the derivation gives the translated derivation. ■

### B.6 Causality-completeness proof

The completeness theorem used here is an external mathematical result. In the finite recursive semi-Markovian setting, the Shpitser–Pearl identification algorithm is sound and complete for ordinary joint interventional distributions, and the rules of do-calculus together with ordinary probability manipulations are complete for identifying such effects [HV06; Pea12; SP06; TS09]. This thesis does not rederive that algorithmic proof. It proves that the result applies to the restricted type-theoretic structural-causal-model record fragment by transport along the two-sided correspondence of Theorem B.5.2.

The theorem-by-theorem scope is as follows. From Shpitser–Pearl, the transferred fragment includes termination and soundness of the finite identification procedure, the hedge failure criterion, completeness of identification for effects of the form  $P_x(y)$ , and completeness of do-calculus with standard probability manipulations for those unconditional effects [SP06]. From Huang–Valtorta, it includes the finite soundness and completeness of the identification algorithm for ordinary causal effects  $P_T(S)$  and the resulting completeness of Pearl’s three do-calculus rules for such effects, after restricting to finite models and applying the finite functionalisation and semi-Markovian correspondence below [HV06]. From Tian–Shpitser, it includes the reviewed completeness result for the unconditional identification of effects and, with the positivity side condition made explicit, the completeness result for the conditional identification of effects  $P_x(y \mid z)$  [TS09]. The counterfactual sections of Tian–Shpitser are not covered here; they require a separate correspondence theorem for counterfactual graphs and multi-world assignments. The route through the transfer is summarised in Table B.2.

## B.6. Causality-completeness proof

| Stage | Result                         | Role in the transfer                                                                                                                        | Depends on                                       | Used in                                 |
|-------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------|
| 1     | External finite completeness   | Fixes the classical result that is imported rather than reproved.                                                                           | Shpitser–Pearl, Huang–Valtorta and Tian–Shpitser | Final transfer theorem                  |
| 2     | Model adequacy                 | Shows that the restricted records and ordinary finite recursive SCMs carry the same variables, mechanisms, interventions and probabilities. | B.5.2                                            | later transport steps                   |
| 3     | Expression and graph transport | Moves finite probability expressions, graph predicates, identifiability claims and do-calculus derivations across the bridge.               | B.5.4, B.5.5, B.5.14, B.5.15                     | do-calculus and identification transfer |
| 4     | Finite negative cases          | Keeps hedge failures inside the finite rational fragment by using finite binary countermodels.                                              | B.6.2                                            | B.6.12                                  |
| 5     | Conditional effects            | Carries conditional identification once the positive-denominator side condition is explicit.                                                | B.6.6, B.6.7                                     | B.6.13                                  |
| 6     | Bayesian-network reduction     | Represents the finite causal-Bayesian-network and latent-projection steps needed for the Huang–Valtorta route.                              | B.6.8, B.6.10                                    | B.6.13                                  |

Table B.2: Route through the causality-completeness transfer.

**Lemma B.6.1** (Disjoint exogenous parents give independence). *Let  $\mathcal{M} = (\mathcal{U}, \mathcal{V}, \mathcal{F}, P_{\mathcal{U}})$  be a finite recursive structural causal model in the sense of Definition A.3.25 whose exogenous distribution factorises as a finite product*

$$P_{\mathcal{U}}(u_1, \dots, u_k) = \prod_{j=1}^k P_{U_j}(u_j),$$

*with each  $P_{U_j} : \text{Prob}_{\mathbb{Q}}(\text{Val}(U_j))$ . Suppose that  $X, Y \in \mathcal{V}$  are endogenous variables and that the finite sets of exogenous variables that functionally feed into the recursive evaluation of  $X$  and  $Y$  are disjoint:*

$$\mathcal{U}_{\text{anc}}(X) \cap \mathcal{U}_{\text{anc}}(Y) = \emptyset.$$

*Here  $\mathcal{U}_{\text{anc}}(V)$  denotes the finite set of exogenous variables that occur in the recursive expansion of the structural equations along the ancestry of  $V$  in the induced graph. Then  $X$  and  $Y$  are independent under the observational distribution induced by  $\mathcal{M}$ , in the sense of Definition A.3.10: for every pair of decidable events  $A$  on  $\text{Val}(X)$  and  $B$  on  $\text{Val}(Y)$ ,*

$$P_{\mathcal{M}}(A(X) \cap B(Y)) = P_{\mathcal{M}}(A(X)) \cdot P_{\mathcal{M}}(B(Y)).$$

*Proof.* Because the model is recursive, the values of  $X$  and  $Y$  under exogenous assignment  $u : \text{Val}(\mathcal{U})$  depend only on the components of  $u$  in  $\mathcal{U}_{\text{anc}}(X)$  and

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$\mathcal{U}_{\text{anc}}(Y)$  respectively. Write  $u = (u_X, u_Y, u_R)$  for the partition into the  $X$ -ancestral block, the  $Y$ -ancestral block, and the remainder. The disjointness assumption guarantees that this partition is well defined. Since  $P_{\mathcal{U}}$  is a finite product distribution, finite Fubini for finite rational sums gives

$$P_{\mathcal{U}}(u_X, u_Y, u_R) = P_{\mathcal{U}_{\text{anc}}(X)}(u_X) \cdot P_{\mathcal{U}_{\text{anc}}(Y)}(u_Y) \cdot P_{\mathcal{U}_R}(u_R),$$

with each factor a finite rational marginal of  $P_{\mathcal{U}}$ .

Let  $\text{val}_X(u_X)$  and  $\text{val}_Y(u_Y)$  denote the values of  $X$  and  $Y$  obtained by recursive evaluation; by the recursive structure these depend only on their respective blocks. Then

$$\begin{aligned} P_{\mathcal{M}}(A(X) \cap B(Y)) &= \sum_u \mathbb{1}_A(\text{val}_X(u_X)) \mathbb{1}_B(\text{val}_Y(u_Y)) P_{\mathcal{U}}(u) \\ &= \left( \sum_{u_X} \mathbb{1}_A(\text{val}_X(u_X)) P_{\mathcal{U}_{\text{anc}}(X)}(u_X) \right) \\ &\quad \cdot \left( \sum_{u_Y} \mathbb{1}_B(\text{val}_Y(u_Y)) P_{\mathcal{U}_{\text{anc}}(Y)}(u_Y) \right) \cdot 1 \\ &= P_{\mathcal{M}}(A(X)) \cdot P_{\mathcal{M}}(B(Y)), \end{aligned}$$

where the second equality is finite distributivity of multiplication over the three independent finite sums and the third equality marginalises the  $u_R$  block (which sums to 1) and recognises the remaining two finite sums as the marginal probabilities of the events  $A(X)$  and  $B(Y)$ . The factorisation extends from pairs to arbitrary finite families of variables with pairwise disjoint exogenous-ancestor sets by an analogous finite Fubini step.  $\blacksquare$

**Lemma B.6.2** (Finite rational hedge countermodels). *Let a finite hedge witness non-identifiability in the Shpitser–Pearl semi-Markovian setting. Then the separating countermodels used for the hedge can be chosen inside the finite rational recursive structural-causal-model fragment: the variables on the hedge are binary, the exogenous variables are independent fair coins, and the structural functions are deterministic parity functions. If the ambient value types are larger than binary, the construction applies after choosing two distinct values for each variable used in the hedge and making the structural functions land only in those chosen values. Thus this support is a finite-value claim for binary value types, or for value types with two distinguished values; singleton value types do not carry the usual hedge countermodel.*

*Proof.* This lemma extracts the finite-rational content of the Shpitser–Pearl hedge countermodel construction [SP06]. Let  $F' \subset F$  be the hedge, with common root set  $R$ . The published proof first reduces the separating query to a root query on an  $R$ -rooted c-forest. On that c-forest, take every observed variable to have value type  $\{0, 1\}$ . For each bidirected latent connection used in the forest, use a binary exogenous root  $U_e : \{0, 1\}$  with the fair-coin distribution, and take all such roots to be independent.

The two countermodels have the same finite exogenous variables and the same exogenous distribution. In the first model, every observed node in  $F$  is the bit parity, i.e. the sum modulo 2, of its observed parents in  $F$  and its incident binary latent parents. In the second model, the same parity equations are used

except that the nodes in  $F'$  disregard parent values contributed by the grown part  $F \setminus F'$ . Outside the hedge, the models are extended in the same way on both sides, so any separation produced on the hedge is preserved in the ambient graph.

The parity argument is the finite counting step in the Shpitser–Pearl construction. In an  $R$ -rooted c-forest, each latent bit in the relevant forest has exactly two downward paths to the root set  $R$ . Hence, before intervention, its contribution to the root parity is counted twice and cancels modulo 2. Thus in the first model the xor of the root values counts each latent bit from  $F$  twice, while in the second model the xor of the root values counts each latent bit from  $F'$  twice. In both cases the possible observed assignments are therefore constrained to have even root parity.

The multiplicities are also the same. If two latent assignments differ, let  $U^*$  be the finite set of latent bits on which they differ. In the c-forest construction there is a topmost observed node  $W$  with an odd number of parents in  $U^*$ ; flipping the bits in  $U^*$  flips the value of  $W$ . Hence the map from latent assignments to observed assignments is injective for the relevant c-forest in each of the two models. Since the binary c-forest construction has  $|U| + 1 = |V|$ , the number of latent assignments equals the number of observed assignments with even root parity. Therefore every even-root-parity observed assignment occurs with the same fair-coin mass in the corresponding forest, and every odd-root-parity assignment has probability zero. The two models agree on the functions and distributions outside the modified part of the hedge, so summing over the same finite fibres gives the same observational distribution on the observed variables.

The hedge condition gives  $F \cap X \neq \emptyset$  and  $F' \cap X = \emptyset$ . After the intervention  $\text{do}(x)$ , the incoming arrows into the intervened variables remove part of the grown structure  $F \setminus F'$ . The double-counting parity cancellation is then broken in one model but not in the other. Consequently there is a root assignment  $r$  such that

$$P_{\mathcal{M}_1, x}(R = r) \neq P_{\mathcal{M}_2, x}(R = r).$$

Since the roots lie among the ancestors of  $Y$  in the graph with arrows into  $X$  deleted, the standard extension step makes downstream variables copy or parity-code the root assignment, giving a difference in the target effect while preserving equality of the observational distribution.

The construction is finite throughout. A product of  $k$  fair coins assigns each exogenous assignment the rational mass  $2^{-k}$ . Thus both separating models lie in the finite rational recursive structural-causal-model fragment. For a larger finite value type, choose an injection of the two bit values into that type and define all structural functions to land in its image. The induced probabilities of assignments outside the image are then zero. This preserves the finite graph witness and the separating binary event. The lemma does not assert a positive-mass countermodel for every ambient assignment; positivity assumptions, when needed for conditional effects, are carried separately in Proposition B.6.6. ■

The restriction to finite rational probabilities is part of the transfer. Successful identification and conditional-identification expressions are built from finite sums, products, marginalisations and quotients of the given observational probabilities, so they remain inside the finite rational probability fragment when

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the input distribution is rational. On the failure side, Lemma B.6.2 gives finite binary countermodels with rational masses for the hedge failures used in the completeness proof. Thus the rational finite subfragment preserves the positive and negative cases needed for the transferred theorem; the claim is not a claim about arbitrary real-valued probability records inside Martin-Löf-style type theory.

This transfer is constructive in the following precise sense. It does not require excluded middle for arbitrary propositions. The case distinctions used by identification, conditional identification and hedge checking are decisions about finite data: membership in a finite set, equality in a finite value type, existence of a finite path, d-separation in a finite graph, containment of finite subgraphs, and positivity or non-zerosness of a rational finite probability. These are decidable predicates in the finite type-theoretic fragment. Thus the algorithmic content imported from the classical proofs can be re-expressed using finite decidable case analysis, not unrestricted classical reasoning.

**Proposition B.6.3** (Constructive finite identification). *In the finite recursive semi-Markovian fragment, the identification procedure can be read as a constructive finite algorithm in the type-theoretic record fragment. It terminates, and each branch condition is decidable.*

*Proof.* The inputs to identification are finite lists or finite indexed families of variables, finite rational probability records, and a finite directed-and-bidirected graph. The branch tests ask whether certain finite sets are empty, whether one finite set is contained in another, whether a finite graph has one or several c-components, whether a variable lies among finite ancestors, and whether a displayed finite product or sum expression is defined. Each of these questions is decidable by finite search.

Termination is by well-founded recursion on an explicit lexicographic measure. For an identification call with input data

$$(G, V, X, Y),$$

where  $G$  is the current finite directed-and-bidirected graph,  $V$  is the finite set of active observed variables, and  $X, Y \subseteq V$  are the finite intervention and outcome sets, define

$$m(G, V, X, Y) := (|V|, |V \setminus X|) \in \mathbb{N}^2,$$

with  $\mathbb{N}^2$  carrying the lexicographic order. The recursive cases of the Shpitser–Pearl identification algorithm [SP06], applied in order so that the first matching case is taken, have the following decrement behaviour:

- if  $X = \emptyset$ , the call returns a finite marginalisation and does not recurse;
- if  $V \setminus \text{An}_G(Y) \neq \emptyset$ , the call recurses on  $V' = \text{An}_G(Y) \subsetneq V$ , strictly decreasing the first component of  $m$ ;
- if  $W := (V \setminus X) \setminus \text{An}_{G_{\overline{X}}}(Y) \neq \emptyset$ , the call recurses with  $X' = X \cup W \supsetneq X$ , leaving  $V$  unchanged and strictly decreasing  $|V \setminus X|$ ;
- if  $C(G[V \setminus X])$  has at least two c-components  $S_1, \dots, S_k$ , the call recurses, for each component, as  $\text{ID}(s_i, v \setminus s_i, P, G)$  with the graph  $G$  and hence  $V$  unchanged; since  $k \geq 2$  and the components partition  $V \setminus X$ , each

$|s_i| < |V \setminus X|$ , so the first component of  $m$  is unchanged and the second component  $|V \setminus X|$  strictly decreases;

- if  $C(G[V \setminus X]) = \{C\}$  with  $C \notin C(G)$  and  $C \subsetneq C'$  for some c-component  $C' \in C(G)$  with  $C' \subsetneq V$ , the call recurses on  $V' = C'$ , strictly decreasing the first component of  $m$ ;
- if  $C(G[V \setminus X]) = \{C\}$  with  $C \in C(G)$ , the call returns a finite product formula and does not recurse;
- if  $C(G[V \setminus X]) = \{C\}$  with  $C(G) = \{V\}$ , the call returns failure with a hedge witness and does not recurse.

Every recursive call therefore has  $m$ -value strictly less than its parent's. The lexicographic order on  $\mathbb{N}^2$  is well founded, so finite induction on  $m$  shows that identification terminates on every input in the fragment. Conditional-identification calls reduce to a finite d-separation search and a finite number of unconditional identification calls on strictly smaller subproblems, so the same lexicographic measure handles them after a single decrement.

No appeal is made to excluded middle for arbitrary propositions: every branch test is a finite decidable predicate. Since Theorem B.5.2 preserves and reflects the finite graph and probability data used by the branch tests, the same finite computation and the same lexicographic measure are available in the type-theoretic record fragment. ■

**Proposition B.6.4** (Identification and conditional-identification trace transport). *Run identification or conditional identification on a finite classical input in the covered fragment, and run the encoded procedure on the corresponding type-theoretic input. The two runs take corresponding branches at every recursive call. If the classical run returns an expression, the type-theoretic run returns the encoded expression with the same rational interpretation. If the classical run returns failure with a hedge witness, the type-theoretic run returns the encoded hedge witness, and conversely for decoded type-theoretic runs.*

*Proof.* A run of identification or conditional identification is a finite well-founded recursion tree whose branch labels are finite graph predicates and finite probability-expression side conditions. Predicate by predicate, the identification branch tests ask whether the intervention set is empty; whether the active node set equals the ancestor set of  $Y$ ; whether a displayed finite set  $W$  is empty; whether the c-component decomposition of a mutilated graph has one component or several; whether the active graph itself is one c-component; whether a component belongs to  $C(G)$ ; and whether there exists a strictly larger c-component containing a given component. These are finite combinations of membership, inclusion, ancestry, mutilation and c-component predicates, all transported by Proposition B.5.5. The conditional-identification branch test asks whether some  $W \in Z$  satisfies the relevant d-separation condition in the appropriate mutilated graph; this is again a finite search over the transported d-separation predicate. The returned probability expressions and their positivity conditions are transported by Proposition B.5.4. Therefore corresponding inputs choose corresponding branches. Induction on the well-founded recursion tree gives correspondence of the returned expression or failure witness. The

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converse direction follows by applying the decoding map  $\text{cl}(-)$  and using the same preservation-and-reflection facts. ■

**Proposition B.6.5** (Constructive hedge witnesses). *In the same fragment, failure of identification transports to an explicit type-theoretic hedge witness, and a type-theoretic hedge witness decodes to an ordinary classical hedge witness.*

*Proof.* A hedge is finite data: two c-forests  $F$  and  $F'$  with the same roots, a finite inclusion  $F' \subset F$ , a finite proof that  $F$  meets the intervention set, a finite proof that  $F'$  does not, and a finite ancestry condition in the graph with arrows into the intervention deleted. By Theorem B.5.2, c-components, c-forests, roots, graph containment, ancestry and graph mutilation are preserved and reflected by the translations. Hence the hedge produced by a classical identification failure encodes as the same finite witness type-theoretically. Conversely, because the type-theoretic side is restricted to records decoded by  $\text{cl}(-)$ , any encoded hedge decodes to the corresponding classical finite graph witness. ■

**Proposition B.6.6** (Conditional-effect correspondence). *Let  $P_x(y \mid z)$  be a finite conditional interventional query in the recursive semi-Markovian fragment. If the denominator  $P_x(z)$  is positive, then*

$$P_{\mathcal{M}_x}(Y = y \mid Z = z) = q \iff P_{\llbracket \mathcal{M} \rrbracket_x}(\llbracket Y = y \rrbracket \mid \llbracket Z = z \rrbracket) = q.$$

*Positivity of the denominator is also preserved and reflected.*

*Proof.* Conditional probability in this thesis is defined only when the conditioning event has positive probability. In the finite rational setting,

$$P_{\mathcal{M}_x}(Y = y \mid Z = z) = \frac{P_{\mathcal{M}_x}(Y = y, Z = z)}{P_{\mathcal{M}_x}(Z = z)}.$$

The numerator and denominator are ordinary finite interventional probabilities, so they are preserved and reflected by Corollary B.5.13. Positivity of the denominator is a decidable order statement about a rational number, and the rational number itself is preserved by the correspondence. Therefore the quotient is defined on one side exactly when it is defined on the other, and the two values agree. ■

**Corollary B.6.7** (Transferred conditional-identification completeness). *The finite conditional-identification completeness result for identifiable conditional effects  $P_x(y \mid z)$  transfers to the type-theoretic record fragment whenever the required conditioning denominators are positive.*

*Proof.* Conditional identification reduces a conditional effect to unconditional identification calls after finite d-separation tests and applications of the action-observation exchange rule. The graph tests are finite decidable predicates preserved by Proposition B.5.5, and the conditional probabilities are preserved by Proposition B.6.6. The unconditional calls are covered by Proposition B.6.3, and the branch-by-branch algorithmic correspondence is Proposition B.6.4. Thus the finite classical conditional-identification derivation transports step by step to the type-theoretic record fragment, with every conditioning step carrying its explicit positive-denominator assumption. ■

**Lemma B.6.8** (Finite rational causal-Bayesian-network functionalisation). *Every finite rational causal Bayesian network on a finite acyclic graph can be represented by a finite recursive structural causal model with independent finite exogenous seeds, deterministic structural functions, and the same induced joint distribution. The same construction applies to the full finite graph before some variables are hidden.*

*Proof.* Fix a topological order  $V_1, \dots, V_n$  of the finite directed acyclic graph. For each variable  $V_i$  and each finite parent assignment  $pa_i$ , the conditional table gives rational numbers

$$P(v_i \mid pa_i)$$

over the finite value type  $\text{Val}(V_i)$ . Since there are only finitely many entries in all rows of this table, choose a common positive denominator  $N_i$  such that every number  $N_i P(v_i \mid pa_i)$  is a natural number. Let the exogenous seed for  $V_i$  be the finite type  $\text{Fin}(N_i)$  with the uniform rational distribution.

For each parent assignment  $pa_i$ , partition  $\text{Fin}(N_i)$  into blocks indexed by  $v_i : \text{Val}(V_i)$ , with the block for  $v_i$  having size  $N_i P(v_i \mid pa_i)$ . Define the structural function  $f_i(pa_i, u_i)$  to be the value whose block contains  $u_i$ . This is a finite deterministic function. The exogenous distribution of the model is the finite product of the uniform seed distributions, so the seeds are independent and rational.

By construction,

$$\Pr(f_i(pa_i, U_i) = v_i) = P(v_i \mid pa_i)$$

for every row of every conditional table. Evaluating the recursive model in the chosen topological order and using independence of the seeds therefore gives

$$\Pr(V_1 = v_1, \dots, V_n = v_n) = \prod_i P(v_i \mid pa_i),$$

which is exactly the Bayesian-network factorisation. All choices made here are finite choices over listed rational tables and finite types, so the construction is available in the finite type-theoretic record fragment. ■

**Definition B.6.9** (Finite latent projection). Let  $G$  be a finite acyclic directed graph whose nodes are partitioned into observed nodes  $V$  and latent nodes  $L$ , in the sense of Definition A.3.24: observed nodes are the endogenous and observed exogenous variables of the model, latent nodes are the unobserved exogenous variables. The finite latent projection of  $G$  onto  $V$  is the directed-and-bidirected graph over  $V$  defined as follows. There is a directed edge  $A \rightarrow B$  between observed nodes when  $G$  contains a directed path from  $A$  to  $B$  whose intermediate nodes, if any, are latent. There is a bidirected edge  $A \leftrightarrow B$  when  $G$  contains a path between  $A$  and  $B$  whose intermediate nodes are latent, whose endpoint arrowheads point into  $A$  and  $B$ , and whose intermediate latent nodes are non-colliders on that path. Equivalently, in the acyclic latent-variable setting used here, the bidirected edge records a finite latent common-cause path between the two observed endpoints.

The canonical edge-latent representative of the projected graph is obtained by replacing each bidirected edge  $A \leftrightarrow B$  by a fresh latent root  $U_{AB}$  with

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arrows  $U_{AB} \rightarrow A$  and  $U_{AB} \rightarrow B$ . This representative is used for the semi-Markovian graph predicates in the completeness transfer. It is not asserted to be judgmentally identical to the original latent causal Bayesian network; the relevant claim is the classical identifiability-preservation result for this finite projection. Since  $G$ ,  $V$  and  $L$  are finite, the path searches and edge tests in the projection are decidable finite predicates and can be represented in the restricted type-theoretic record fragment.

**Proposition B.6.10** (Finite semi-Markovian projection). *The finite latent projection of Definition B.6.9, followed by its canonical edge-latent semi-Markovian representative, is representable in the type-theoretic fragment, and its classical identifiability-preservation theorem transports across the correspondence.*

*Proof.* By Lemma B.6.8, the finite rational causal Bayesian network can first be represented as a finite recursive structural causal model on the full graph, including latent variables. The graph step is then exactly the finite latent projection of Definition B.6.9: retain observed variables, add directed edges for directed latent-mediated paths, and add bidirected edges for latent common-cause paths. Its canonical edge-latent representative adds one fresh finite latent root for each bidirected edge. In the type-theoretic setting, all of these are finite list or finite indexed-family operations with decidable edge and path predicates.

The claim needed here is not that the original latent model and the canonical edge-latent representative are judgmentally identical, nor that the thesis has reproved the Huang–Valtorta reduction. The claim is that the finite projection and the target query are represented on both sides of the correspondence, so the classical identifiability-preservation theorem used by Huang–Valtorta can be transported [HV06]. Since the functionalised structural causal model, the finite projected graph, its canonical edge-latent representative and the relevant interventional expressions are finite data preserved by Theorem B.5.2, the preservation result applies to the restricted type-theoretic record fragment. ■

**Theorem B.6.11** (Transferred do-calculus completeness). *In the finite, discrete, recursive structural-causal-model fragment covered by Theorem B.5.2, every identifiable encoded ordinary interventional distribution  $\llbracket P_x(y) \rrbracket$ , with  $P_x(y)$  read as  $P(Y = y \mid \text{do}(X = x))$ , has a type-theoretic do-calculus derivation reducing it to an encoded observational expression.*

*Proof.* Suppose  $\llbracket P_x(y) \rrbracket$  is identifiable in the type-theoretic record fragment. By Proposition B.5.14, the corresponding classical query  $P_x(y)$  is identifiable in the classical finite structural-causal-model fragment. By the published Shpitser–Pearl and Huang–Valtorta completeness results, there exists a finite classical derivation, using do-calculus and standard probability manipulations, that reduces  $P_x(y)$  to an expression containing only observational quantities.

By Proposition B.5.15, this finite derivation translates step by step into the type-theoretic setting. The final expression is observational classically, and observability is preserved by the encoding because the expression contains no remaining intervention operator. Hence the translated derivation reduces  $\llbracket P_x(y) \rrbracket$  to an encoded observational expression. That is exactly completeness of do-calculus for the encoded finite fragment. ■

**Corollary B.6.12** (Transferred failure witnesses). *If the classical identification algorithm fails because a hedge witnesses non-identifiability, then the encoded hedge witnesses non-identifiability in the type-theoretic fragment.*

*Proof.* Hedges are finite graph-theoretic objects built from c-forests, ancestry, containment and the intervention-mutilated graph. These operations are preserved and reflected by Proposition B.5.5. Therefore a classical hedge translates into an encoded hedge with the same root, containment and ancestry properties. Since identifiability itself is transported by Proposition B.5.14, and since identification failure traces are transported by Proposition B.6.4, the encoded hedge witnesses the corresponding encoded failure of identifiability. ■

**Theorem B.6.13** (Finite causal-completeness transfer). *In the restricted finite rational recursive type-theoretic structural-causal-model fragment of Definition B.5.1, the ordinary Pearl–Tian–Shpitser completeness results covered in this section carry over in the stated transfer sense: completeness of identification for finite ordinary interventional distributions, completeness of do-calculus for those effects, hedge witnesses for identification failure, completeness of conditional identification for finite conditional effects with positive denominators, and the finite functionalisation and semi-Markovian projection steps used to reduce the Huang–Valtorta setting.*

*Proof.* The underlying model classes are equivalent by Proposition B.5.6. Observational and interventional probabilities are preserved by Theorem B.5.2, composite probability expressions by Proposition B.5.4, graph predicates by Proposition B.5.5, and the identification and conditional-identification branch traces by Proposition B.6.4. Identifiability itself is transported by Proposition B.5.14. The finite rational negative cases are supported by Lemma B.6.2, and the causal-Bayesian-network reduction is supported by Lemma B.6.8 and Proposition B.6.10. The published finite semi-Markovian completeness theorems can therefore be applied on the classical side and then transported back to the restricted type-theoretic side. Conditional effects use Proposition B.6.6, so their positive-denominator assumptions are part of the transported statement. ■

The scope of this transfer should be kept explicit. It covers the finite ordinary interventional-distribution fragment used above, quantified over the restricted type-theoretic structural-causal-model records that decode to ordinary finite recursive structural causal models. It also covers finite conditional effects once their positive-denominator side conditions are made explicit, and it covers the finite causal-Bayesian-network functionalisation and semi-Markovian projection steps used by the Huang–Valtorta reduction. Finite rational probability is the intended target of the theorem, not an approximation to a hidden real-valued result. The transfer does not by itself cover cyclic or non-recursive structural causal models, arbitrary type-theoretic causal records outside the decoded fragment, general counterfactual identification, transportability, meta-synthesis or path-specific effects. Those results may be treated similarly only after their additional objects and operations receive their own correspondence theorems.



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*“Contrarily, it is characteristic of a truly wise man that, being just a mortal, he doesn’t want to transgress any boundaries with his wisdom and together with humanity either turns a blind eye, or, out of courtesy, wanders along”*

Folly in *The Praise of Folly*

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